

Cone bipolar cell synapses generate transient versus sustained signals in parallel ON pathways of the mouse retina

Reviewed Preprint

v2 • August 21, 2025

Revised by authors

Reviewed Preprint

v1 • August 20, 2024

Sidney P Kuo, Wan-Qing Yu, Prerna Srivastava, Haruhisa Okawa, Luca Della Santina, David M Berson, Gautam B Awatramani, Rachel OL Wong, Fred Rieke 

Department of Neurobiology and Biophysics, University of Washington, Seattle, United States • Department of Neuroscience, University of Minnesota, Minneapolis, United States • Department of Biology, University of Victoria, Victoria, Canada • Department of Neuroscience, Brown University, Providence, United States

 https://en.wikipedia.org/wiki/Open_access

 Copyright information

eLife Assessment

This **important** study shows that retinal bipolar cell subtype-specific differences in the size of synaptic ribbon-associated vesicle pools contribute to the transient versus sustained kinetics of the responses of retinal ganglion cells. The data are extensive and **compelling**. This work will be of broad interest to researchers working on synaptic transmission, retinal signal processing, and sensory neurobiology.

<https://doi.org/10.7554/eLife.98817.2.sa4>

Abstract

Parallel processing is a fundamental organizing principle in the nervous system and understanding how parallel neural circuits generate distinct outputs from common inputs is a key goal of neuroscience. In the mammalian retina, divergence of cone signals into multiple feedforward bipolar cell pathways forms the initial basis for parallel retinal circuits dedicated to specific visual functions. Here, we used patch-clamp electrophysiology, electron microscopy and two photon imaging of a fluorescent glutamate sensor to examine how kinetically-distinct responses arise in transient versus sustained ON alpha RGCs (ON-T and ON-S RGCs) of the mouse retina. We directly compared the visual response properties of these RGCs with their presynaptic bipolar cell partners, which we identified using 3D electron microscopy reconstruction. Different ON bipolar cell subtypes (type 5i, type 6 and type 7) had indistinguishable light-driven responses whereas extracellular glutamate signals around RGC dendrites and postsynaptic excitatory currents measured in ON-T and ON-S RGCs in response to the identical stimuli used to probe bipolar cells were kinetically distinct. Anatomical examination of the bipolar cell axon terminals presynaptic to ON-T and ON-S RGCs suggests that bipolar subtype-specific differences in the size of synaptic ribbon-associated vesicle pools may contribute to transient versus sustained kinetics. Our findings indicate that feedforward bipolar cell synapses are a primary point of divergence in kinetically distinct visual pathways.

Introduction

Divergence into parallel pathways allows neural circuits to generate diverse outputs from a common set of inputs. This functional diversity can be quite striking; in vision, for example, light inputs initially encoded by the same collection of cone photoreceptors mediate sensitivity to several kinds of motion, to color and to spatial structure, all in the same region of space. These diverse functional sensitivities arise through parallel processing of light inputs, first in the retina and subsequently in downstream visual areas. Here we focus on the origin of transient and sustained responses – a key differentiating characteristic of retinal output cells across species (Awatramani and Slaughter, 2000 [↗](#); Baden et al., 2016 [↗](#); Bae et al., 2018 [↗](#); Cleland et al., 1973 [↗](#); Ikeda and Wright, 1972 [↗](#); Roska and Werblin, 2001 [↗](#); Werblin, 1970 [↗](#)).

The retina provides a unique opportunity to study the mechanisms that differentiate signals in parallel pathways. We have a near-complete list of retinal cell types and substantial information about which of these cell types participate in specific parallel pathways (Helmstaedter et al., 2013 [↗](#); Masland, 2012 [↗](#)). Under daylight conditions, retinal processing starts with cone photoreceptors transducing light inputs and providing divergent output to ~13 types of bipolar cell (Euler et al., 2014 [↗](#); Greene et al., 2016 [↗](#); Helmstaedter et al., 2013 [↗](#); Shekhar et al., 2016 [↗](#); Tsukamoto and Omi, 2017 [↗](#)). The resulting bipolar cell signals are transmitted to ~20-40 retinal ganglion cell (RGC) types (Baden et al., 2016 [↗](#); Bae et al., 2018 [↗](#); Goetz et al., 2022 [↗](#)). RGC responses are also strongly shaped by the activity of a large and diverse collection of amacrine cells that receive input from bipolar cells and provide presynaptic inhibition to bipolar cells and postsynaptic inhibition to RGCs (Diamond, 2017 [↗](#); Jazdzinsky and Baccus, 2013 [↗](#)). Finally, RGC responses are shaped by feedforward synaptic properties, particularly synaptic depression and facilitation (DeVries, 2000 [↗](#); Grabner et al., 2016 [↗](#); Jarsky et al., 2011 [↗](#); Lagnado and Schmitz, 2015 [↗](#); Nikolaev et al., 2013 [↗](#); Oesch and Diamond, 2011 [↗](#)).

These mechanisms collectively produce highly distinct responses across RGC types - with cells that sample inputs on a variety of spatial and temporal scales, cells with distinct chromatic properties, cells that sense the direction of moving objects, and cells that separate local from global motion (reviewed by Golisch and Meister, 2010 [↗](#); Kerschensteiner, 2022 [↗](#)). Several mechanisms contribute to these functional differences, starting with the division of signals into ON and OFF pathways at the cone-to-bipolar synapse (Euler et al., 2014 [↗](#); Slaughter and Miller, 1981 [↗](#)). Temporal properties of responses of different bipolar cells can also differ for some stimuli (Awatramani and Slaughter, 2000 [↗](#); Baden et al., 2013 [↗](#); Euler et al., 2014 [↗](#); Euler and Masland, 2000 [↗](#); Ichinose et al., 2014 [↗](#); Ichinose and Hellmer, 2016 [↗](#); Wu et al., 2000 [↗](#)). In the inner retina, integration of excitatory and inhibitory synaptic inputs produces directionally-selective responses in specific RGC types (Taylor and Vaney, 2002 [↗](#); Vaney et al., 2012 [↗](#)) and differences in spatial weighting functions in other RGCs (Cafaro and Rieke, 2013 [↗](#); Roska and Werblin, 2001 [↗](#); Zhang et al., 2012 [↗](#)). Properties of dendritic integration and RGC spike generation also differentiate responses of some RGCs (Margolis et al., 2010 [↗](#); Ran et al., 2020 [↗](#); Wienbar and Schwartz, 2022 [↗](#)).

A fundamental difference among RGCs is whether their responses are sustained or transient (Awatramani and Slaughter, 2000 [↗](#); Baden et al., 2016 [↗](#); Bae et al., 2018 [↗](#); Cleland et al., 1973 [↗](#); Ikeda and Wright, 1972 [↗](#); Roska and Werblin, 2001 [↗](#)). Cone photoreceptors respond to changes in light intensity with graded potentials that have both a transient and sustained component (Angueyra and Rieke, 2013 [↗](#); Cangiano et al., 2012 [↗](#); Szikra et al., 2014 [↗](#)). Both kinetic components of the cone response are maintained in the firing rates of sustained cells, which track the prevailing contrast. However, only the initial component of the cone response is preserved in the spiking response of transient cells, which therefore respond primarily to changes in contrast. Where and how these differences originate is not clear. Here, we show that the distinct kinetics of

responses of ON-sustained alpha and ON-transient alpha RGCs (1) persist for stimuli that elicit near-identical responses among the relevant bipolar cells, (2) persist without amacrine cell-mediated inhibition and, (3) are correlated with anatomical differences in output synapses of bipolar cell subtypes that are presynaptic to these RGCs. These findings highlight the importance of feedforward bipolar cell output synapses in producing kinetically distinct retinal outputs.

Results

We focused on two ON RGC cell types, ON-transient alpha (ON-T) and ON-sustained alpha (ON-S) RGCs, that could be readily identified in a flat-mount preparation of the isolated mouse retina (**Fig. 1A**) by their characteristic spiking response to a step increment in light intensity (**Fig. 1B**) (Bae et al., 2018; Farrow et al., 2013; Goetz et al., 2022; Krieger et al., 2017; Kuo et al., 2016). Based on their light response, dendritic field size and stratification pattern in the inner plexiform layer (IPL) (see below; **Fig. S4**), the ON-T RGCs we studied here likely correspond to the PV2 RGC type of Farrow et al. (2013), 6sw RGC type of Bae et al. (2018) and the medium-field ON-transient RGC of Goetz et al. (2022). ON-S RGCs correspond to the PV1 RGC type of Farrow et al. (2013) and 8w RGC type of Bae et al. (2018).

We characterized the temporal spiking response properties of ON-T and ON-S RGCs by using an extracellular cell-attached patch electrode to measure action potential firing from either cell type while presenting a spatially uniform, randomly fluctuating light stimulus (Gaussian noise) that allowed us to efficiently probe a variety of stimulus contrasts and temporal frequencies.

These and all subsequent physiological recordings were conducted using a photopic mean background light level (~ 3300 R*/rod/s; ~ 2000 P*/S-cone/s; ~ 1650 P*/M-cone/s) to suppress rod photoreceptor responses and focus on cone-mediated signaling (Grimes et al., 2018; Ke et al., 2014). **Figure 1C** shows the spiking response of example ON-T and ON-S RGCs to multiple repetitions of the same random stimulus contrast sequence; these RGC types clearly responded differently to temporally modulated light input. In particular, noise-evoked ON-T RGC spike responses were comprised primarily of large amplitude, less frequent spike bursts whereas ON-S RGC responses were less discrete and consisted of events with a wider range of amplitudes (**Fig. 1C** and Fig S1A).

We summarized the response of each ON-T and ON-S RGC by computing the average stimulus preceding each recorded spike (the spike-triggered average or STA stimulus) when presenting a non-repeated Gaussian noise stimulus (**Fig. 1D**) (Chichilnisky, 2001). We quantified response kinetics from these STA waveforms by measuring the time of zero-crossing (between trough and peak) as well as the ratio of amplitudes of the trough and peak (biphasic index). ON-T RGC STAs were significantly more biphasic (biphasic index = 0.63 ± 0.07 ; mean $\pm$ S.D.; $n = 15$) than ON-S RGC STAs (biphasic index = 0.43 ± 0.11 ; mean $\pm$ S.D.; $n = 13$) (**Fig. 1E**, right), consistent with their more transient responses to step stimuli (**Fig. 1B**). ON-T RGC responses were also slightly delayed (zero-cross time = -0.080 ± 0.0063 sec; mean $\pm$ S.D.; $n = 15$) relative to those of ON-S RGCs (zero-cross time = -0.072 ± 0.0052 sec; mean $\pm$ S.D.; $n = 13$) (**Fig. 1C**, bottom and 1E, left). The STA provides a convenient summary of RGC temporal response properties, but it does not capture all differences between the RGC spike responses – e.g. the discreteness of the ON-T responses in **Figure 1C**. Nonetheless, the combination of direct observation of the spike responses and STA kinetics illustrate how ON-S and ON-T responses differ.

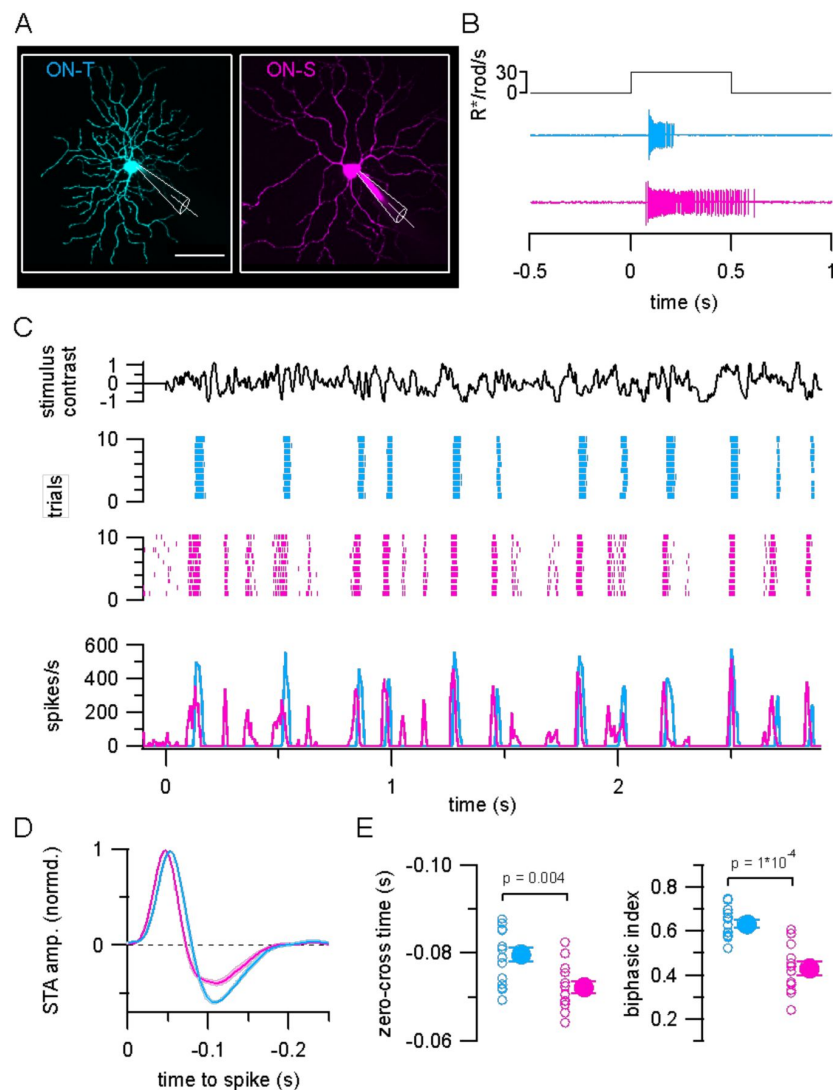


Figure 1.

Distinct temporal spiking response properties of ON-transient (ON-T) versus ON-sustained (ON-S) RGCs.

(A) Maximum intensity projection z-stack fluorescence images of ON-T (left, cyan) and ON-S (right, magenta) RGCs dialyzed with Alexa Fluor 594 during whole-cell patch-clamp recording. Scale bar 50 μm . (B) Characteristic spiking responses to a light increment for example ON-T and ON-S RGCs. (C) Example spiking responses to repeated trials of spatially uniform (52.0 μm spot diameter) Gaussian noise stimulus (top black trace) presented with a mean luminance of ~ 20.00 $P^*/\text{S-cone/s}$ (~ 33.00 $R^*/\text{rod/s}$; ~ 1650 $P^*/\text{M-cone/s}$). ON-T (cyan) and ON-S (magenta) RGCs were recorded sequentially in the same piece of retina. Average spike responses (bottom) calculated from 20 repetitions of the noise stimulus. (D) ON-T and ON-S STAs derived from responses to non-repeated noise stimuli. Colored lines and shaded areas show mean $\pm$ SEM STA waveforms from ON-T ($n = 15$) and ON-S ($n = 13$) RGCs. The STA time axis is reversed to facilitate comparison with linear filters in **Figures 2** and **4**. The mean line obscures the SEM shading for much of the filter in both cell types. (E) Quantification of STA kinetics in different RGCs. Open circles show measurements from individual cells and filled circles with error bars show mean $\pm$ SEM values. Significance values from Wilcoxon rank sum tests.

Bipolar cell to RGC synapses are an important site of signal diversification in ON pathways

A variety of mechanisms could contribute to the distinct temporal response characteristics of ON-T versus ON-S RGCs, including differences in excitatory and/or inhibitory input conveyed by presynaptic circuits. Parallel visual channels are initially established by the divergence of cone signals to different cone bipolar cell subtypes, which have axons that terminate at distinct depths within the IPL and can correspondingly contact distinct RGC types (Euler et al., 2014 [↗](#); Wässle et al., 2009 [↗](#)). Prior studies have demonstrated IPL layer-specific light-evoked glutamate release kinetics from different bipolar cell synaptic terminals (Borghuis et al., 2013a [↗](#); Franke et al., 2017 [↗](#)). We therefore sought to understand to what extent kinetic differences in bipolar cell-mediated excitatory synaptic input could account for the transient versus sustained responses of these RGCs.

To do this, we used somatic whole-cell voltage-clamp recordings to measure excitatory synaptic currents from both RGC types while stimulating the retina with the same spatially uniform Gaussian noise stimulus that we used to characterize RGC spike output (Fig. 2A [↗](#)). We used these measurements to construct linear-nonlinear (LN) models comprised of a linear temporal filter (Fig. 2B [↗](#) and Fig. S2A [↗](#)) and a time-invariant (static) nonlinearity (Fig. S2B [↗](#)) that together provide a simple estimate of the transformation between visual input and excitatory synaptic input to each RGC type (Kim and Rieke, 2001 [↗](#)). LN models for each RGC type accurately predicted measured responses to noise stimuli that were not used to construct the models, although LN models of ON-S RGC responses performed better than those for ON-T RGCs (correlation coefficient between average response to repeated noise stimulus and model prediction: 0.61 ± 0.06 (ON-S, $n = 9$) and 0.50 ± 0.13 (ON-T, $n = 11$), mean $\pm$ S.D.; $p = 0.028$ Wilcoxon rank-sum test). ON-T stimulus-evoked excitatory currents were biased towards large amplitude events compared to those of ON-S RGCs (Fig. 2A [↗](#), Fig. S1B [↗](#)), like the distinct amplitude distributions of spike events in these cell types noted above (Fig. 1C [↗](#) and Fig. S1A [↗](#)).

The linear filter captures the kinetics of the RGC response and is analogous to the time-reversed STA derived from spike responses (Chichilnisky, 2001 [↗](#)). As with the STAs in Figure 1D [↗](#), excitatory linear filters were slower and more biphasic in ON-T (zero-cross time = 0.10 ± 0.013 s; biphasic ind. = 0.45 ± 0.090 ; mean $\pm$ S.D.; $n = 13$) compared to ON-S RGCs (zero-cross time = 0.083 ± 0.0074 s; biphasic ind. = 0.31 ± 0.10 ; mean $\pm$ S.D.; $n = 15$) (Fig. 2C [↗](#)). This latter difference is consistent with more transient light-evoked excitatory synaptic input to ON-T RGCs. Indeed, light step-evoked excitatory currents to ON-T RGCs decayed to baseline levels within a few hundred milliseconds, even in the continued presence of light input, whereas excitatory inputs to ON-S RGCs had a sustained component that persisted for the duration of the light step (Fig. 2D-E [↗](#)).

These results show that transient versus sustained kinetics are already present in the excitatory pathways that provide input to ON-T and ON-S RGCs. However, these findings do not rule out additional contributions to the distinct spiking behavior of these RGC types from amacrine cell-mediated inhibitory synaptic input to the RGCs or from different intrinsic spiking properties. To examine whether postsynaptic inhibition contributes importantly to kinetic differences between ON-T and ON-S RGCs, we measured both excitatory and inhibitory synaptic currents in response to Gaussian noise stimuli in a subset of RGC recordings (Fig. S3 [↗](#)). Like excitatory currents, light-evoked inhibitory currents were more transient in ON-T RGCs (linear filter biphasic ind. = 0.34 ± 0.12 ; mean $\pm$ S.D.; $n = 8$) compared to those in ON-S RGCs (biphasic ind. = 0.16 ± 0.09 ; mean $\pm$ S.D.; $n = 8$) (Fig. S3B-C [↗](#); $p = 0.01$, Wilcoxon rank sum test). We also quantified the relative strength of excitatory and inhibitory synaptic inputs by calculating the ratio between the largest noise-evoked excitatory and inhibitory conductances (G_{exc} and G_{inh}; derived from the measured currents) in response to a repeated noise stimulus in each RGC (Fig. S3A [↗](#), D-E). This ratio did not differ significantly between the two cell types (ON-T RGCs: peak G_{exc} / peak G_{inh} = 1.46 ± 0.65 ; mean $\pm$

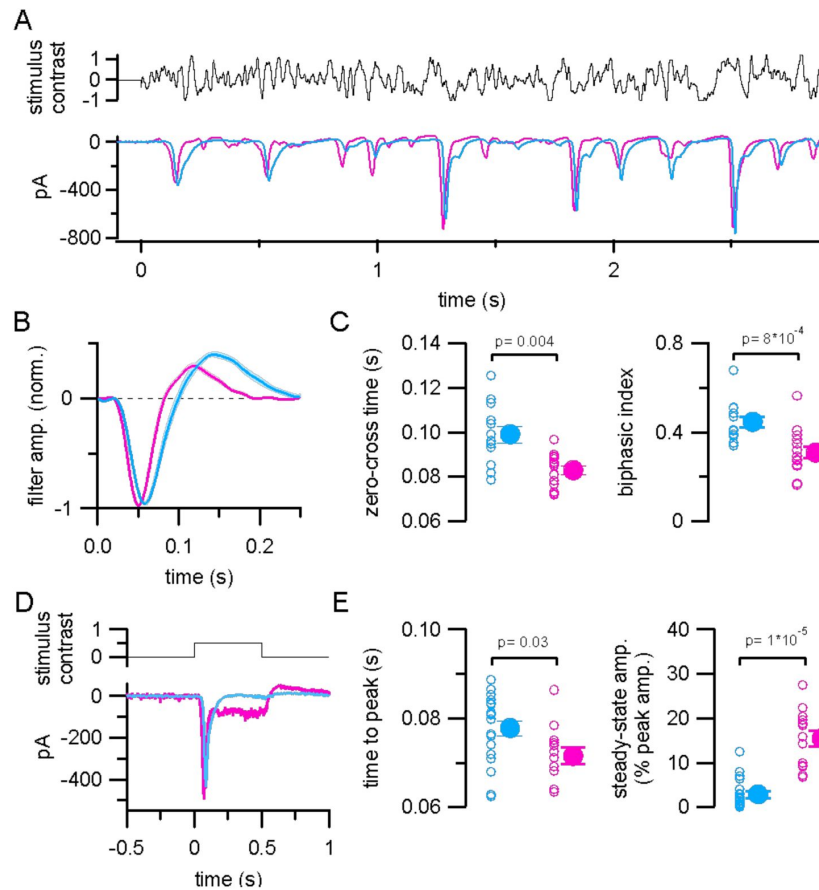


Figure 2.

Distinct excitatory synaptic input to ON-T and ON-S RGCs.

(A) Excitatory synaptic currents measured in example ON-T (cyan) and ON-S (pink) RGCs in response to the same Gaussian noise stimulus used in **Fig. 1**. Traces show average currents from ten repetitions of the same stimulus and are baseline subtracted to facilitate comparison. (B) Linear filters computed from responses to non-repeated noise stimulus presentation. Lines and shaded areas show mean $\pm$ SEM of responses from ON-T ($n = 13$) and ON-S ($n = 15$) RGCs. The mean line obscures the SEM shading for much of the filter in both cell types. (C) Quantification of filter kinetics in different RGCs. (D) Excitatory currents measured from example ON-T and ON-S RGCs in response to a 0.5 second step increase in light intensity (50% contrast; mean response to five repetitions of stimulus). (E) Quantification of step response kinetics ($n = 19$ ON-T RGCs; $n = 13$ ON-S RGCs). In (C) and (E) open circles show measurements from individual cells and filled circles with error bars show mean $\pm$ SEM values. Significance values from Wilcoxon rank sum tests.

S.D.; $n = 6$; ON-S RGCs: peak Gexc / peak Ginh = 4.04 ± 2.48 ; mean $\pm$ S.D.; $n = 7$; $p=0.10$, Wilcoxon rank sum test). The relatively small inhibitory inputs together with the clear differences we observed in excitatory synaptic input between ON-T and ON-S RGCs indicate that bipolar cell pathways have a central role in establishing transient versus sustained kinetics.

ON-T and ON-S RGCs receive input from distinct bipolar types

Retinal circuits dedicated to transient versus sustained signaling are generally located at the center and borders of the IPL (Baden et al., 2016 [↗](#); Bae et al., 2018 [↗](#); Beaudoin et al., 2019 [↗](#); Borghuis et al., 2013b [↗](#); Franke et al., 2017 [↗](#); Roska and Werblin, 2001 [↗](#)). We compared stratification patterns of ON-T and ON-S RGCs in a subset of whole-cell recordings by filling cells with a fluorescent dye and imaging the dendrites of physiologically-characterized RGCs using two-photon microscopy. We assessed IPL depth in live tissue by using retinas from a transgenic mouse line in which type 5 bipolar cells strongly express EGFP (*Gjd2-EGFP* mice). These EGFP+ bipolar cells express calcium binding protein 5 (CaBP5; **Fig. S3B-D** [↗](#)) and have axons that occupy a narrow band just above and overlapping with the dendrites of ON starburst amacrine cells (**Fig. S3A** [↗](#)) (Ghosh et al., 2004 [↗](#)). Based on the regular spacing of their somas, these cells likely correspond to one of the three subtypes of type 5 bipolar cells previously identified in mouse retina (Greene et al., 2016 [↗](#); Rodieck, 1991 [↗](#)) (**Fig. S3E-F** [↗](#)). Imaging of RGCs together with EGFP+ axon terminals in *Gjd2-EGFP* mice revealed that ON-T and ON-S RGC dendrites occupy distinct, mostly non-overlapping regions of the IPL (**Fig. S4** [↗](#)).

The distinct IPL stratification of ON-T and ON-S RGCs indicates they receive input from different bipolar cell populations. Recent serial block-face electron microscopy (SBFSEM) connectomic analyses found that type 6 and type 7 bipolar cells account for ~50-52% and 46% of the excitatory synapses formed upon ON-S RGC dendrites (Sabbah et al., 2018 [↗](#); Swygart et al., 2024 [↗](#)). However, the bipolar cell subtype(s) presynaptic to ON-T RGCs have not previously been identified. Our observation that ON-T RGC dendrites occupy a narrow band in the IPL that closely matches the stratification pattern of EGFP+ type 5 bipolar cell axons in *Gjd2-EGFP* mice (**Fig. S5** [↗](#)) suggests that these bipolar cells likely provide a substantial fraction of the excitatory input to ON-T RGCs. To test this possibility, we transfected retinas from *Gjd2-EGFP* mice with a plasmid encoding PSD-95 fused to the red fluorescent reporter mCherry to label postsynaptic sites of excitatory input (**Fig. S5** [↗](#)). We used confocal microscopy to quantify the fraction of putative synapses, defined as sites with overlapping PSD-95 and EGFP fluorescent signals, that arose from *Gjd2-EGFP*+ axon terminals (Schwartz et al., 2012 [↗](#)). From this analysis, we concluded that the labeled type 5 bipolar cells in *Gjd2-EGFP* mice provide ~40% of the ribbon synaptic input to ON-T RGCs (**Fig. S5D** [↗](#)).

We next performed correlative fluorescence imaging and SBFSEM to identify which type 5 bipolar subtype is labeled in *Gjd2-EGFP* mice and to further characterize what other bipolar cell types provide the remaining input to ON-T RGCs. We first used 2P microscopy to image the dendritic morphology of two dye-filled ON-T RGCs that were identified from their light-evoked spike responses in flat-mounted retina from *Gjd2-EGFP* transgenic mice (**Fig. 3A** [↗](#)). After 2P imaging, we fixed the retina and used the near infrared branding (NIRB) technique to create fiducial marks in the tissue around the dye-filled ON-T RGCs (Bishop et al., 2011 [↗](#)). After acquiring the SBFSEM volume, we located the somas of the two ON-T RGCs using these fiducial marks and reconstructed the dendritic arbors of both cells (e.g. **Fig. 3B** [↗](#)). The dendritic morphologies from our EM reconstruction matched those from the light microscopy images. This suggests that we successfully identified the two ON-T RGCs recorded physiologically in our EM volume. We chose the two longest dendritic arbors from the RGC soma to the dendritic tip and identified all the synapses from bipolar cells along these dendrites (**Fig. 3C** [↗](#)). We then reconstructed all the presynaptic bipolar cells and classified them according to their axon morphology and stratification (**Fig. 3D** [↗](#)). Careful examination of axonal tiling among type 5i, 5o and 5t subtypes provided additional

confirmation of the classification of these type 5 bipolar cells (**Fig. S7A**) (Greene et al., 2016). When comparing the EM images with light microscopy images, it is clear that *GJD2-GFP*⁺ bipolar cells are type 5i bipolar cells (**Fig. S7B**).

Results from the SBFSEM reconstructions agreed closely with those from the light microscopy approach described above. Type 5i bipolar cell axon terminals provided the largest fraction of synapses onto the two ON-T RGCs within our SBFSEM volume (~30-35%). Other inputs arose from type 5o (~10%), type 5t (~5-15%), XBC (~15-30%), and, somewhat surprisingly given the differences in stratification, from type 6 (~20%) bipolar cells (**Fig. 3E**). We confirmed this connectivity pattern by additionally analyzing bipolar cell synaptic inputs to a putative ON-T RGC (RGC7089) in the same publicly available SBFSEM volume (Ding et al., 2016) that was used by Sabbah et al., (2018) to map inputs to ON-S RGCs (**Fig. 3E**).

Bipolar cell light responses differ minimally across types

Are transient versus sustained signals already differentiated in the voltage responses of the different bipolar cells presynaptic to ON-T and ON-S RGCs? Or is the distinct excitatory synaptic input that these RGCs receive primarily due to differences at the bipolar cell to RGC synapse? To answer these questions, we compared the light-driven response properties of the bipolar cell subtypes that comprise most of the excitatory input to ON-T or ON-S RGCs. We made targeted electrophysiological recordings from type 5i, type 6 and type 7 bipolar cells in retinal slices from three transgenic mouse lines in which these different bipolar cell subtypes express fluorescent proteins (see Methods). In these experiments, we used somatic whole-cell perforated-patch recordings to limit perturbation of the G-protein signaling cascade that mediates ON bipolar cell light responses.

Somatic resting membrane potentials did not differ significantly among these bipolar cell types (mean $\pm$ S.D. V_{rest} : -38 ± 6.3 mV in type 5i ($n = 11$); -33.7 ± 6.3 mV in type 6 ($n = 7$); -36.5 ± 4.7 mV in type 7 ($n = 6$); Kruskal-Wallis test, $p = 0.35$; values are not corrected for the liquid junction potential since that is hard to estimate in perforated-patch recordings). We characterized light responses using the same spatially-uniform Gaussian noise stimulus (**Fig. 4B**) and linear-nonlinear model analysis (**Fig. 4C** and **Fig. S7**) as in RGC recordings to facilitate direct comparison of results from bipolar cells and RGCs (Kim and Rieke, 2001; Rieke, 2001). LN models of bipolar cell responses accurately predicted responses to noise stimuli (correlation coefficient between average response to repeated noise stimulus and LN model prediction = 0.75 ± 0.15 for $n = 18$ bipolar cells ($n = 13$ type 5; $n = 3$ type 6; $n = 2$ type 7)). Unlike for ON-T and ON-S RGCs, temporal characteristics of type 5i, type 6 and type 7 bipolar cell linear filters were indistinguishable from each other under our recording conditions (**Fig. 4B-D**). Although there was some variability in the temporal response properties across the population of bipolar cells we recorded, these differences were not associated with morphologically and molecularly defined bipolar cell types. Rather, we observed variability in response kinetics within each bipolar cell subtype and temporal responses thus overlapped substantially across different type 5i, type 6 and type 7 bipolar cells (**Fig. 4** and **Fig. S7**).

Responses to light steps measured in a subset of type 5i and type 6 bipolar cell recordings confirmed our conclusion from Gaussian noise stimuli that different bipolar cell types exhibit nearly identical light response kinetics under our recording conditions (**Fig. 4E-F**). Unlike the clear differences in light step-evoked excitatory currents we measured in ON-T and ON-S RGCs (**Fig. 2D-E**), type 5i and type 6 bipolar cells both had sustained responses to light steps. These findings indicate that mechanisms upstream of the generation of the cone bipolar cell voltage response, such as synaptic transmission from cones to cone bipolar cells and intrinsic response properties of the cone bipolar cells, cannot explain the kinetically distinct responses in parallel ON channels mediated by type 5i, type 6 or type 7 bipolar cells. Consistent with this conclusion, ON-T and ON-S RGCs exhibit kinetically distinct light step-evoked spiking responses under scotopic conditions (not shown), where much of the RGC response originates from signals carried from

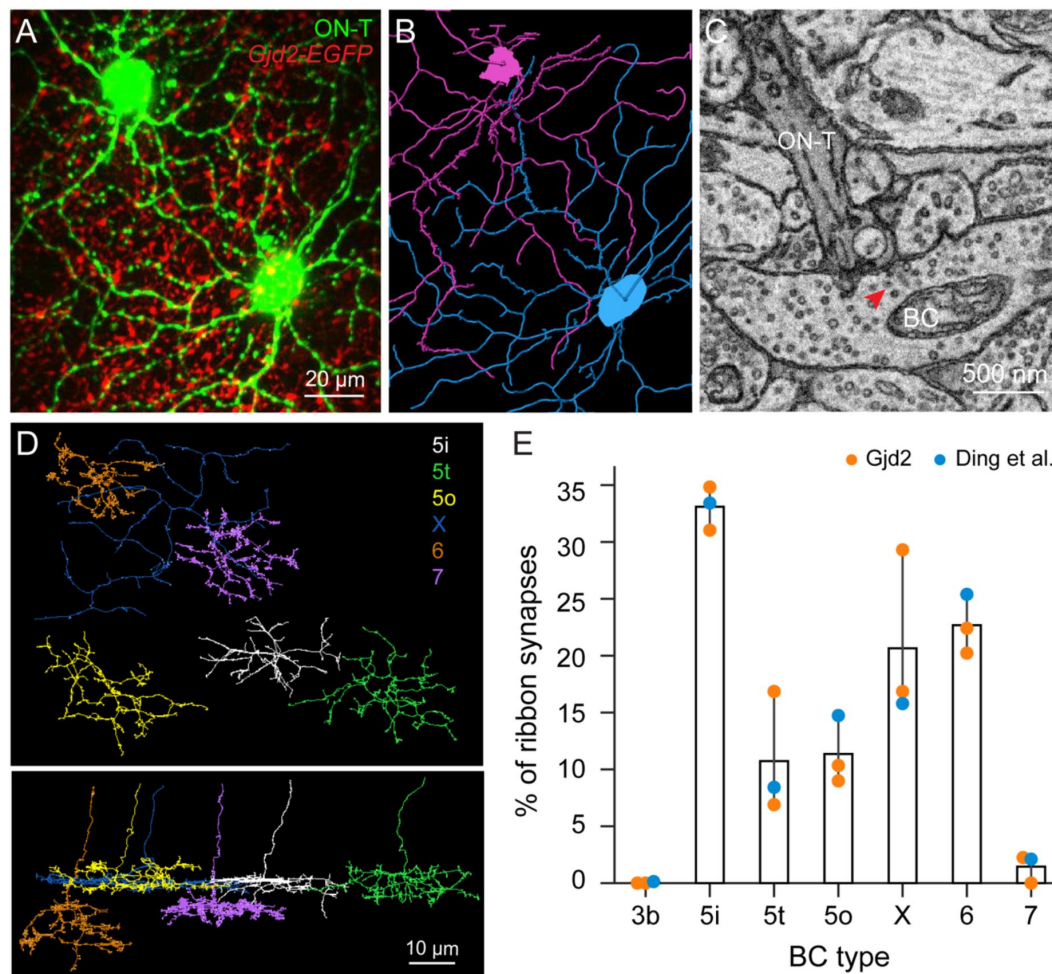


Figure 3.

Identification of bipolar cell subtypes presynaptic to ON-T RGCs.

(A) Maximum-projection z-stacks of two dye-filled ON-T RGCs (green) in wholemount retina from Gjd2-EGFP (red) transgenic mice. (B) Traced skeletons of dendritic arbors of the two ON-T RGCs from EM volume. (C) An EM micrograph showing a ribbon synapse (arrowhead) on a BC axonal terminal onto the dendrite of an ON-T RGC. (D) An example skeleton for each presynaptic BC type (types 5i, 5t, 5o, X, 6 and 7) of ON-T RGCs. Top: whole-mount view, Bottom: side view. (E) Bar plot showing the proportion of input synapses to ON-T RGCs from each BC type. Mean $\pm$ SEM (n=3 RGCs, 2 from this study and one from Ding et al. (2016)).

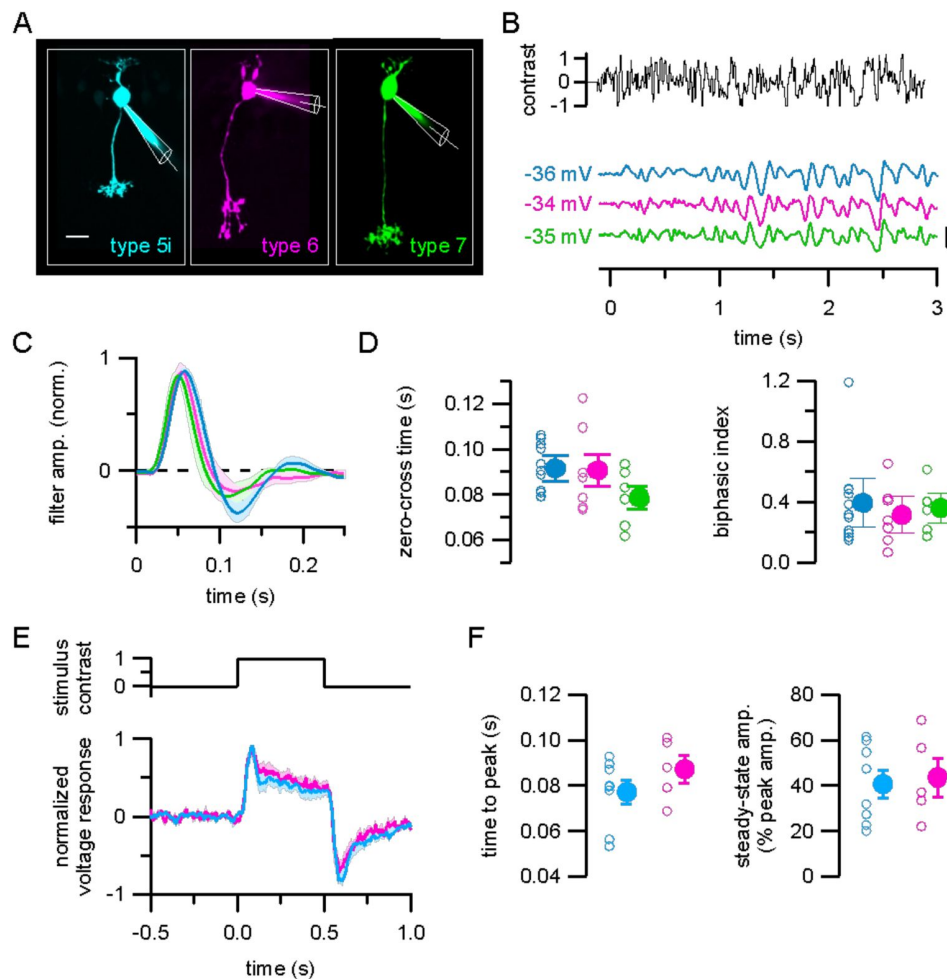


Figure 4.

Bipolar cell subtypes presynaptic to ON-T and ON-S RGCs have indistinguishable temporal response properties.

(A) Maximum-projection z-stacks of fluorescent dye-filled bipolar cells targeted for electrophysiological recording in retinal slices from Gjd2-EGFP (left), Grm6-tdTomato (middle), and Gus8.4-EGFP (right) transgenic mice. (B) Responses to spatially uniform Gaussian noise stimulation (same as in **Figs. 1** and **2**) in example type 5i (cyan), type 6 (magenta) and type 7 (green) bipolar cells. Voltage traces are average responses to ten repetitions of the noise stimulus. Scale bar = 5 mV. (C) Linear filters computed from responses to non-repeated noise stimulus presentation. Lines and shaded areas show mean $\pm$ SEM of responses (type 5i, $n = 11$; type 6, $n = 7$; type 7, $n = 6$). (D) Quantification of response kinetics from linear filters. No significant differences across types ($p = 0.19$ zero-cross times; $p = 0.89$ biphasic index; Kruskal-Wallis test). (E) Responses to step light increment. Lines and shaded areas show peak-normalized mean $\pm$ SEM responses across type 5i bipolar cells ($n = 8$) and type 6 bipolar cells ($n = 5$). (F) Quantification of step response kinetics. Time to peak and steady-state response amplitude were not different between type 5i and type 6 bipolar cells ($p = 0.72$ and $p = 0.35$, respectively; Wilcoxon rank sum test).

outer to inner retina via the rod bipolar cell pathway (**Fig. 1B**) (Grimes et al., 2014; Ke et al., 2014). Because AII amacrine cells distribute rod bipolar cell signals directly to cone bipolar cell axon terminals, differences observed under scotopic conditions likely arise at and/or downstream of cone bipolar cell synaptic output.

Presynaptic inhibition does not establish kinetic differences in bipolar cell-mediated excitation

Bipolar cells participate in diverse inhibitory circuits with amacrine cells, which are comprised of >40 morphologically and functionally distinct cell types (Diamond, 2017; Helmstaedter et al., 2013). Past work suggests that amacrine cell-mediated presynaptic inhibition is critical to bipolar cell subtype-specific patterns of light-evoked glutamate release (Franke et al., 2017). Our bipolar cell voltage recordings, which should reflect a combination of light-evoked excitatory and inhibitory input, argue that presynaptic inhibition is not required for generating differential synaptic output between type 5i, type 6 and type 7 bipolar cells. However, the slicing procedure we used to prepare tissue for bipolar cell recordings may damage some inhibitory connections, especially long-range amacrine cell inputs. Distance-dependent attenuation of voltage signals between bipolar cell axon terminals and the somatic recording site could also limit our ability to accurately measure the influence of presynaptic inhibition in these recordings. We therefore performed additional anatomical and pharmacological experiments to investigate a possible role for presynaptic inhibition in generating kinetically distinct excitation to ON-T and ON-S RGCs.

Many bipolar cell output synapses are organized into a “dyad” arrangement in which a bipolar cell ribbon synapse releases glutamate onto the processes of two different postsynaptic cells. At dyads, bipolar cell output synapses can be presynaptic to the processes of two amacrine cells, two RGCs, or an amacrine cell and a RGC (Cohen and Sterling, 1990; Dowling and Boycott, 1966; McGuire et al., 1984; Yu et al., 2023). Dyad synapses comprised of an amacrine cell and a RGC may shape bipolar cell glutamate release dynamics onto the RGC partner because glutamatergic excitation of the postsynaptic amacrine cell can elicit release of inhibitory neurotransmitters (GABA or glycine) from the amacrine cell processes that feed back onto the bipolar cell axon terminal. This form of inhibitory feedback control of bipolar cell output, termed reciprocal inhibition, could potentially contribute to differences in glutamate release onto ON-T and ON-S RGCs.

We analyzed bipolar cell dyads with postsynaptic amacrine cell processes and either ON-T or ON-S RGC dendrites in SBFSEM volumes to examine whether reciprocal inhibition is likely to generate more transient glutamate release onto ON-T RGCs. We defined putative reciprocal synapses as bipolar cell/amacrine cell/RGC dyads in which we observed clustered vesicles within the amacrine cell process that were in close proximity to the bipolar cell ribbon synapse (**Fig. 5A**). Our ultrastructural analysis revealed that the prevalence of reciprocal synapses was similar at dyad synapses with postsynaptic ON-T versus ON-S RGCs (**Fig. 5A-B**). This finding is not consistent with an important role for feedback/reciprocal inhibition in generating differential bipolar cell output onto the different RGC types.

We also performed pharmacological experiments to further examine what role, if any, inhibition may have in establishing transient glutamate release onto ON-T RGC dendrites. We used ACSF in place of Ames solution for the dissection and maintenance of tissue for these experiments. ACSF is often used in retina experiments, including the glutamate release measurements describe below, and we wanted to check that our results generalized across conditions. Light step-evoked currents in both ON-T and ON-S RGCs were more sustained than responses measured using Ames’ solution (**Fig. S9A-B**). Nonetheless, the difference between sustained and transient RGC types was maintained across different experimental conditions.

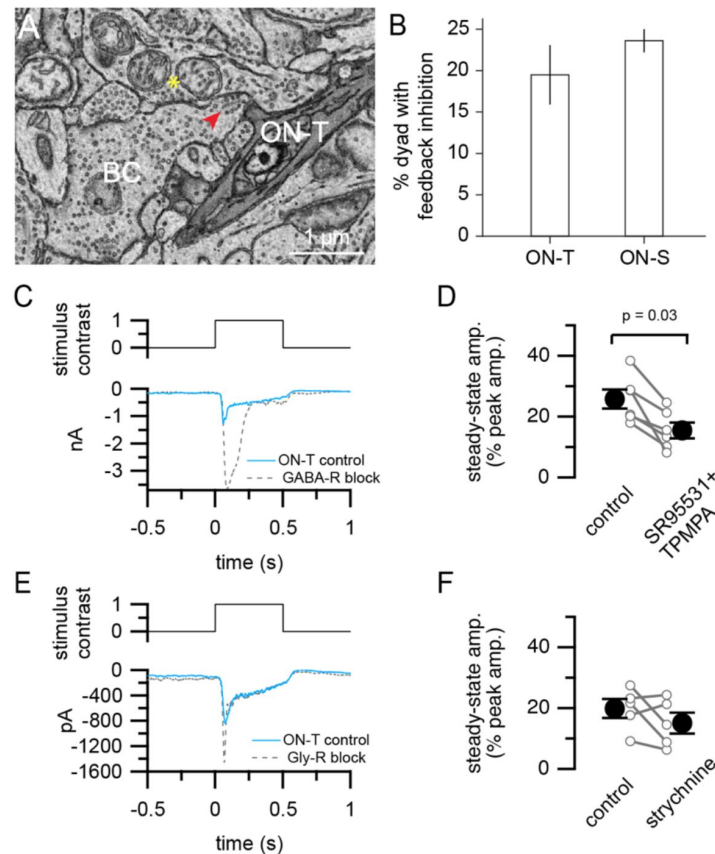


Figure 5.

Presynaptic inhibition is not required for kinetic differences between bipolar cell inputs to ON-T versus ON-S RGCs.

(A) Example EM micrograph showing a ribbon dyad (red arrowhead) between a BC and an ON-T RGC and an amacrine cell, with an adjacent feedback inhibitory synapse (yellow asterisk). (B) Bar plots showing the percentage of RGC/AC ribbon dyads with feedback inhibition. Mean $\pm$ SD ($n=2$ ON-S RGCs 85 dyads analyzed, 2 ON-T RGCs with 83 dyads analyzed) (C) Excitatory currents measured in response to light step increment in control conditions (cyan) and in the presence of GABA receptor antagonists (10 μ M SR95531 and 50 μ M TPMPA; dashed gray) in an example ON-T RGC. Traces are mean response from six repeats of stimulus presentation. (D) Summary of steady-state amplitudes measured from step responses in control or with GABA receptor antagonists. Step-evoked currents were significantly more transient when GABA/C receptors were blocked ($p=0.031$; Wilcoxon signed rank test; $n=6$ RGCs) (E) Control excitatory currents (cyan) or with addition of strychnine (1 μ M; dashed gray) to bath in an example ON-T RGC. (F) Summary of strychnine experiments. No significant difference between conditions ($p=0.44$; Wilcoxon signed rank test; $n=5$ RGCs). In (D) and (F), open circles with lines show measurements from individual ON-T RGCs. Filled circles with error bars show mean $\pm$ SEM.

In ON-T RGCs, blockade of GABAA and GABAC receptors via bath co-application of SR95531 and TPMPA enhanced the peak amplitude of the initial transient component of light-step evoked excitatory synaptic currents (-2023 ± 372.7 pA; mean $\pm$ S.D.) compared to control (-833.6 ± 103.2 pA) ($p = 0.031$, Wilcoxon sign rank test; $n = 6$) whereas the sustained component at the end of the light step was not significantly affected (control: -200.7 ± 14.71 pA; GABA-R block: -290.9 ± 43.69 ; $p = 0.16$, Wilcoxon sign rank test; $n = 6$) (**Fig. 5C**). The strong enhancement of the initial phase of the response we observed is consistent with prior work showing that GABAC receptor-mediated inhibition limits multivesicular release from bipolar cells (Sagdullaev et al., 2006). Block of GABA receptors thus made excitatory currents in ON-T RGCs more transient compared to control conditions (**Fig. 5D**). Note that this is opposite the expected effect if presynaptic GABA receptors contribute to generating transient responses in these cells; GABA receptor antagonists should have made excitatory currents more sustained if presynaptic GABAergic inhibition was generating transient bipolar cell input to ON-T RGCs. Thus, GABAergic inhibition appears to control response gain and time course of bipolar cell synaptic output to ON-T RGCs but is not responsible for the transient kinetics of excitatory input to these cells.

We also found that glycinergic inhibition is not likely to contribute importantly to the more transient glutamate release from bipolar cells presynaptic to ON-T RGCs; bath application of the glycine receptor antagonist strychnine did not significantly alter the kinetics of light step-evoked excitatory currents in these cells (**Fig. 5E-F**). Taken together with our ultrastructural analysis, these results show that amacrine cell-mediated GABAergic or glycinergic inhibition of bipolar cell axon terminals does not explain kinetic differences in excitatory synaptic input to ON-T and ON-S RGCs.

Bipolar cell glutamate release differs across types

Taken together, the bipolar cell and RGC recordings in **Figures 2** and **4** indicate that bipolar cell synapses have a key role in the separation of visual signals into transient or sustained channels within ON pathways in the retina. Presynaptic and/or postsynaptic mechanisms could contribute to the distinct kinetics of bipolar cell-mediated excitatory input to ON-T and ON-S RGCs. To investigate whether presynaptic differences in glutamate release from bipolar cell axon terminals shape RGC response kinetics, we expressed the fluorescent glutamate reporter iGluSnFR on the dendrites of ON-T and ON-S RGCs (see Methods). Using two-photon glutamate sensor imaging, we found that light-evoked glutamate signals had significantly faster kinetics at ON-T RGC dendrites (steady-state amplitude as % of peak amplitude = 21 ± 13 ; mean $\pm$ S.D.; $n = 224$ ROIs) compared to ON-S RGC dendrites (steady-state amp. as % of peak amp. = 44 ± 12 ; mean $\pm$ S.D.; $n = 307$ ROIs) (**Fig. 6**).

The experiments summarized in **Figure 5** suggest that presynaptic inhibitory input to bipolar axon terminals is not required for the difference between ON-T and ON-S responses. The glutamate release measurements provided a further test. We compared the kinetics of release before and after suppressing spiking amacrine cell activity (with bath application of the voltage-gated Na^+ channel blocker TTX) or all amacrine cells (with bath application of the glutamate receptor antagonist NBQX). This approach avoided the confounding changes in excitatory signals we encountered with inhibitory receptor blockers (e.g. the increase in excitatory input to ON-T RGCs in **Figure 5C** and oscillatory bursts of excitation in ON-S RGCs (not shown)). The more transient glutamate release kinetics at ON-T vs ON-S dendrites remained even when amacrine cell responses were suppressed (control: steady-state amp. as % of peak amp. 13 ± 10 ; mean $\pm$ S.D.; $n = 189$ ROIs/4 FOVs for ON-T dendrites vs. 40 ± 12 ; mean $\pm$ S.D.; $n = 287$ ROIs/8 FOVs for ON-S dendrites; TTX: 16 ± 9 ; mean $\pm$ S.D.; $n = 50$ ROIs/3 FOVs for ON-T dendrites vs. 35 ± 11 ; mean $\pm$ S.D.; $n = 146$ ROIs/6 FOVs for ON-S dendrites; $*p < 0.001$; t-test; NBQX: 7 ± 3 ; mean $\pm$ S.D.; $n = 112$ ROIs/1 FOV for ON-T dendrites vs 24 ± 9 ; mean $\pm$ S.D.; $n = 97$ ROIs/2 FOVs for ON-S dendrites; $*p < 0.001$; t-test; **Figure 7**).

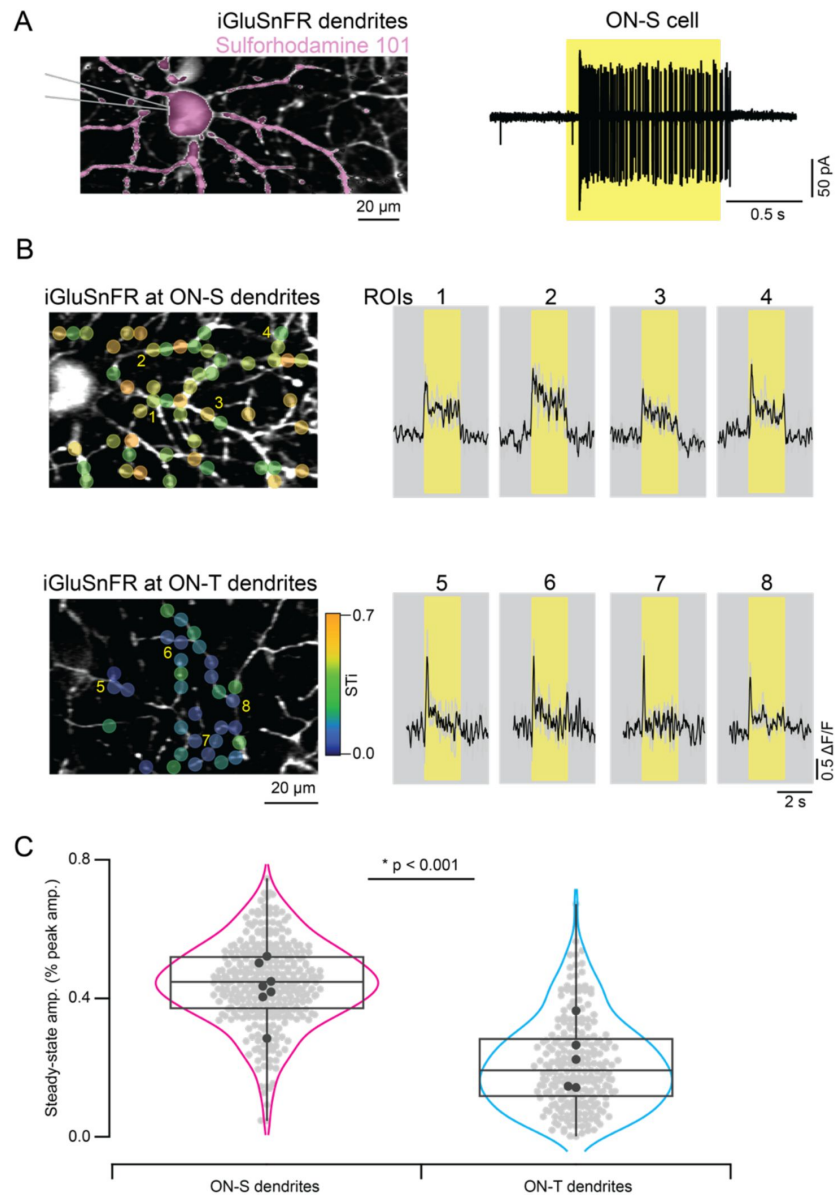


Figure 6.

Distinct glutamate input kinetics at ON-S vs ON-T RGC dendrites.

(A) ON-S RGC dendrites expressing iGluSnFR in a *Kcng4-Cre* mouse were filled with sulforhodamine 101 (left) after measuring their spiking response to a 200 μm spot extracellularly (Right). Yellow bands indicate stimulus duration. (B) iGluSnFR expressing ON-S and ON-T dendrites at different depths of the inner-plexiform layer (Left) were imaged using two-photon microscopy. Example iGluSnFR signals extracted from small regions of interest (ROIs 1-8) from ON-S and ON-T RGC dendrites illustrate the kinetic differences in their respective inputs. Black, mean responses; gray, individual trials. (C) Distribution of steady-state response amplitudes (mean fluorescence during last 1 s of stimulus) of the individual (gray) and average (black) ROIs from ON-S and ON-T dendrites evoked by a 200 μm static spot ($n = 7$ FOVs, 6 retinas for ON-S and 5 FOVs, 5 retinas for ON-T RGC dendrites, * $p < 0.001$; t-test between ROIs).

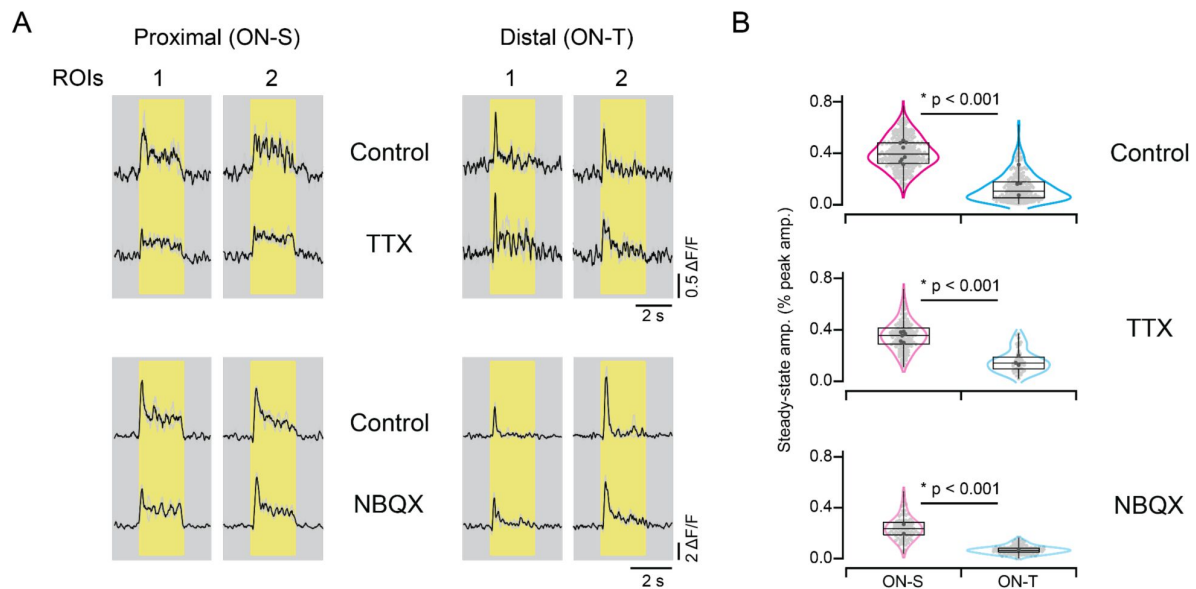


Figure 7.

Distinction between sustained and transient responses remains when amacrine cell activity is blocked.

(A) iGluSnFR measurements of glutamate release in control conditions and in the presence of TTX (which blocks spike-dependent amacrine cell output); or in the presence of NBQX, which blocks AMPA receptor mediated input to all amacrine cells (as well as horizontal cells). (B) Ratio of steady state to transient response across ROIs before (top) and during (bottom) application of TTX/NBQX.

These results show that differences in glutamate release from presynaptic bipolar cells contribute to transient versus sustained ON RGC responses and that the differences in release are retained without amacrine feedback to bipolar cells. These experiments do not, however, rule out the possibility that additional postsynaptic mechanisms, for example differential glutamate receptor expression, contribute to kinetic differences between RGC types.

Although whole-cell excitatory currents in ON-T and ON-S RGCs (**Fig. 2**, **Fig. S8**) and iGluSnFR signals measured from ON-T and ON-S RGC dendrites (**Fig. 6**) both exhibited clear kinetic differences between the RGC types, there were quantitative differences in the response to light steps between the different measurement approaches. As described above, these differences appear to arise from differences in the external solution (Ames vs ACSF). The iGluSnFR experiments used an ACSF solution, like the patch-clamp experiments in **Figs. 5** and **S8**. Both iGluSnFR responses and excitatory synaptic inputs measured in ACSF showed a larger sustained component of the step response at both ON-T and ON-S dendrites compared to currents measured in Ames (compare **Fig. 2D-E**, **Fig. S8A-B** and **Fig. 6B-C**). This observation highlights the importance of comparing responses between different RGC types, as well as between RGCs and bipolar cells, under identical recording conditions.

Importantly, the kinetics of glutamate release measured using iGluSnFR resemble the excitatory synaptic inputs to ON-T and ON-S RGCs more closely than the relevant bipolar voltage responses (**Fig. 4**). This indicates that the difference between the ON-T and ON-S responses originates at least in part in the conversion of bipolar voltage to glutamate release.

Ribbon size and vesicle replenishment

Prior work has shown that bipolar cells can exhibit profound short-term depression of glutamate release due to use-dependent depletion of the readily-releasable pool of synaptic vesicles (Burrone and Lagnado, 2000; Singer and Diamond, 2006, 2003; von Gersdorff and Matthews, 1997). Strong vesicle depletion at the cone bipolar cell synapses onto ON-T RGCs could generate more transient responses in these cells compared to ON-S RGCs. To examine this possibility, we measured excitatory synaptic inputs to ON-T and ON-S RGCs in response to a paired flash stimulus in which two brief light flashes were separated by varying time intervals. As in **Figures 1**, **2** and **4**, we used Ames' medium for tissue preparation and maintenance in these experiments. These experiments used the same photopic background light level (~ 3300 R*/rod/s; ~ 2000 P*/Scone/s; ~ 1650 P*/Mcone/s) as used for Gaussian noise and step stimuli.

Flash-evoked excitatory synaptic currents were suppressed in both RGC types at short inter-flash intervals, consistent with incomplete recovery from vesicle depletion at the time of the second flash (**Fig. 8A** top/middle, B). This suppression was much stronger for currents measured in ON-T RGCs compared to ON-S RGCs. We quantified the extent of suppression using a paired flash ratio, which was calculated as the peak amplitude of the response to the second flash divided by the peak amplitude of the response to the first flash. At the shortest interval we tested (50 ms), the response to a second flash was almost completely suppressed in ON-T RGCs (paired flash ratio = 0.04 ± 0.03 ; mean $\pm$ S.D.; $n = 6$) whereas the response to a second flash in ON-S RGCs was suppressed considerably less (paired flash ratio = 0.29 ± 0.08 ; $n = 5$). To rule out the possibility that a reduction in bipolar cell voltage response to the second flash could account for the strong paired flash suppression of excitatory input to RGCs, we also measured type 5i bipolar cell responses to the paired flash stimuli. The amplitude of flash-evoked voltage responses in type 5i bipolar cells was unchanged by a preceding flash at all time intervals (**Fig. 8A** bottom, C), which argues that paired flash suppression arises at the bipolar output synapse. These results are consistent with the idea that differences in the extent of stimulus-evoked vesicle pool depletion in bipolar cells could underlie transient versus sustained excitatory input to ON-T and ON-S RGCs.

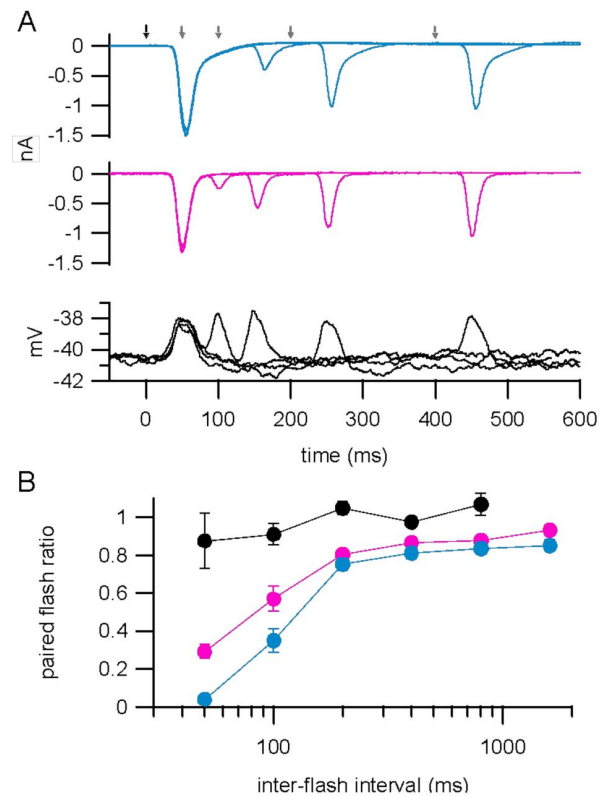


Figure 8.

Stronger flash-evoked suppression of excitatory currents in ON-T versus ON-S RGCs.

(A) Paired flash-evoked excitatory currents measured at different inter-flash intervals (50, 100, 200, 400 ms) in an example ON-T (top; cyan) and an ON-S (middle, magenta) RGC. Bottom black traces show voltage responses to same stimulus measured in an example type 5i bipolar cell. Arrows at top indicate timing of first flash (black) and second flashes (gray). Flashes were 400% contrast and 10 ms duration. Traces show mean responses to five repeats of each paired flash interval. (B) Filled circles with error bars show mean $\pm$ SEM paired-flash ratios at different inter-flash intervals across all ON-T RGCs (cyan; $n=6$), ON-S RGCs (magenta; $n=5$) and type 5i bipolar cells (black; $n=6$).

We next looked for ultrastructural correlates of differential synapse function in bipolar cell axon terminals reconstructed from SBFSEM volumes. Interestingly, type 6 bipolar cell synaptic ribbons were ~2-3 fold larger than those of types 5i bipolar cells ([Fig. 9A-B](#), [Fig. S10A](#)). These differences did not depend on the identity of the postsynaptic RGC partner ([Fig. 9D](#), [Fig. S10B](#)). Ribbons in XBCs, type 5o and type 5t bipolar cells were also smaller than type 6 bipolar cell ribbons ([Fig. 9D](#), [Fig. S10B](#)). This bipolar cell subtype-specific difference in ribbon size and the difference in ratio of type 6 to type 5i input between RGC types meant that ribbons were on average smaller at synapses made on ON-T compared to ON-S dendrites ([Fig. 9C-D](#), [Fig. S10B](#)).

Discussion

We sought to understand how kinetically distinct visual pathways are established in the retina by directly comparing the visual response properties of transient and sustained ON RGCs with their presynaptic bipolar cells. Our results show that bipolar cell synapses have a key role in establishing different temporal channels in ON circuits of the retina. Further, amacrine cell-mediated presynaptic inhibition does not appear to account for transient versus sustained bipolar cell synaptic output. Rather, bipolar cell subtype-specific differences in synaptic release properties likely generate kinetically distinct output to postsynaptic targets. Our anatomical analysis suggests that these differences may at least in part be due to differences in synaptic vesicle pool size.

A key aspect of our work is that we measured RGC and bipolar cell responses using identical stimuli and under the same experimental recording conditions. To our knowledge, a direct comparison of RGC responses with those of their presynaptic bipolar cells has not previously been reported. The importance of measuring RGC and bipolar cell responses under the same conditions is highlighted by our observation that excitatory synaptic currents measured in ON alpha RGCs exhibited quantitatively different kinetics, although relative differences between types were preserved, when measurements were made from retinas maintained in ACSF versus Ames' bath solution (compare [Fig. 2](#) (Ames') and [Figs. 5](#) and [S9](#) (ACSF)).

We found that the bipolar cell subtypes that provide the major input to ON-T (type 5i) or ON-S RGCs (types 6 and 7) had indistinguishable light-evoked voltage responses ([Figure 4](#)). This was not entirely surprising given that most or all ON cone bipolar cell subtypes are indirectly coupled to other nearby ON cone bipolar cells via electrical synapses between bipolar cell axon terminals and the distal processes of AII amacrine cells. These electrical synapses form the basis for the primary rod pathway that mediates vision under dim light conditions, but lateral signals can also flow among neighboring bipolar cells under conditions where cone signaling dominates ([Demb and Singer, 2012](#)). Indeed, a substantial fraction (~30-50%) of light-evoked responses in the bipolar cell types we investigated here can be attributed to lateral signals that arrive via electrical synapses at photopic light levels ([Kuo et al., 2016](#)). Each AII amacrine cell is electrically coupled to multiple ON cone bipolar subtypes as well as to other AII amacrine cells ([Marc et al., 2014](#); [Tsukamoto and Omi, 2017](#); [Veruki and Hartveit, 2002a](#), [2002b](#)). Different ON cone bipolar cell subtypes may also be directly coupled via gap junctions to each other ([Sigulinsky et al., 2020](#); [Tsukamoto and Omi, 2017](#)).

Direct or indirect (via AII) coupling may limit the extent to which different bipolar cells can generate subtype-specific voltage responses. These findings differ from a previous study that found some differences among mouse ON cone bipolar cell subtypes. [Ichinose et al., \(2014\)](#) reported that a subset of type 5 bipolar cells, which the authors termed type 5f, and type 7 bipolar cells had more transient responses than type 6 bipolar cells. Methodological differences may account for the discrepancy between our results and the previous work by [Ichinose et al., \(2014\)](#). For example, Ichinose et al. included inhibitory receptor antagonists in their experimental solutions and used 100 μ m diameter spots to characterize bipolar cell responses

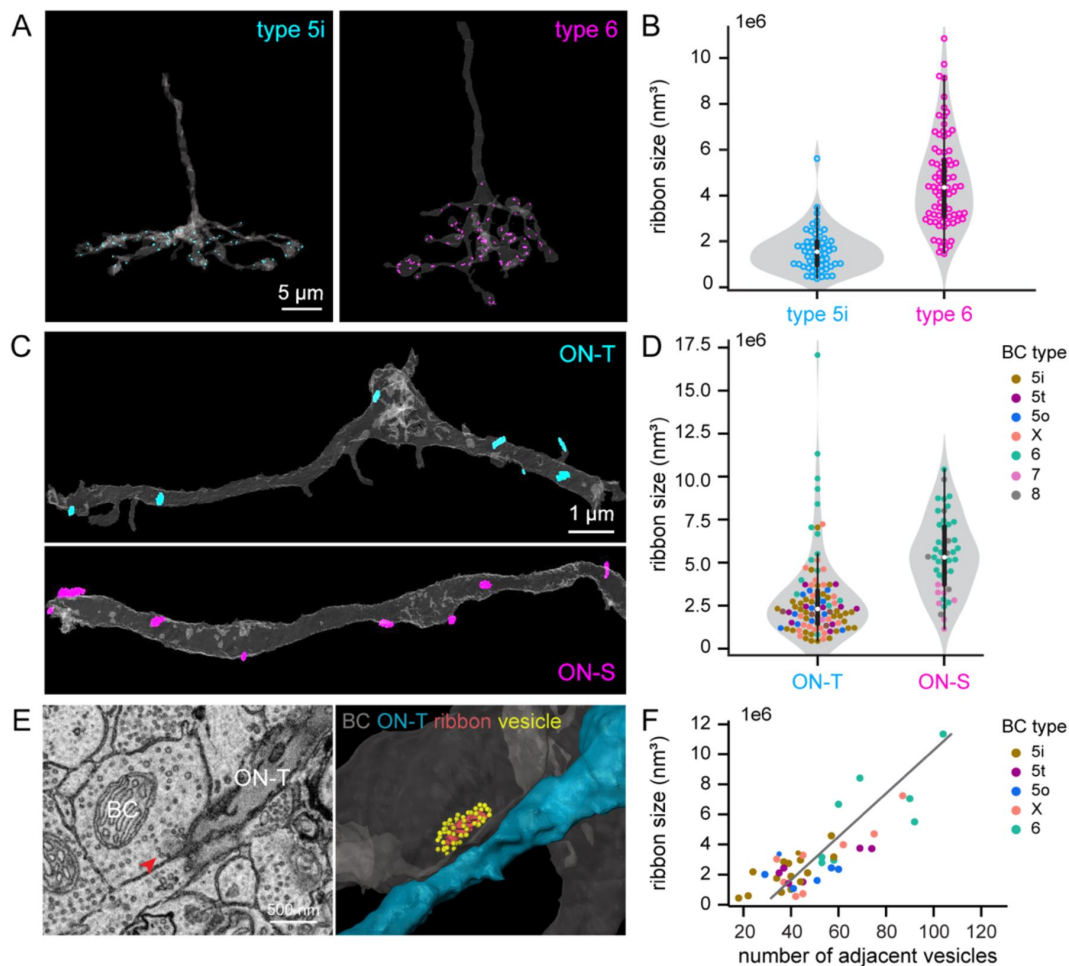


Figure 9.

Bipolar cell subtype-specific differences in ribbon size (volume).

(A) 3D reconstructions showing segmented axon stalk and axonal terminal (gray) and all the synaptic ribbons within the axonal terminal for a type 5i (cyan) and a type 6 (magenta) BC. (B) Violin plot showing distribution of ribbon volumes (each data point is a ribbon) for the type 5i and type 6 BC in (A). The boxes indicate the interquartile range (thick lines, 25%-75%), median value (white dot), and 1.5 times interquartile range (thin line). Shaded gray areas represent rotated kernel density plots. (C) 3D reconstructions showing segmented dendritic arbor (gray) and all the synaptic ribbons presynaptic to these segments of ON-T (cyan) or ON-S (magenta) RGC dendrites. (D) Violin plot with scattered dots showing ribbon volumes for each ribbon presynaptic to a stretch of reconstructed ON-T or ON-S RGC dendrite. Each ribbon (dot) is colored differently for each BC type. (E) An EM micrograph (left) and 3D reconstruction (right) of a ribbon and its adjacent vesicles. The ribbon is indicated by the red arrow in the EM image. In the 3D reconstruction, yellow dots indicate the adjacent vesicles surrounding the presynaptic ribbon (red) in the BC terminals (gray). Postsynaptic ON-T RGC is in blue. (F) Scatter plot showing ribbon size in volume in relationship with the number of adjacent vesicles. Gray line represents the linear regression model fitted to the data. Linear regression: $\beta = 6.98 \times 10^{-6}$, $p < 0.001$, $r^2 = 0.66$.

whereas we left inhibitory signaling intact and used larger (520 μm diameter) spots to match the stimuli we used for RGC recordings. Interestingly, a recent investigation of spatial and temporal response properties of light-evoked intracellular calcium signals across bipolar cell types suggests that ON bipolar cells can exhibit type-specific response kinetics for small stimuli ($< \sim 200 \mu\text{m}$ diameter) but not larger spot sizes, presumably due to surround inhibition (Hsiang et al., 2024). Regardless of the reasons for the different findings, our observation that bipolar cell subtypes exhibited similar responses under the same recording and light stimulus conditions for which we observed obvious differences among postsynaptic RGCs argues that mechanisms upstream of bipolar cell glutamate release (e.g. cone to bipolar cell synapses or intrinsic voltage response properties of bipolar cells) are unlikely to have a major role in shaping RGC type-specific responses for the stimuli we used here.

A limitation of our work is that we did not record from every one of the bipolar cell subtypes that are presynaptic to ON-T RGCs. Instead, we measured responses from type 5i and type 6 bipolar cells, which together comprise $\sim 50\%$ of the synaptic input to ON-T RGC dendrites. It is possible that the other presynaptic bipolar cell types (type 5o, type 5t and XBC) may have more transient light-evoked voltage responses that could contribute to the transient excitation measured in ON-T RGCs. In fact, a previous study in mouse retina found that XBCs and a subset of type 5 bipolar cells express voltage-gated Na^+ channels (Hellmer et al., 2016). These channels may enhance response transience or might even contribute to the generation of light-driven all-or-none spiking responses in these bipolar cells (Ichinose et al., 2014; Puthussery et al., 2013; Saszik and DeVries, 2012).

Even if transient intrinsic response properties of type 5o, 5t and XBC bipolar cells contribute to the generation of transient responses in ON-T RGCs, this alone is unlikely to account for the kinetics of the excitatory synaptic currents we measured in these RGCs. Type 5i and type 6 bipolar cells have a significant sustained component of their light step-evoked voltage response (Fig. 4E-F), whereas excitatory currents measured for the same stimulus and under the same conditions in ON-T RGCs had little to no sustained component (Fig. 2D-E). We would expect to observe some sustained excitatory input to ON-T RGCs if type 5i and type 6 bipolar cell synaptic output follows the voltage response of these bipolar cells. Thus, a marked transformation of the sustained bipolar cell membrane voltage response to a transient synaptic current must take place. Our measurements of extracellular glutamate at ON-T and ON-S dendrites (Fig. 5) suggest that much of this transformation occurs at the level of bipolar cell glutamate release. It is interesting in this context that ON-T and ON-S RGCs both receive input from type 6 bipolar cells (comprising $\sim 20\%$ and $\sim 50\%$ of total synaptic input, respectively). Due to differences in dendrite stratification, type 6 bipolar cell synapses onto ON-T RGCs arise from a region of the axon that is located more proximal to the bipolar cell soma than the synapses formed onto ON-S RGCs, which are found within the distal axon lobules. However, we did not observe any differences in ribbon size at distal versus proximal locations in type 6 bipolar cell axons (Figure 9D). Whether postsynaptic target-specific synapse function, as has previously been reported in salamander retinal bipolar cells (Asari and Meister, 2014, 2012) and mouse starburst amacrine cells (Pottackal et al., 2021), contributes to signal diversification in circuits involving type 6 bipolar cells will require further investigation.

Our electrophysiological and glutamate imaging results indicate that amacrine cell-mediated inhibition at bipolar cell synaptic terminals is not necessary for generating transient glutamate release kinetics onto ON-T RGC dendrites. This is seemingly at odds with prior work that has indicated an essential role for surround inhibition in establishing type-specific bipolar cell glutamate release dynamics (Franke et al., 2017). However, we note that our analysis here focused only on whether presynaptic inhibition shapes the kinetics of excitatory inputs to ON-T RGCs. Our anatomical and pharmacological experiments together argue that presynaptic inhibition is not an important factor in the transformation from sustained bipolar cell voltage responses to transient excitatory synaptic currents in ON-T RGCs. This is not to say that amacrine cell-mediated inhibition does not have a role in generating subtype-specific response properties

for other aspects of bipolar cell function. Indeed, inhibition does have an important role in establishing differential spatial tuning of excitatory inputs to the RGC types we studied; excitatory synaptic currents in ON-T RGCs exhibit stronger surround inhibition than ON-S RGCs, and this surround inhibition is reduced by bath application of GABA receptor antagonists (Fig. S9C). Interestingly, recent work revealed postsynaptic target-specific surround inhibition at neighboring type 6 bipolar cell output synapses within the same axon terminal (Swygart et al., 2024). Whether target-specific presynaptic inhibition of type 6 bipolar cells also contributes to differential surround inhibition between ON-T and ON-S RGCs or is primarily due to differences across bipolar cell types (e.g. type 5i/5t/XBC vs. type 6/7), remains to be determined.

We conclude that transient versus sustained excitatory inputs to ON-T and ON-S RGCs most likely arise from bipolar cell subtype-specific differences in synaptic release properties. Excitatory inputs to ON-T RGCs exhibited stronger paired flash depression than those to ON-S RGCs (Fig. 8), which indicates distinct vesicle cycling dynamics at the bipolar cell axon terminals presynaptic to the different RGC types. Our ultrastructural analysis suggests that bipolar subtype-dependent differences in ribbon size may be a factor in establishing transient versus sustained glutamate release kinetics (Fig. 9). It is important to note, however, that type 7 bipolar cells (~45% of excitatory input to ON-S RGCs) have small synaptic ribbons, yet iGluSnFR imaging of type 7 bipolar cell axon terminals found that these cells exhibit sustained glutamate release (Srivastava et al., 2022).

We found that ribbon size is strongly correlated with the total number of vesicles that are in close contact with the synaptic ribbon across different ON bipolar cell types (Fig. 9E-F, Fig. S10C). Thus, there are fewer ribbon-associated vesicles at glutamatergic synapses formed on ON-T RGC dendrites compared to those at ON-S RGCs. Ribbon-associated vesicles are thought to correspond to the releasable pool of vesicles, comprised of both a readily-releasable pool that are docked at the base of the ribbon, at the presynaptic active zone, and a reserve (alternatively referred to as ‘releaseable’) pool of vesicles that are bound to the ribbon at more distal sites (Grabner et al., 2023; Lagnado and Schmitz, 2015; von Gersdorff et al., 1996; Wan and Heidelberger, 2011). The number of ribbon-associated vesicles, as well as the rate at which the readily-releasable and reserve vesicle pools are replenished following exocytosis, together define the amplitude and time course of stimulus-evoked vesicle release (Jackman et al., 2009; Lagnado and Schmitz, 2015; Neves and Lagnado, 1999; Oesch and Diamond, 2011; Singer and Diamond, 2006; von Gersdorff and Matthews, 1997). For example, a small pool of readily-releasable vesicles and relatively slow replenishment in rod bipolar cell axon terminals contributes to transient release kinetics from these cells (Singer and Diamond, 2006). Our observation that the average excitatory synaptic input to ON-T RGCs is smaller and that most excitatory input to ON-T cells originates from bipolar synapses with fewer ribbon-associated vesicles are consistent with a role for vesicle depletion in establishing transient glutamate release onto these RGCs. The larger pool of ribbon-associated vesicles in type 6 bipolar cell terminals, which comprise ~ 55% of the synaptic input to ON-S RGCs, may support sustained synaptic transmission onto these RGCs. However, additional mechanisms, such as bipolar cell type-specific differences in the rate of refilling of the ribbon-associated vesicles could also contribute to sustained versus transient glutamate release.

Direct measurements of exocytosis at ribbon synapses show that release rates increase with increasing ribbon size and the number of vesicles bound to the ribbon. Evidence for this dependence includes a tight correspondence between functional measures of the releasable vesicle pool and anatomical measurements of the number of vesicles bound to the ribbon (Grabner et al., 2023; von Gersdorff et al., 1996). Ribbon size and the number of ribbon-associated vesicles vary across the day-night cycle; these morphological changes are correlated with diurnal changes in maintained vesicle fusion rates (Babai et al., 2016; Dembla et al., 2020; Hull et al., 2006). Finally, release rates decrease when ribbon size is reduced (Mehta et

al., 2013 [↗](#); Wakeham et al., 2023 [↗](#)), or when ribbons are ablated either abruptly or genetically (Grabner and Moser, 2021 [↗](#); Maxeiner et al., 2016 [↗](#); Mehta et al., 2013 [↗](#); Snellman et al., 2011 [↗](#)).

These biophysical studies are consistent with the correlation that we observe here between ribbon size and the extent to which transmitter release is transient vs. sustained. A note of caution in making this comparison is that exocytosis is sensitive to the voltage trajectory (Snellman et al., 2009 [↗](#)), and the biophysical studies often use more rapid and larger voltage changes than those elicited by physiological light inputs. In fact, genetic deletion of ribbons does not strongly alter light-evoked excitatory currents in ON-S RGCs (Okawa et al., 2019 [↗](#)). Why ribbon deletion has relatively little effect on light-evoked responses in ON-S RGCs remains to be determined. It is interesting to note that ~30% of synapses formed by type 6 bipolar cells onto ON-S RGC dendrites lack ribbons even in wildtype retina (Okawa et al., 2019 [↗](#)). Future studies will be needed to investigate to what extent such ribbonless synapses could contribute to bipolar cell type-specific differences in glutamate release such as we observed here.

The dynamics of transmitter release are also impacted by vesicle depletion and the time required for refilling of releasable vesicle pools following strong stimulation. In particular, the sustained transmitter release rate should be equal to the sustained refilling rate. Direct measurements of refilling at the cone output synapse following strong activation show that it is quite slow – with a time constant near 1 sec (Grabner et al., 2016 [↗](#); Jackman et al., 2009 [↗](#); Rabi et al., 2006 [↗](#)). This is far too slow to explain the recovery from paired-flash depression that we observe here. Several factors may contribute to this difference. First, postsynaptic receptor saturation can contribute to paired-pulse depression, and it can recover from such depression more quickly than presynaptic release (Grabner et al., 2016 [↗](#)). Second, the paired-flash depression that we measure in RGC synaptic inputs is largely absent in the bipolar voltage responses. This suggests that it is dominated by a history-dependence at the cone bipolar output synapses and refilling of vesicle pools may proceed more quickly at bipolar synapses than cone synapses.

Mechanisms that we did not directly investigate here may also contribute to the difference between transient and sustained responses. For example, presynaptic resting potential has a strong effect on steady-state vesicle release and thus stimulus-evoked release dynamics.

Exocytosis that occurs when bipolar cells rest at voltages near release threshold can lead to a depletion of readily-releasable vesicles that limits the amount of release that occurs upon further light-induced depolarizations (Grimes et al., 2014 [↗](#); Jarsky et al., 2011 [↗](#); Oesch and Diamond, 2011 [↗](#)). Thus, an initial phasic component of release can be reduced by depolarized resting potentials whereas hyperpolarized potentials can facilitate greater occupancy of vesicles in the releasable pool, which can enhance response transience. Bipolar subtype-dependent differences in release threshold due to differential Ca^{2+} channel expression/regulation (Pangrsic et al., 2018 [↗](#)) and/or coupling between Ca^{2+} channel activity and vesicle release (Burrone et al., 2002 [↗](#); Burrone and Lagnado, 2000 [↗](#)) could also impact release dynamics. Several aspects of excitatory synaptic currents in the two RGC types we studied here are consistent with a situation in which bipolar cell inputs to ON-T RGCs have a higher threshold for glutamate release compared to those at ON-S RGCs; ON-T RGCs receive less spontaneous excitatory synaptic input than ON-S RGCs (Kuo et al., 2016 [↗](#)) and stimulus-evoked EPSCs are slower, sparser and more biased towards larger amplitude events in ON-T RGCs compared to those in ON-S RGCs (**Fig. 2B**, **Fig. S1B** [↗](#)). Further investigation will be required to determine whether this is due to differences in axon terminal resting potential and/or other presynaptic factors. We did not find evidence for subtype-specific differences in resting membrane potential among type 5i, type 6 and type 7 bipolar cells, but our somatic voltage recordings likely have limited ability to resolve small voltage differences at the bipolar cell axon terminals.

Our results do not rule out additional roles for postsynaptic mechanisms in establishing transient versus sustained RGC spike output. For example, delayed amacrine cell-mediated inhibitory synaptic input to RGC dendrites could truncate spiking responses (Nirenberg and Meister, 1997 [↗](#)). We did not observe a significant difference in the light-evoked excitation/inhibition amplitude ratio between ON-S and ON-T RGCs (Fig. S3E [↗](#)), but our data does not exclude a role for postsynaptic inhibition in generating transient responses. RGC cell-intrinsic mechanisms, such as RGC subtype-specific differences in voltage-gated Na⁺ channel properties and/or expression pattern could also contribute to distinct response kinetics (Brombas et al., 2022 [↗](#); Chang et al., 2023 [↗](#); Werginz et al., 2020 [↗](#)). While a combination of pre- and postsynaptic factors may ultimately act in concert to generate transient versus sustained RGC spike output (Warwick et al., 2018 [↗](#); Werginz et al., 2020 [↗](#)), our results establish a fundamental role for bipolar cell synapses in the segregation of visual input into kinetically distinct channels in ON pathways of the retina.

Materials and Methods

Animals

Experiments used 5- to 15-week-old wild-type, *Grm6-tdTomato* (Morgan et al., 2011 [↗](#)), *Gus8.4-EGFP* (Huang et al., 1999 [↗](#)), *Gjd2-EGFP*, or *Kcng4-Cre* (Duan et al., 2014 [↗](#)) transgenic mice.

Transgenic mouse strains were backcrossed into the same genetic background as wild-type mice (C57BL/6J; Jackson Laboratory). Animal care and handling followed procedures approved by the Institutional Animal Care and Use Committee of the University of Washington or the University of Victoria Animal Care Committee.

Tissue preparation

Retinal tissue was prepared for physiological recordings as previously described (Kuo et al., 2016 [↗](#); Srivastava et al., 2022 [↗](#)). Briefly, mice were killed by cervical dislocation or decapitation and retinas were dissected from both eyes in oxygenated (95% O₂, 5% CO₂), bicarbonate-buffered Ames' (Sigma) or artificial cerebrospinal fluid (ACSF) solution and either flat-mounted with the vitreal surface facing up onto a poly-D-lysine coated glass coverslip (Corning) (RGC patch-clamp recordings and iGluSnFR imaging) or sliced in an orientation perpendicular to the flat-mount plane using a vibratome (200 μm slice thickness; bipolar cell recordings) before being transferred to an upright 2P fluorescence microscope. ACSF contained (in mM): 125 NaCl, 2.5 KCl, 1.2 CaCl₂, 1.5 MgCl₂, 1.25 NaH₂PO₄, 20 glucose, 0.5 L-glutamine, 0.1 sodium ascorbate, and 26 NaHCO₃ (pH 7.4, 310mOsm/L). For preparation of retinal slices, isolated retina was surrounded with melted agarose (maintained at ~40°C) followed by rapid cooling using an ice block. The embedded retina was then positioned in front of the vibratome blade in an orientation orthogonal to the flat mount plane for sectioning. All experiments exclusively used tissue from ventral retina.

Electrophysiology

During recordings, retinal tissue was continuously perfused (~6-8 mL/min) with oxygenated, bicarbonate-buffered Ames' solution maintained at 30-34 °C. In a subset of electrophysiological recordings (Figs. 6C-F [↗](#) and S8), tissue was perfused with ACSF rather than Ames' solution.

Electrical signals were acquired at 10 kHz and low-pass filtered at 3 kHz using a Multiclamp 700B or Multiclamp 700A amplifier (Molecular Devices) and Symphony Data Acquisition software (<http://symphony-das.github.io> [↗](#)). The somas of different ON bipolar cell subtypes were targeted for gramicidin perforated-patch current-clamp recordings using 2P laser scanning excitation (960nm) in retinal slices prepared from transgenic mouse lines in which bipolar cell subtypes expressed fluorescent proteins. Tissue from *Grm6-tdTomato* mice, in which a sparse subset of ON bipolar cell types (type 6, 7 and 8) express tdTomato was used to target recordings to type 6 bipolar cells

(Morgan et al., 2011 [↗](#); Schwartz et al., 2012 [↗](#)). Type 7 bipolar cells were targeted for recording in *Gus8.4-EGFP* mice, which express EGFP in a type 7 bipolar cells and rod bipolar cells (Huang et al., 2003 [↗](#)). Type 5i bipolar cells were targeted in retinas from *Gjd2-EGFP* mice, which we found express EGFP selectively in these bipolar cells (**Fig. S6** [↗](#)) as well as less robustly in an uncharacterized amacrine cell type (not shown). For bipolar cell recordings, we minimized light exposure from the 2P laser by restricting scan regions to the inner retina, away from photoreceptors, and by keeping laser exposure brief ($< \sim 30$ sec/scan region; < 2 mW post-objective laser power) (Kuo et al., 2016 [↗](#)). After identification of fluorescent cell bodies, we acquired recordings using infrared differential interference contrast microscopy. Bipolar recordings used an internal solution containing 123 mM K-aspartate, 10 mM KCl, 10 mM HEPES, 1 mM MgCl₂, 1 mM CaCl₂, 2 mM EGTA, 4 mM Mg-ATP, 0.5 mM Tris-GTP, 100 μ M Alexa Fluor 594-hydrazide and 25–30 μ g/mL gramicidin D (280 mOsm/L; pH 7.2). Pipettes (~ 7 –9 Ω tip resistance) were front-filled with gramicidin-free internal solution to facilitate formation of high resistance (> 1 G Ω) seals between patch pipettes and the cell membrane. After seal formation, series resistance (R_s) was monitored in voltage-clamp mode by applying a 10 mV voltage step across the pipette tip. Experiments were initiated once R_s declined to < 50 M Ω , which typically took 20–30 min. During recordings, we occasionally confirmed the integrity of the cell membrane under the recording electrode by monitoring Alexa Fluor 594 fluorescence using 2P excitation; if any dye was visible in the cell body, we abandoned the recording. No holding current was applied during current-clamp recordings from bipolar cells. At the conclusion of each bipolar cell recording, the membrane under the patch pipette was intentionally ruptured and fluorescence images of Alexa Fluor 594-filled cells were acquired using 2P microscopy to confirm bipolar cell subtype identities (Wässle et al., 2009 [↗](#)).

RGC recordings were performed in a flat-mount preparation of the retina from wild-type mice or *Gjd2-EGFP* transgenic mice following previously described methods (Kuo et al., 2016 [↗](#)). Excitatory synaptic currents were isolated in whole-cell voltage-clamp recordings by holding RGCs near the reversal potential for Cl[−]-mediated conductances (-68.5 mV). Patch pipettes (2.5–4 M Ω) were filled with an internal solution containing (in mM): 105 Cs-methanesulfonate, 10 Tetraethylammonium-Cl, 20 HEPES, 10 EGTA, 2 QX-314, 5 Mg-ATP, 0.5 Tris-GTP (~ 280 mOsm; pH 7.2). Series resistance (< 15 M Ω) was compensated by 70% using the amplifier circuitry. Inhibitory receptor antagonists (SR-95531, TPMPA, strychnine) were obtained from Tocris or Sigma Aldrich and added to the bath solution from concentrated aqueous stock solutions.

iGluSnFR expression and two-photon imaging

Intravitreal injection of a cre-dependent viral vector (pAAV.hSyn.Flex.iGluSnFR.WPRE.SV40; Addgene Catalog #98929-AAV1) was performed into *Kcng4-cre* mice to selectively express the glutamate sensor, iGluSnFR, in alpha RGCs, including ON-T and ON-S RGCs (Krieger et al., 2017 [↗](#)). All injection procedures, tissue preparations, and subsequent two-photon imaging were performed as previously described in Srivastava et al., 2022 [↗](#). Briefly, for iGluSnFR imaging, the laser was tuned to 920 nm and all recordings were acquired at a frame rate of 58.25 Hz (256 x 256 pixels). In *KCNG4-Cre* mice, ON-S alpha RGCs were first identified by their spiking properties using extracellular cell-attached recordings. To identify the dendrites of ON-S RGCs, cells were filled with sulphorhodamine 101. ON-T RGC dendrites were then recorded at the same location but deeper into the retina (Krieger et al., 2017 [↗](#)).

Visual stimulation

Visual stimuli were presented from a photopic mean background illumination ($\sim 3,300$ rhodopsin isomerizations per rod per second ($R^*/\text{rod/s}$)) to focus specifically on cone-mediated signaling (Grimes et al., 2018 [↗](#)). Visual stimuli for electrophysiological recordings were generated using Stage Visual Stimulation Software (<http://stage-vss.github.io> [↗](#)). For all electrophysiological recordings except those in **Figs. 6C–F** [↗](#) and S8, light stimuli consisted of a spatially uniform circular spot (520 μ m diameter) delivered from a LED (405 nm peak output; Hosfelt) that was

focused through the microscope substage condenser (Kuo et al., 2016). In the experiments shown in **Figs. 6C-F** and S8, a Digital Light Processing (DLP) projector driven by a 405 nm LED (Wintech Digital) was used to project spots of various sizes through the microscope objective (Kuo et al., 2020). For iGluSnFR imaging (**Fig. 5**), background illuminance was kept around 1000 photons/ $\mu\text{m}^2/\text{s}$. For experiments, static spots of different sizes were generated and presented using StimGen software (<https://github.com/benmurphybaum/StimGen>). Stimuli were presented via a DLP through the substage condenser focused onto the photoreceptor layer of the retina. All visual stimuli were presented 4 s after the start of the recording for a duration of 2 s.

Data analysis and statistics

Spike triggered average (STA) waveforms (**Fig. 1**) and linear-nonlinear (LN) models of light-driven RGC (**Fig. 2**) and bipolar cell (**Fig. 4**) responses were derived from responses to randomly fluctuating light stimuli (Gaussian distribution of light intensities with standard deviation = 50% of mean intensity; 0-60 Hz bandwidth) as described previously (Chichilnisky, 2001; Kim and Rieke, 2001; Rieke, 2001). For LN model construction, the linear filter was calculated by computing the cross-correlation of the light stimulus with the measured voltage (bipolar cell) or current (RGC) response and dividing the result by the power spectrum of the stimulus. The static nonlinearity was constructed from a point-by-point comparison of the predicted filter output with the measured response and binning the result along the prediction axis. In almost all cases, the linear filter had a biphasic shape with initial and secondary lobes that are opposite polarity. A small number of bipolar cells had triphasic filters (**Fig. S7**). We quantified the kinetics of each STA or linear filter by measuring the time at which the STA/filter waveform crossed zero (“zero-cross time”; e.g. **Fig. 1C**, left) between the first and second lobes of the filter (e.g. between peak and trough of STA waveforms) and by calculating a biphasic index = $|A| / |B|$, where $|B|$ is the absolute peak amplitude of the initial lobe of the filter and $|A|$ is the amplitude of the peak amplitude of the secondary lobe (e.g. **Fig. 2C**, right). We quantified the kinetics of responses to light step stimuli by calculating the time between light step onset and the peak amplitude response (“time to peak”; e.g. **Fig. 2E**, left) and dividing the mean steady-state response (calculated from the last 100 ms light step increment) by the peak amplitude initial response at step onset (and multiplying by 100) (e.g. **Fig. 2E**, right). We note that alternative measures of step response or linear filter kinetics than those presented in **Fig. 4** were also not statistically different between bipolar cell subtypes (not shown). For light step-evoked responses, steady-state measurements were obtained from the mean current/voltage over the last 100 ms of the 500 ms light steps (electrophysiology) or the mean fluorescence over the last 1 sec of the 2 sec light step (iGluSnFR). Statistical tests were performed using MATLAB (Mathworks) or Igor Pro (WaveMetrics) software.

Biolistic transfection

DNA plasmids encoding mTFP1-myc (24 μg) and PSD95-mCherry (12 μg) under the control of the cytomegalovirus (CMV) promoter were coated onto gold particles (1.6 μm diameter, 12.5 mg, Bio-Rad). Using a Helios gene gun (40 psi, Bio-Rad), these particles were then delivered to retinal ganglion cells in whole-mount retinas. Subsequently, the transfected retinas were incubated overnight in mACSF within a humid, oxygenated chamber maintained at 33°C.

Immunohistochemistry

Retinal sections (60 μm thick) were prepared using a vibratome (Leica) from retinal whole mount retinas of *Gjd2-EGFP* mice. Both whole mount retinas and retinal sections were fixed in 4% PFA for 30 minutes. Retinal sections were embedded in low melting point agarose (Sigma-Aldrich). Retinal tissues were first incubated with blocking solution (5% normal donkey serum, 0.5% Triton-X100 in PBS, Sigma-Aldrich) for 2 hours at 4°C, then incubated overnight at 4°C with primary antibodies dissolved in blocking solution. Excess primary antibody was rinsed 3×10 minutes with PBS, then sections were incubated for 2 hours with secondary antibodies dissolved in PBS. Excess secondary

antibody was rinsed 3×10 minutes with PBS and sections were mounted on standard microscopy glass slides using Vectashield mounting medium (Vector). Primary antibody used in the study were: goat anti-ChAT diluted 1:1000 Chemicon AB144P, mouse anti-CaBP5 diluted 1:1000 courtesy of Dr. Haeseleer, mouse anti-myc diluted at 1:1000, chicken anti-GFP diluted at 1:1000. Secondary antibody were donkey anti-goat IgG Dylight-568 diluted 1:1000, donkey anti-mouse Dylight-647 diluted 1:1000, donkey anti-mouse Dylight-405 diluted at 1:1000, donkey anti-chicken Alexa 488, the nuclear dye TOPRO3 (diluted 1:200, Thermo-Fisher).

Near-infrared branding (NIRB) and serial block face scanning electron microscopy (SBFSEM)

Two neighboring ON-T RGCs were physiologically identified in *Gjd2-EGFP* mice. After identifying ON-T RGCs via their characteristic step-evoked extracellular spike response (e.g. **Figure 1B** [↗](#)), they were filled with Alexa 488 or Alexa 555. Dendritic arbors and T5i-BC axonal arbors were imaged using a confocal microscope. The retina was then fixed with 4% glutaraldehyde for 30 minutes at room temperature (RT). The retina was washed with 0.1M Na⁺ Cacodylate buffer for three times and flat-mounted with 0.1M Na⁺ Cacodylate buffer. To relocate the two RGCs under electron microscopy, we applied the near-infrared branding (NIRB) technique previously published to burn fiducial markers in the retina ([Bishop et al., 2011](#) [↗](#)). After NIRBing, the retina tissue was unmounted and fixed in 4% glutaraldehyde overnight. The retina was processed for SBFSEM subsequently according to the protocol described previously ([Della Santina et al., 2016](#)). Briefly, the tissue was initially treated with reduced osmium for 1 hour in the refrigerator. Subsequently, after washing with double-distilled H₂O (ddH₂O), the tissue was placed in a freshly prepared thiocarbohydrazide solution for 20 minutes at RT. Following another round of ddH₂O rinsing, the tissue was incubated in 2% osmium tetroxide for 30 minutes at RT. After rinsing, the sample was incubated in 1% uranyl acetate at 4°C overnight. Post-wash, the tissue was stained with Walton's lead aspartate for 30 minutes at 60°C. Following another wash, the retinal piece underwent dehydration using a graded alcohol series (20%, 50%, 70%, 90%, 100%, 100%), each step lasting 5 minutes, followed by two changes of 100% propylene oxide at RT for 10 minutes each. Finally, the tissue was embedded in Epon. The resulting block was then trimmed and mounted in the SBEM microscope (GATAN/Zeiss SIGMA, 3View), and image stacks were acquired at a voxel size of 5 × 5 × 50 nm.

Supplemental Information

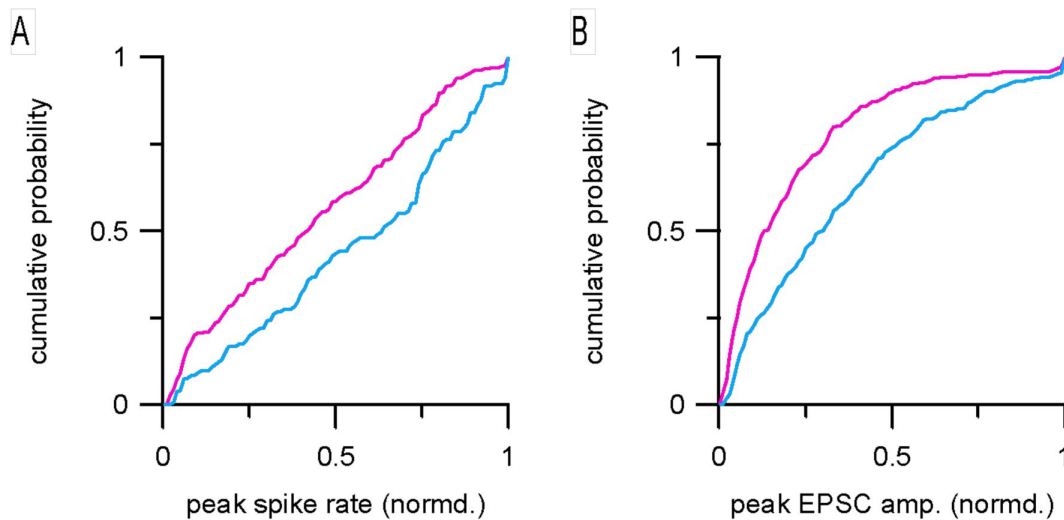


Figure S1.

Noise-evoked responses are biased towards large amplitude events in ON-T compared to ON-S RGCs.

(A) Spike event peak spike rate cumulative distributions for ON-T (cyan; $n = 6$ cells; 131 spike events) and ON-S RGC (magenta; $n = 7$ cells; 300 spike events). Firing rates from average responses to a repeated noise stimulus (see [Fig. 1B-C](#)) were normalized to the maximal firing rate in each cell before combining events across cells. (B) EPSC amplitude cumulative distributions for ON-T ($n = 13$ cells; 275 EPSCs) and ON-S ($n = 10$ cells; 365 EPSCs) RGCs. Same stimulus and normalization procedure as in (A).

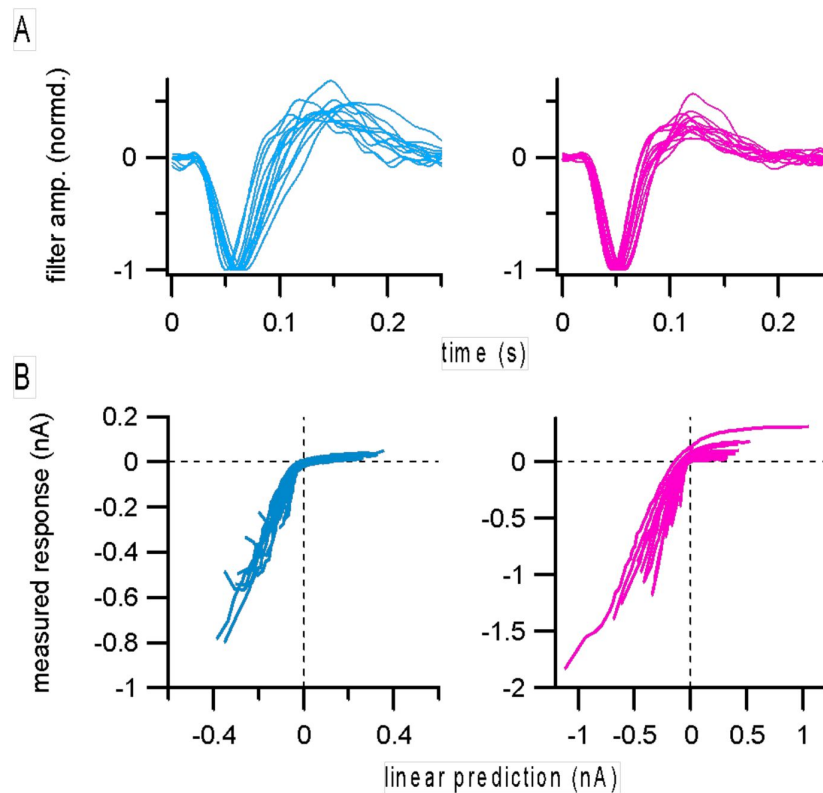


Figure S2.

Linear-nonlinear models of excitatory synaptic input to ON-T and ON-S RGCs.

(A) Linear filters computed from responses to non-repeated noise stimulus presentation. Lines show filters from individual ON-T (cyan; $n=13$) and ON-S (magenta; $n=15$) RGCs. Filters were normalized to the peak amplitude of the primary inward component to facilitate visual comparison of temporal responses. Same cells as shown in average filters in [Fig. 2B](#). (B) Nonlinearities derived by plotting the predicted response (convolution of linear filter and stimulus) against the measured excitatory currents for each RGC shown in (A). Linear predictions were calculated using non-normalized filters.

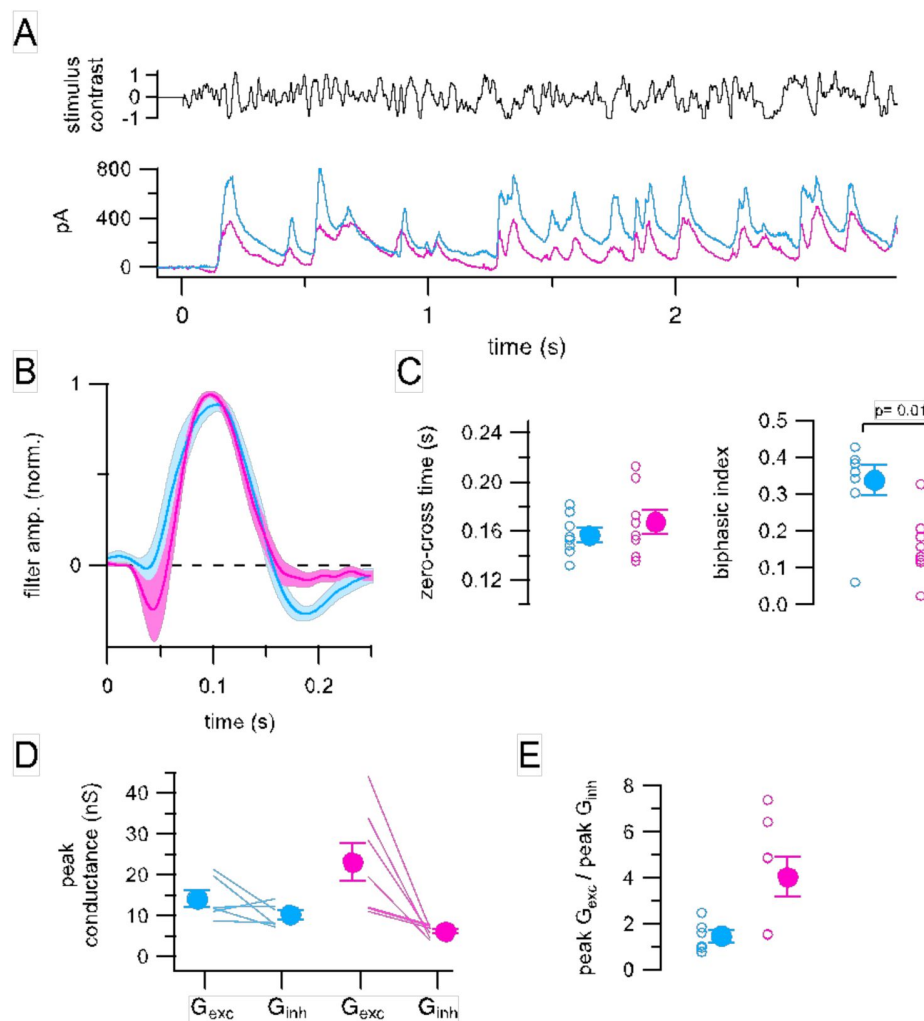


Figure S3.

Inhibitory synaptic input to ON-T and ON-S RGCs.

(A) Inhibitory synaptic currents measured at $V_{hold} = +10$ mV (reversal potential for excitation) in example ON-T (cyan) and ON-S (pink) RGCs in response to same Gaussian noise stimulus as in **Figs. 1**–**2**. Traces show average currents from ten repetitions of the same stimulus and are baseline subtracted to facilitate comparison. (B) Linear filters computed from responses to non-repeated noise stimulus presentation. Lines and shaded areas show mean $\pm$ SEM of responses from ON-T ($n = 8$) and ON-S ($n = 8$) RGCs. (C) Quantification of filter kinetics in different RGCs. Zero-cross times were not significantly different ($p = 0.57$). Significance values from Wilcoxon rank sum test. (D) Peak amplitudes of excitatory and inhibitory conductances (G_{exc} and G_{inh} , respectively) calculated from average current responses to repeated noise stimuli such as shown in **Figs. 2A** and **S3A**. Lines connect peak G_{exc} and G_{inh} values measured from the same RGCs. Circles with error bars show mean $\pm$ SEM across ON-T RGCs ($n = 6$) and ON-S RGCs ($n = 7$) (E) Summary of peak excitatory/inhibitory conductance ratios. Open circles show measurements from individual cells and filled circles with error bars show mean $\pm$ SEM values. No significant difference between ON-T and ON-S RGCs; $p = 0.10$, Wilcoxon rank sum test.

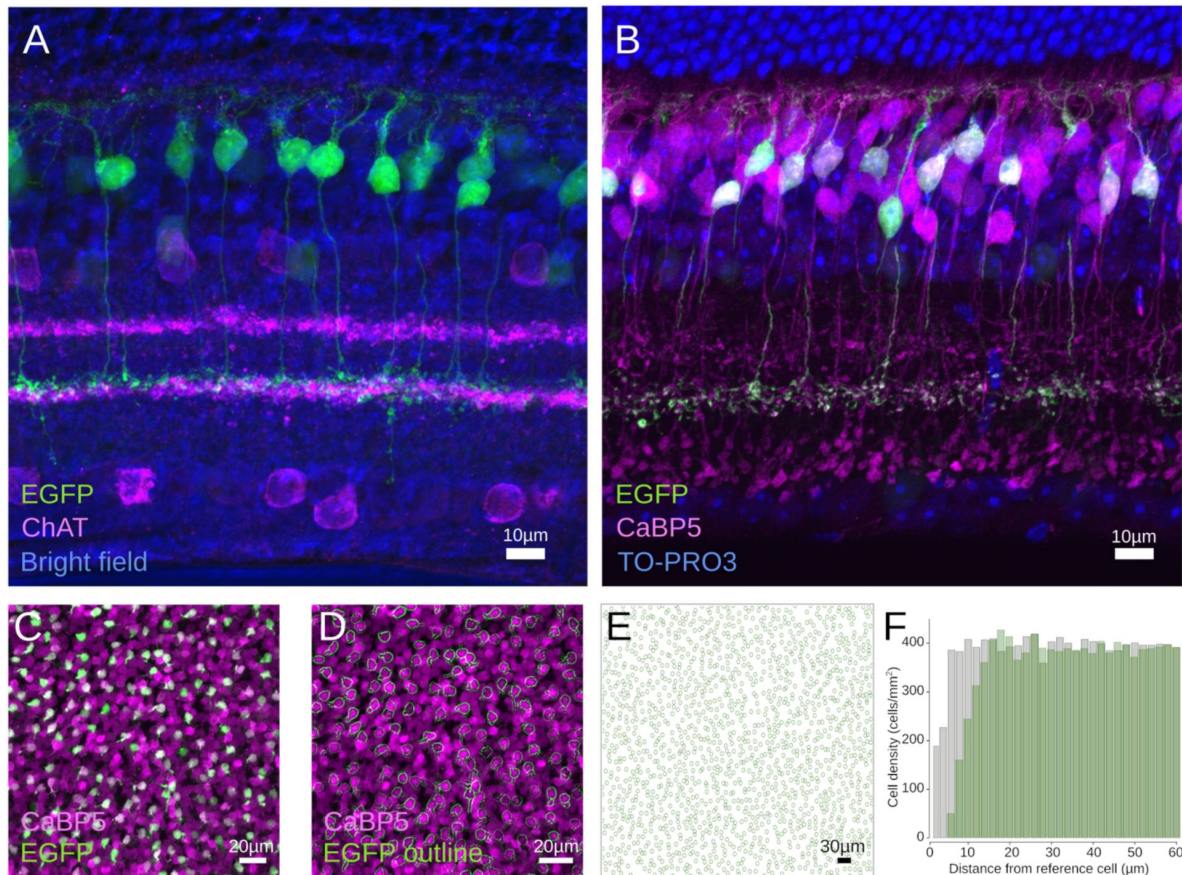


Figure S4.

Characterization of Gjd2-EGFP+ bipolar cells.

(A) Vertical section of the Gjd2-EGFP mouse line showing EGFP positive bipolar cells axons (green) stratifying in the ON sublamina of the IPL, co-labeled with cholinergic amacrine cell (magenta, ChAT antibody). Transmitted light image of the retina in blue. (B) Vertical section of the Gjd2-EGFP mouse line showing EGFP positive bipolar cells (green) colocalized with the T5 marker CaBP5 (magenta). (C) En-face image of a whole mount retina from the GJD2 mouse line showing EGFP positive bipolar cells bodies (green) co-labeled with the T5 bipolar cell marker CaBP5 (magenta). (D) Same field as in (C) with cell bodies of EGFP positive bipolar cells outlined (green) by application of a digital edge outlining filter in ImageJ. (E) Position of cell body centroids for EGFP positive bipolar cells in a larger field of view (20X objective) outlined by green circles. (F) Density recovery profile analysis showing the density profile as a function of distance from the reference neuron for EGFP positive bipolar cells (green histograms) and for an equal amount of neurons randomly positioned in the same space (grey histograms) showing the presence of a characteristic exclusion distance for EGFP positive bipolar cells, hallmark of a retinal mosaic.

Figure S5.

ON-T and ON-S RGC dendrites ramify in distinct IPL sublaminae.

(A) Side projections from 2P z-stack images of dye-filled RGCs (ON-T: top, cyan; ON-S: bottom, magenta) and EGFP signal (yellow) in retinas from Gjd2-EGFP mice. RGCs are same cells as shown in en face view in [Fig. 1A](#) (B) Quantification of fluorescence signals from dye filled RGCs in Gjd2-EGFP mouse retina as in (A). Colored lines and shaded areas show mean $\pm$ SEM fluorescence intensity with respect to distance from peak EGFP (yellow) fluorescence intensity (n= 6 ON-T, cyan and n= 4 ON-S, magenta, RGCs).

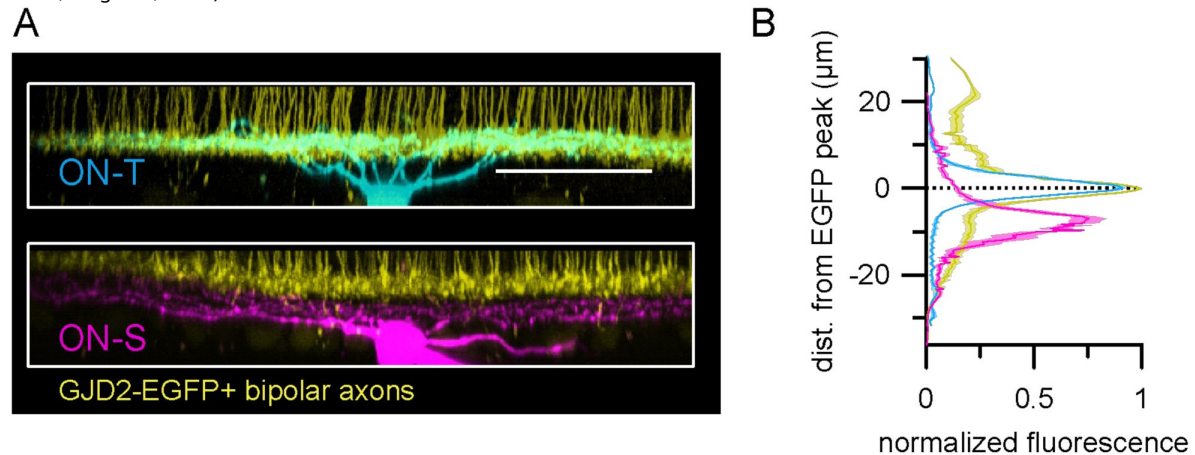
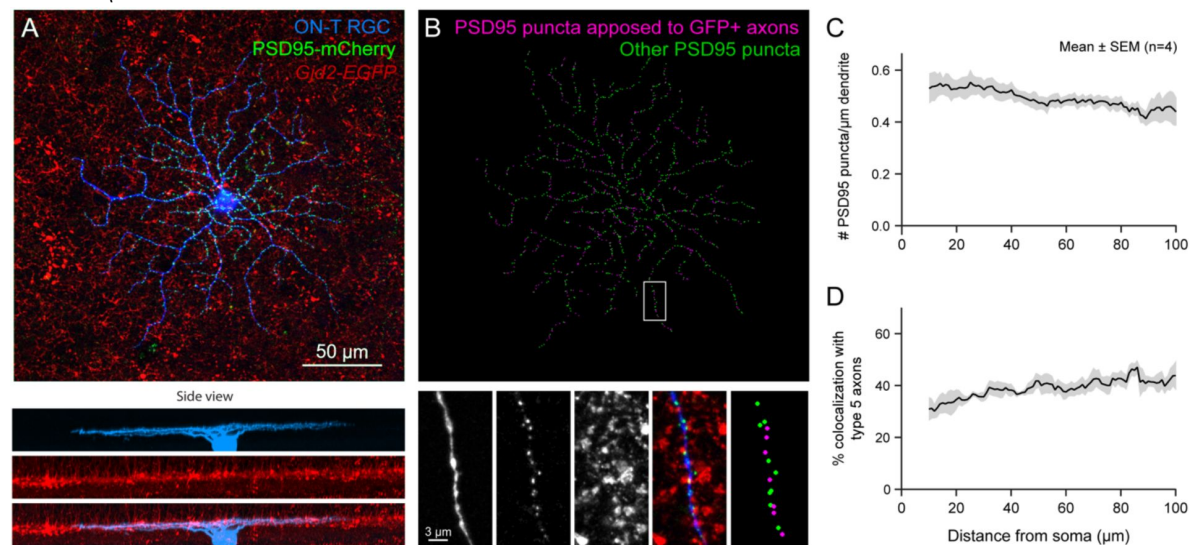


Figure S6.

Gjd2-EGFP+ type 5 bipolar cells provide ~40% of total synapses onto ON-T RGCs.

(A) An example ON-T RGC biolistically labeled by Cerulean Fluorescent Protein (blue) and PSD95-mCherry (green) in retina from a Gjd2-EGFP (red) transgenic mouse. Top: en face view, Bottom: side views. (B) Top: Dot map of PSD95-mCherry apposed (magenta) or not apposed (green) to GFP+ BC axon terminals. Bottom: Magnified view of the raw images and the dot map for a segment of dendritic arbor in the white box. (C) Quantification of total excitatory synapse density across the dendritic arbor of ON-T RGCs. (D) Quantification of excitatory synapses in ON-T dendrites that are adjacent to GFP+ axon terminals.



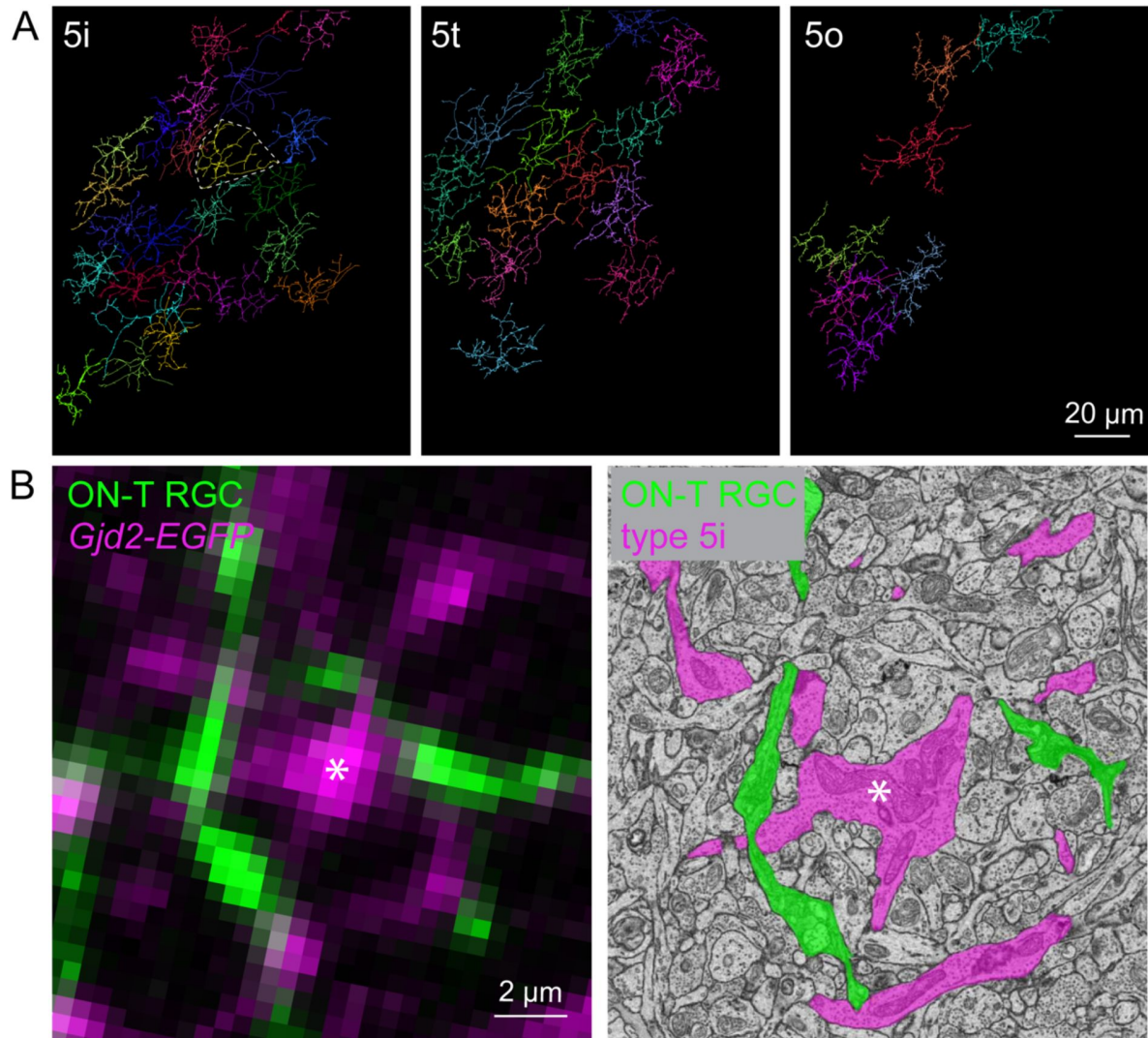


Figure S7.

Classification of three subtypes of type 5 bipolar cells.

(A) Wholemount views of the axon terminals of type 5i, type 5t and type 5o bipolar cells in the SBFSEM volume in [Fig. 3](#). The tiling of the space validates the categorization of the three subtypes of type 5 bipolar cells. (B) Identification of Gjd2-EGFP+ bipolar cells as type 5i bipolar cells, supported by correlated light (left) and electron (right) microscopy images. The reconstructed skeleton of the type 5i bipolar cell in (B) is delineated by dashed lines in (A).

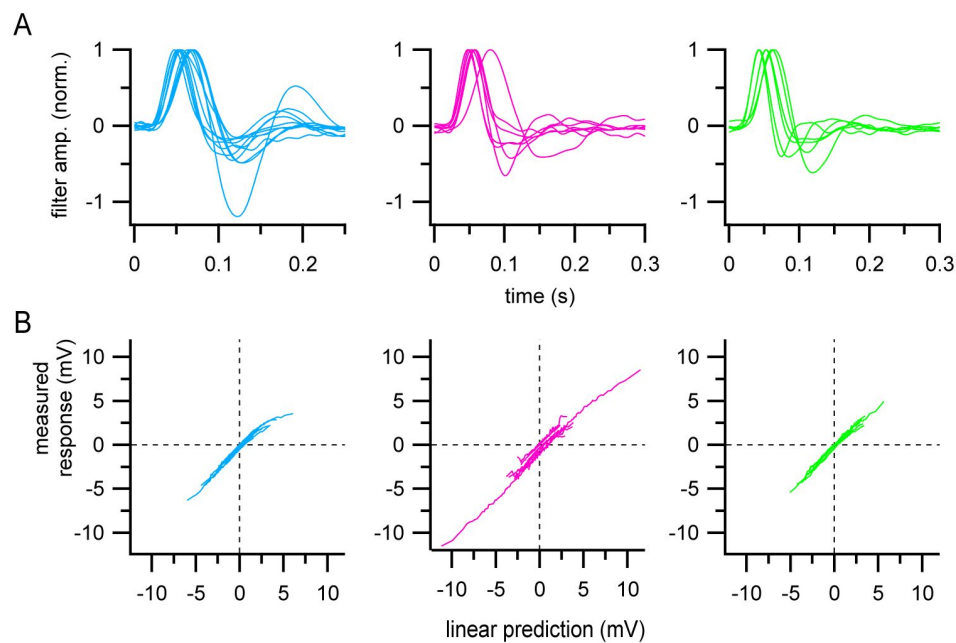


Figure S8.

Linear-nonlinear models of bipolar cell voltage responses.

(A) Individual linear filters for all type 5i (cyan; $n = 11$), type 6 (magenta; $n = 7$), and type 7 (green; $n = 6$) bipolar cells computed from responses to non-repeated noise stimulus presentation. Filters were normalized to the peak amplitude of the primary inward component to facilitate visual comparison of temporal responses. Same cells as shown in average filters in [Fig. 4B](#). (B) Nonlinearities derived by plotting the predicted response (convolution of linear filter and stimulus) against the measured voltage response for each bipolar cell shown in (A). Linear predictions were calculated using non-normalized filters.

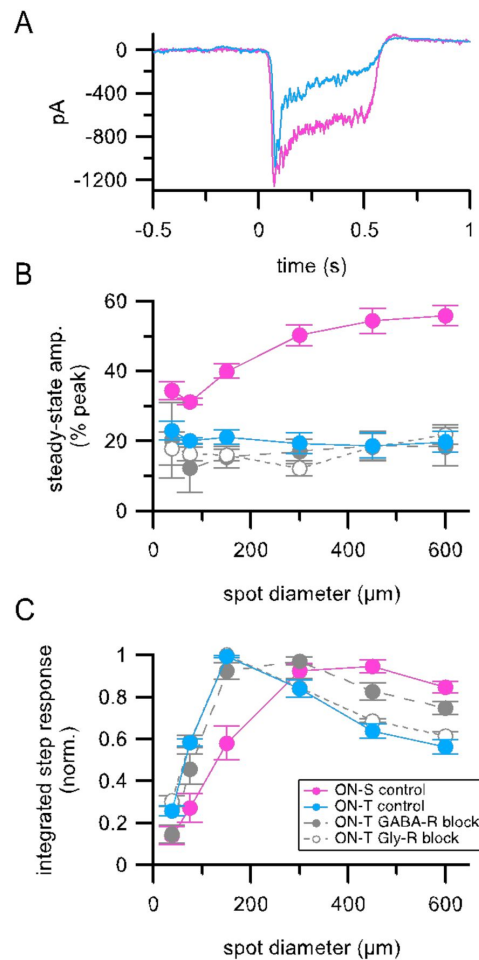


Figure S9.

Presynaptic inhibition does not affect response kinetics but does affect spatial integration for excitatory inputs to ON-T RGCs.

(A) Step-evoked (100% contrast, 52.0 μm spot diameter, 500 ms duration) excitatory synaptic currents in example ON-T (cyan) and ON-S RGCs measured in ACSF. Traces show mean responses to ten trials and are baseline subtracted to facilitate comparison. (B) Summary of light step-evoked steady-state response amplitude for different stimulus sizes measured in ACSF. Note that excitatory inputs in ON-T are more transient than those in ON-S RGCs across all stimulus sizes, but responses in both RGC types are more sustained than when measurements were performed using Ames' solution (compare to **Fig. 2D-E**). (C) Summary of total response, measured as the integrated excitatory current response during the light step, versus stimulus spot size. Responses were normalized to the maximal response in each cell prior to averaging. Excitation in ON-T RGCs is tuned to smaller spot sizes than in ON-S RGCs and this tuning relies in part on GABAergic inhibition. Symbols with error bars in (B) and (C) show mean ± SEM (ON-T control n = 12; ON-S control n = 6; ON-T GABA-R block n = 5; ON-T Gly-R block n = 5).

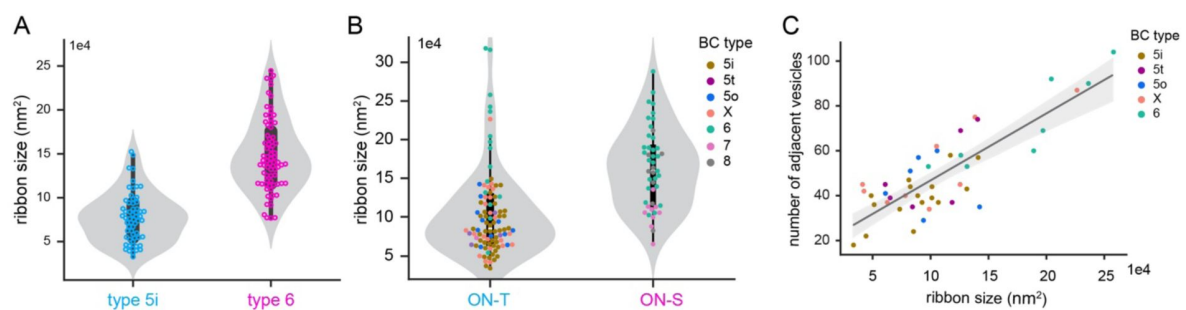


Figure S10.

Bipolar cell subtype-specific differences in ribbon surface areas.

(A) Violin plot showing distribution of ribbon surface areas (each data point is a ribbon) for the type 5i and type 6 BC in (Fig 9A). (B) Violin plot with scattered dots showing ribbon surface areas for each ribbon presynaptic to a stretch of reconstructed ON-T or ON-S RGC dendrite. (C) Scatter plot showing ribbon size in surface area in relationship with the number of adjacent vesicles. Gray line represents the linear regression model fitted to the data. Linear regression: $\beta = 0.0003$, $p < 0.001$, $r^2 = 0.69$.

References

1. Angueyra JM, Rieke F (2013) **Origin and effect of phototransduction noise in primate cone photoreceptors** *Nat Neurosci* **16**:1692–1700 <https://doi.org/10.1038/nn.3534> | Google Scholar
2. Asari H, Meister M (2014) **The projective field of retinal bipolar cells and its modulation by visual context** *Neuron* **81**:641–652 <https://doi.org/10.1016/j.neuron.2013.11.029> | Google Scholar
3. Asari H, Meister M (2012) **Divergence of visual channels in the inner retina** *Nat Neurosci* **15**:1581–1589 <https://doi.org/10.1038/nn.3241> | Google Scholar
4. Awatramani GB, Slaughter MM (2000) **Origin of transient and sustained responses in ganglion cells of the retina** *J Neurosci* **20**:7087–7095 <https://doi.org/10.1523/JNEUROSCI.20-18-07087.2000> | Google Scholar
5. Babai N, Sendelbeck A, Regus-Leidig H, Fuchs M, Mertins J, Reim K, Brose N, Feigenspan A, Brandstätter JH (2016) **Functional Roles of Complexin 3 and Complexin 4 at Mouse Photoreceptor Ribbon Synapses** *J Neurosci* **36**:6651–6667 <https://doi.org/10.1523/JNEUROSCI.4335-15.2016> | Google Scholar
6. Baden T, Berens P, Bethge M, Euler T (2013) **Spikes in mammalian bipolar cells support temporal layering of the inner retina** *Curr Biol* **23**:48–52 <https://doi.org/10.1016/j.cub.2012.11.006> | Google Scholar
7. Baden T, Berens P, Franke K, Román Rosón M, Bethge M, Euler T (2016) **The functional diversity of retinal ganglion cells in the mouse** *Nature* **529**:345–350 <https://doi.org/10.1038/nature16468> | Google Scholar
8. Bae JA, Mu S, Kim JS, Turner NL, Tartavull I, Kemnitz N, Jordan CS, Norton AD, Silversmith WM, Prentki R, Sorek M, David C, Jones DL, Bland D, Sterling ALR, Park J, Briggman KL, Seung HS, Eyewirers (2018) **Digital Museum of Retinal Ganglion Cells with Dense Anatomy and Physiology** *Cell* **173**:1293–1306 <https://doi.org/10.1016/j.cell.2018.04.040> | Google Scholar
9. Beaudoin DL, Kupershtok M, Demb JB (2019) **Selective synaptic connections in the retinal pathway for night vision** *J Comp Neurol* **527**:117–132 <https://doi.org/10.1002/cne.24313> | Google Scholar
10. Bishop D, Nikić I, Brinkoetter M, Knecht S, Potz S, Kerschensteiner M, Misgeld T (2011) **Near-infrared branding efficiently correlates light and electron microscopy** *Nat Methods* **8**:568–570 <https://doi.org/10.1038/nmeth.1622> | Google Scholar
11. Borghuis BG, Marvin JS, Looger LL, Demb JB (2013a) **Two-photon imaging of nonlinear glutamate release dynamics at bipolar cell synapses in the mouse retina** *J Neurosci* **33**:10972–10985 <https://doi.org/10.1523/JNEUROSCI.1241-13.2013> | Google Scholar

12. Borghuis BG, Marvin JS, Looger LL, Demb JB (2013b) **Two-photon imaging of nonlinear glutamate release dynamics at bipolar cell synapses in the mouse retina** *J Neurosci* **33**:10972–10985 <https://doi.org/10.1523/JNEUROSCI.1241-13.2013> | Google Scholar
13. Brombas A, Zhou X, Williams SR (2022) **Light-evoked dendritic spikes in sustained but not transient rabbit retinal ganglion cells** *Neuron* **110**:2802–2814 <https://doi.org/10.1016/j.neuron.2022.06.009> | Google Scholar
14. Burrone J, Lagnado L (2000) **Synaptic depression and the kinetics of exocytosis in retinal bipolar cells** *J Neurosci* **20**:568–578 <https://doi.org/10.1523/JNEUROSCI.20-02-00568.2000> | Google Scholar
15. Burrone J, Neves G, Gomis A, Cooke A, Lagnado L (2002) **Endogenous calcium buffers regulate fast exocytosis in the synaptic terminal of retinal bipolar cells** *Neuron* **33**:101–112 [https://doi.org/10.1016/s0896-6273\(01\)00565-7](https://doi.org/10.1016/s0896-6273(01)00565-7) | Google Scholar
16. Cafaro J, Rieke F (2013) **Regulation of spatial selectivity by crossover inhibition** *J Neurosci* **33**:6310–6320 <https://doi.org/10.1523/JNEUROSCI.4964-12.2013> | Google Scholar
17. Cangiano L, Asteriti S, Cervetto L, Gargini C (2012) **The photovoltage of rods and cones in the dark-adapted mouse retina** *J Physiol* **590**:3841–3855 <https://doi.org/10.1113/jphysiol.2011.226878> | Google Scholar
18. Chang L, Ran Y, Yang M, Auferkorte O, Butz E, Hüsler L, Haverkamp S, Euler T, Schubert T (2023) **Spike desensitisation as a mechanism for high-contrast selectivity in retinal ganglion cells** *Front Cell Neurosci* **17**:1337768 <https://doi.org/10.3389/fncel.2023.1337768> | Google Scholar
19. Chichilnisky EJ (2001) **A simple white noise analysis of neuronal light responses** *Network* **12**:199–213 [Google Scholar](https://doi.org/10.1016/S0959-4702(01)00003-6)
20. Cleland BG, Levick WR, Sanderson KJ (1973) **Properties of sustained and transient ganglion cells in the cat retina** *J Physiol* **228**:649–680 <https://doi.org/10.1113/jphysiol.1973.sp010105> | Google Scholar
21. Cohen E, Sterling P (1990) **Demonstration of cell types among cone bipolar neurons of cat retina** *Philos Trans R Soc Lond B Biol Sci* **330**:305–321 <https://doi.org/10.1098/rstb.1990.0201> | Google Scholar
22. Demb JB, Singer JH (2012) **Intrinsic properties and functional circuitry of the AII amacrine cell** *Vis Neurosci* **29**:51–60 <https://doi.org/10.1017/S0952523811000368> | Google Scholar
23. Dembla E, Dembla M, Maxeiner S, Schmitz F (2020) **Synaptic ribbons foster active zone stability and illumination-dependent active zone enrichment of RIM2 and Cav1.4 in photoreceptor synapses** *Sci Rep* **10**:5957 <https://doi.org/10.1038/s41598-020-62734-0> | Google Scholar
24. DeVries SH (2000) **Bipolar cells use kainate and AMPA receptors to filter visual information into separate channels** *Neuron* **28**:847–856 [https://doi.org/10.1016/s0896-6273\(00\)00158-6](https://doi.org/10.1016/s0896-6273(00)00158-6) | Google Scholar
25. Diamond JS (2017) **Inhibitory Interneurons in the Retina: Types, Circuitry, and Function** *Annu Rev Vis Sci* **3**:1–24 <https://doi.org/10.1146/annurev-vision-102016-061345> | Google Scholar

26. Ding H, Smith RG, Poleg-Polsky A, Diamond JS, Briggman KL (2016) **Species-specific wiring for direction selectivity in the mammalian retina** *Nature* **535**:105–110 <https://doi.org/10.1038/nature18609> | [Google Scholar](#)
27. Dowling JE, Boycott BB (1966) **Organization of the primate retina: electron microscopy** *Proc R Soc Lond B Biol Sci* **166**:80–111 <https://doi.org/10.1098/rspb.1966.0086> | [Google Scholar](#)
28. Duan X, Krishnaswamy A, De la Huerta I, Sanes JR (2014) **Type II cadherins guide assembly of a direction-selective retinal circuit** *Cell* **158**:793–807 <https://doi.org/10.1016/j.cell.2014.06.047> | [Google Scholar](#)
29. Euler T, Haverkamp S, Schubert T, Baden T (2014) **Retinal bipolar cells: elementary building blocks of vision** *Nat Rev Neurosci* **15**:507–519 <https://doi.org/10.1038/nrn3783> | [Google Scholar](#)
30. Euler T, Masland RH (2000) **Light-evoked responses of bipolar cells in a mammalian retina** *J Neurophysiol* **83**:1817–1829 <https://doi.org/10.1152/jn.2000.83.4.1817> | [Google Scholar](#)
31. Farrow K, Teixeira M, Szikra T, Viney TJ, Balint K, Yonehara K, Roska B (2013) **Ambient illumination toggles a neuronal circuit switch in the retina and visual perception at cone threshold** *Neuron* **78**:325–338 <https://doi.org/10.1016/j.neuron.2013.02.014> | [Google Scholar](#)
32. Franke K, Berens P, Schubert T, Bethge M, Euler T, Baden T (2017) **Inhibition decorrelates visual feature representations in the inner retina** *Nature* **542**:439–444 <https://doi.org/10.1038/nature21394> | [Google Scholar](#)
33. Ghosh KK, Bujan S, Haverkamp S, Feigenspan A, Wässle H (2004) **Types of bipolar cells in the mouse retina** *J Comp Neurol* **469**:70–82 <https://doi.org/10.1002/cne.10985> | [Google Scholar](#)
34. Goetz J, Jessen ZF, Jacobi A, Mani A, Cooler S, Greer D, Kadri S, Segal J, Shekhar K, Sanes JR, Schwartz GW (2022) **Unified classification of mouse retinal ganglion cells using function, morphology, and gene expression** *Cell Rep* **40**:111040 <https://doi.org/10.1016/j.celrep.2022.111040> | [Google Scholar](#)
35. Gollisch T, Meister M (2010) **Eye smarter than scientists believed: neural computations in circuits of the retina** *Neuron* **65**:150–164 <https://doi.org/10.1016/j.neuron.2009.12.009> | [Google Scholar](#)
36. Grabner CP, Futagi D, Shi J, Bindokas V, Kitano K, Schwartz EA, DeVries SH (2023) **Mechanisms of simultaneous linear and nonlinear computations at the mammalian cone photoreceptor synapse** *Nat Commun* **14**:3486 <https://doi.org/10.1038/s41467-023-38943-2> | [Google Scholar](#)
37. Grabner CP, Moser T (2021) **The mammalian rod synaptic ribbon is essential for Cav channel facilitation and ultrafast synaptic vesicle fusion** *eLife* **10**:e63844 <https://doi.org/10.7554/eLife.63844> | [Google Scholar](#)
38. Grabner CP, Ratliff CP, Light AC, DeVries SH (2016) **Mechanism of High-Frequency Signaling at a Depressing Ribbon Synapse** *Neuron* **91**:133–145 <https://doi.org/10.1016/j.neuron.2016.05.019> | [Google Scholar](#)
39. Greene MJ, Kim JS, Seung HS, Eyewirers (2016) **Analogous Convergence of Sustained and Transient Inputs in Parallel On and Off Pathways for Retinal Motion Computation** *Cell Rep* **14**:1892–1900 <https://doi.org/10.1016/j.celrep.2016.02.001> | [Google Scholar](#)

40. Grimes WN, Baudin J, Azevedo AW, Rieke F (2018) **Range, routing and kinetics of rod signaling in primate retina** *eLife* **7**:e38281 <https://doi.org/10.7554/eLife.38281> | Google Scholar
41. Grimes WN, Schwartz GW, Rieke F (2014) **The synaptic and circuit mechanisms underlying a change in spatial encoding in the retina** *Neuron* **82**:460–473 <https://doi.org/10.1016/j.neuron.2014.02.037> | Google Scholar
42. Hellmer CB, Zhou Y, Fyk-Kolodziej B, Hu Z, Ichinose T (2016) **Morphological and physiological analysis of type-5 and other bipolar cells in the Mouse Retina** *Neuroscience* **315**:246–258 <https://doi.org/10.1016/j.neuroscience.2015.12.016> | Google Scholar
43. Helmstaedter M, Briggman KL, Turaga SC, Jain V, Seung HS, Denk W (2013) **Connectomic reconstruction of the inner plexiform layer in the mouse retina** *Nature* **500**:168–174 <https://doi.org/10.1038/nature12346> | Google Scholar
44. Hsiang J-C, Shen N, Soto F, Kerschensteiner D (2024) **Distributed feature representations of natural stimuli across parallel retinal pathways** *Nat Commun* **15**:1920 <https://doi.org/10.1038/s41467-024-46348-y> | Google Scholar
45. Huang L, Max M, Margolskee RF, Su H, Masland RH, Euler T (2003) **G protein subunit G gamma 13 is coexpressed with G alpha o, G beta 3, and G beta 4 in retinal ON bipolar cells** *J Comp Neurol* **455**:1–10 <https://doi.org/10.1002/cne.10396> | Google Scholar
46. Huang L, Shanker YG, Dubauskaite J, Zheng JZ, Yan W, Rosenzweig S, Spielman AI, Max M, Margolskee RF (1999) **Ggamma13 colocalizes with gustducin in taste receptor cells and mediates IP3 responses to bitter denatonium** *Nat Neurosci* **2**:1055–1062 <https://doi.org/10.1038/15981> | Google Scholar
47. Hull C, Studholme K, Yazulla S, von Gersdorff H (2006) **Diurnal changes in exocytosis and the number of synaptic ribbons at active zones of an ON-type bipolar cell terminal** *J Neurophysiol* **96**:2025–2033 <https://doi.org/10.1152/jn.00364.2006> | Google Scholar
48. Ichinose T, Fyk-Kolodziej B, Cohn J (2014) **Roles of ON cone bipolar cell subtypes in temporal coding in the mouse retina** *J Neurosci* **34**:8761–8771 <https://doi.org/10.1523/JNEUROSCI.3965-13.2014> | Google Scholar
49. Ichinose T, Hellmer CB (2016) **Differential signalling and glutamate receptor compositions in the OFF bipolar cell types in the mouse retina** *J Physiol* **594**:883–894 <https://doi.org/10.1113/JP271458> | Google Scholar
50. Ikeda H, Wright MJ (1972) **Receptive field organization of “sustained” and “transient” retinal ganglion cells which subserve different function roles** *J Physiol* **227**:769–800 <https://doi.org/10.1113/jphysiol.1972.sp010058> | Google Scholar
51. Jackman SL, Choi S-Y, Thoreson WB, Rabl K, Bartoletti TM, Kramer RH (2009) **Role of the synaptic ribbon in transmitting the cone light response** *Nat Neurosci* **12**:303–310 <https://doi.org/10.1038/nn.2267> | Google Scholar
52. Jadzinsky PD, Baccus SA (2013) **Transformation of visual signals by inhibitory interneurons in retinal circuits** *Annu Rev Neurosci* **36**:403–428 <https://doi.org/10.1146/annurev-neuro-062012-170315> | Google Scholar

53. Jarsky T, Cembrowski M, Logan SM, Kath WL, Rieke H, Demb JB, Singer JH (2011) **A Synaptic Mechanism for Retinal Adaptation to Luminance and Contrast** *J Neurosci* **31**:11003–11015 <https://doi.org/10.1523/JNEUROSCI.2631-11.2011> | Google Scholar
54. Ke J-B, Wang YV, Borghuis BG, Cembrowski MS, Rieke H, Kath WL, Demb JB, Singer JH (2014) **Adaptation to background light enables contrast coding at rod bipolar cell synapses** *Neuron* **81**:388–401 <https://doi.org/10.1016/j.neuron.2013.10.054> | Google Scholar
55. Kerschensteiner D (2022) **Feature Detection by Retinal Ganglion Cells** *Annu Rev Vis Sci* **8**:135–169 <https://doi.org/10.1146/annurev-vision-100419-112009> | Google Scholar
56. Kim KJ, Rieke F (2001) **Temporal contrast adaptation in the input and output signals of salamander retinal ganglion cells** *J Neurosci* **21**:287–299 <https://doi.org/10.1523/JNEUROSCI.21-01-00287.2001> | Google Scholar
57. Krieger B, Qiao M, Rousso DL, Sanes JR, Meister M (2017) **Four alpha ganglion cell types in mouse retina: Function, structure, and molecular signatures** *PLoS One* **12**:e0180091 <https://doi.org/10.1371/journal.pone.0180091> | Google Scholar
58. Kuo SP, Chiang P-P, Nippert AR, Newman EA (2020) **Spatial Organization and Dynamics of the Extracellular Space in the Mouse Retina** *J Neurosci* **40**:7785–7794 <https://doi.org/10.1523/JNEUROSCI.1717-20.2020> | Google Scholar
59. Kuo SP, Schwartz GW, Rieke F (2016) **Nonlinear Spatiotemporal Integration by Electrical and Chemical Synapses in the Retina** *Neuron* **90**:320–332 <https://doi.org/10.1016/j.neuron.2016.03.012> | Google Scholar
60. Lagnado L, Schmitz F (2015) **Ribbon Synapses and Visual Processing in the Retina** *Annu Rev Vis Sci* **1**:235–262 <https://doi.org/10.1146/annurev-vision-082114-035709> | Google Scholar
61. Marc RE, Anderson JR, Jones BW, Sigulinsky CL, Lauritzen JS (2014) **The AII amacrine cell connectome: a dense network hub** *Front Neural Circuits* **8**:104 <https://doi.org/10.3389/fncir.2014.00104> | Google Scholar
62. Margolis DJ, Gartland AJ, Euler T, Detwiler PB (2010) **Dendritic Calcium Signaling in ON and OFF Mouse Retinal Ganglion Cells** *J Neurosci* **30**:7127–7138 <https://doi.org/10.1523/JNEUROSCI.5694-09.2010> | Google Scholar
63. Masland RH (2012) **The Neuronal Organization of the Retina** *Neuron* **76**:266–280 <https://doi.org/10.1016/j.neuron.2012.10.002> | Google Scholar
64. Maxeiner S, Luo F, Tan A, Schmitz F, Südhof TC (2016) **How to make a synaptic ribbon: RIBEYE deletion abolishes ribbons in retinal synapses and disrupts neurotransmitter release** *EMBO J* **35**:1098–1114 <https://doi.org/10.15252/embj.201592701> | Google Scholar
65. McGuire BA, Stevens JK, Sterling P (1984) **Microcircuitry of bipolar cells in cat retina** *J Neurosci* **4**:2920–2938 <https://doi.org/10.1523/JNEUROSCI.04-12-02920.1984> | Google Scholar
66. Mehta B, Snellman J, Chen S, Li W, Zenisek D (2013) **Synaptic ribbons influence the size and frequency of miniature-like evoked postsynaptic currents** *Neuron* **77**:516–527 <https://doi.org/10.1016/j.neuron.2012.11.024> | Google Scholar
67. Morgan JL, Soto F, Wong ROL, Kerschensteiner D (2011) **Development of cell type-specific connectivity patterns of converging excitatory axons in the retina** *Neuron* **71**:1014–

1021 <https://doi.org/10.1016/j.neuron.2011.08.025> | Google Scholar

68. Neves G, Lagnado L (1999) **The kinetics of exocytosis and endocytosis in the synaptic terminal of goldfish retinal bipolar cells** *J Physiol* **515**:181–202 <https://doi.org/10.1111/j.1469-7793.1999.181ad.x> | Google Scholar
69. Nikolaev A, Leung K-M, Odermatt B, Lagnado L (2013) **Synaptic mechanisms of adaptation and sensitization in the retina** *Nat Neurosci* **16**:934–941 <https://doi.org/10.1038/nn.3408> | Google Scholar
70. Nirenberg S, Meister M (1997) **The light response of retinal ganglion cells is truncated by a displaced amacrine circuit** *Neuron* **18**:637–650 [https://doi.org/10.1016/s0896-6273\(00\)80304-9](https://doi.org/10.1016/s0896-6273(00)80304-9) | Google Scholar
71. Oesch NW, Diamond JS (2011) **Ribbon synapses compute temporal contrast and encode luminance in retinal rod bipolar cells** *Nat Neurosci* **14**:1555–1561 <https://doi.org/10.1038/nn.2945> | Google Scholar
72. Okawa H, Yu W-Q, Matti U, Schwarz K, Odermatt B, Zhong H, Tsukamoto Y, Lagnado L, Rieke F, Schmitz F, Wong ROL (2019) **Dynamic assembly of ribbon synapses and circuit maintenance in a vertebrate sensory system** *Nat Commun* **10**:2167 <https://doi.org/10.1038/s41467-019-10123-1> | Google Scholar
73. Pangrsic T, Singer JH, Koschak A (2018) **Voltage-Gated Calcium Channels: Key Players in Sensory Coding in the Retina and the Inner Ear** *Physiol Rev* **98**:2063–2096 <https://doi.org/10.1152/physrev.00030.2017> | Google Scholar
74. Pottackal J, Singer JH, Demb JB (2021) **Computational and Molecular Properties of Starburst Amacrine Cell Synapses Differ With Postsynaptic Cell Type** *Front Cell Neurosci* **15**:660773 <https://doi.org/10.3389/fncel.2021.660773> | Google Scholar
75. Puthusseri T, Venkataramani S, Gayet-Primo J, Smith RG, Taylor WR (2013) **NaV1.1 channels in axon initial segments of bipolar cells augment input to magnocellular visual pathways in the primate retina** *J Neurosci* **33**:16045–16059 <https://doi.org/10.1523/JNEUROSCI.1249-13.2013> | Google Scholar
76. Rabl K, Cadetti L, Thoreson WB (2006) **Paired-pulse depression at photoreceptor synapses** *J Neurosci* **26**:2555–2563 <https://doi.org/10.1523/JNEUROSCI.3667-05.2006> | Google Scholar
77. Ran Y, Huang Z, Baden T, Schubert T, Baayen H, Berens P, Franke K, Euler T (2020) **Type-specific dendritic integration in mouse retinal ganglion cells** *Nat Commun* **11**:2101 <https://doi.org/10.1038/s41467-020-15867-9> | Google Scholar
78. Rieke F (2001) **Temporal contrast adaptation in salamander bipolar cells** *J Neurosci* **21**:9445–9454 | Google Scholar
79. Rodieck RW (1991) **The density recovery profile: a method for the analysis of points in the plane applicable to retinal studies** *Vis Neurosci* **6**:95–111 <https://doi.org/10.1017/s095252380001049x> | Google Scholar

80. Roska B, Werblin F (2001) **Vertical interactions across ten parallel, stacked representations in the mammalian retina** *Nature* **410**:583–587 <https://doi.org/10.1038/35069068> | Google Scholar
81. Sabbah S, Papendorp C, Koplas E, Beltoja M, Etebari C, Gunesch AN, Carrete L, Kim MT, Manoff G, Bhatia-Lin A, Zhao T, Schreck D, Dowling H, Briggman KL, Berson DM (2018) **Synaptic circuits for irradiance coding by intrinsically photosensitive retinal ganglion cells (preprint)** *Neuroscience* <https://doi.org/10.1101/442954> | Google Scholar
82. Sagdullaev BT, McCall MA, Lukasiewicz PD (2006) **Presynaptic inhibition modulates spillover, creating distinct dynamic response ranges of sensory output** *Neuron* **50**:923–935 <https://doi.org/10.1016/j.neuron.2006.05.015> | Google Scholar
83. Saszik S, DeVries SH (2012) **A mammalian retinal bipolar cell uses both graded changes in membrane voltage and all-or-nothing Na⁺ spikes to encode light** *J Neurosci* **32**:297–307 <https://doi.org/10.1523/JNEUROSCI.2739-08.2012> | Google Scholar
84. Schwartz GW, Okawa H, Dunn FA, Morgan JL, Kerschensteiner D, Wong RO, Rieke F (2012) **The spatial structure of a nonlinear receptive field** *Nat Neurosci* **15**:1572–1580 <https://doi.org/10.1038/nn.3225> | Google Scholar
85. Shekhar K, Lapan SW, Whitney IE, Tran NM, Macosko EZ, Kowalczyk M, Adiconis X, Levin JZ, Nemesh J, Goldman M, McCarroll SA, Cepko CL, Regev A, Sanes JR (2016) **Comprehensive Classification of Retinal Bipolar Neurons by Single-Cell Transcriptomics** *Cell* **166**:1308–1323 <https://doi.org/10.1016/j.cell.2016.07.054> | Google Scholar
86. Sigulinsky CL, Anderson JR, Kerzner E, Rapp CN, Pfeiffer RL, Rodman TM, Emrich DP, Rapp KD, Nelson NT, Lauritzen JS, Meyer M, Marc RE, Jones BW (2020) **Network Architecture of Gap Junctional Coupling among Parallel Processing Channels in the Mammalian Retina** *J Neurosci* **40**:4483–4511 <https://doi.org/10.1523/JNEUROSCI.1810-19.2020> | Google Scholar
87. Singer JH, Diamond JS (2006) **Vesicle depletion and synaptic depression at a mammalian ribbon synapse** *J Neurophysiol* **95**:3191–3198 <https://doi.org/10.1152/jn.01309.2005> | Google Scholar
88. Singer JH, Diamond JS (2003) **Sustained Ca²⁺ entry elicits transient postsynaptic currents at a retinal ribbon synapse** *J Neurosci* **23**:10923–10933 <https://doi.org/10.1523/JNEUROSCI.23-34-10923.2003> | Google Scholar
89. Slaughter MM, Miller RF (1981) **2-amino-4-phosphonobutyric acid: a new pharmacological tool for retina research** *Science* **211**:182–185 <https://doi.org/10.1126/science.6255566> | Google Scholar
90. Snellman J, Mehta B, Babai N, Bartoletti TM, Akmentin W, Francis A, Matthews G, Thoreson W, Zenisek D (2011) **Acute destruction of the synaptic ribbon reveals a role for the ribbon in vesicle priming** *Nat Neurosci* **14**:1135–1141 <https://doi.org/10.1038/nn.2870> | Google Scholar
91. Snellman J, Zenisek D, Nawy S (2009) **Switching between transient and sustained signalling at the rod bipolar-AII amacrine cell synapse of the mouse retina** *J Physiol* **587**:2443–2455 <https://doi.org/10.1113/jphysiol.2008.165241> | Google Scholar
92. Srivastava P, de Rosenroll G, Matsumoto A, Michaels T, Turple Z, Jain V, Sethuramanujam S, Murphy-Baum BL, Yonehara K, Awatramani GB (2022) **Spatiotemporal properties of**

glutamate input support direction selectivity in the dendrites of retinal starburst amacrine cells *eLife* **11**:e81533 <https://doi.org/10.7554/eLife.81533> | Google Scholar

93. Swygart D, Yu W-Q, Takeuchi S, Wong ROL, Schwartz GW (2024) **A presynaptic source drives differing levels of surround suppression in two mouse retinal ganglion cell types** *Nat Commun* **15**:599 <https://doi.org/10.1038/s41467-024-44851-w> | Google Scholar
94. Szikra T, Trenholm S, Drinnenberg A, Jüttner J, Raics Z, Farrow K, Biel M, Awatramani G, Clark DA, Sahel J-A, da Silveira RA, Roska B (2014) **Rods in daylight act as relay cells for cone-driven horizontal cell-mediated surround inhibition** *Nat Neurosci* **17**:1728–1735 <https://doi.org/10.1038/nn.3852> | Google Scholar
95. Taylor WR, Vaney DI (2002) **Diverse synaptic mechanisms generate direction selectivity in the rabbit retina** *J Neurosci* **22**:7712–7720 [Google Scholar](https://doi.org/10.1523/JNEUROSCI.22-24-10558.2002)
96. Tsukamoto Y, Omi N (2017) **Classification of Mouse Retinal Bipolar Cells: Type-Specific Connectivity with Special Reference to Rod-Driven AII Amacrine Pathways** *Front Neuroanat* **11**:92 <https://doi.org/10.3389/fnana.2017.00092> | Google Scholar
97. Vaney DI, Sivyer B, Taylor WR (2012) **Direction selectivity in the retina: symmetry and asymmetry in structure and function** *Nat Rev Neurosci* **13**:194–208 <https://doi.org/10.1038/nrn3165> | Google Scholar
98. Veruki ML, Hartveit E (2002a) **Electrical synapses mediate signal transmission in the rod pathway of the mammalian retina** *J Neurosci* **22**:10558–10566 <https://doi.org/10.1523/JNEUROSCI.22-24-10558.2002> | Google Scholar
99. Veruki ML, Hartveit E (2002b) **AII (Rod) amacrine cells form a network of electrically coupled interneurons in the mammalian retina** *Neuron* **33**:935–946 [https://doi.org/10.1016/s0896-6273\(02\)00609-8](https://doi.org/10.1016/s0896-6273(02)00609-8) | Google Scholar
100. von Gersdorff H, Matthews G (1997) **Depletion and replenishment of vesicle pools at a ribbon-type synaptic terminal** *J Neurosci* **17**:1919–1927 <https://doi.org/10.1523/JNEUROSCI.17-06-01919.1997> | Google Scholar
101. von Gersdorff H, Vardi E, Matthews G, Sterling P. (1996) **Evidence that vesicles on the synaptic ribbon of retinal bipolar neurons can be rapidly released** *Neuron* **16**:1221–1227 [https://doi.org/10.1016/s0896-6273\(00\)80148-8](https://doi.org/10.1016/s0896-6273(00)80148-8) | Google Scholar
102. Wakeham CM, Shi Q, Ren G, Haley TL, Duvoisin RM, von Gersdorff H, Morgans CW (2023) **Trophoblast glycoprotein is required for efficient synaptic vesicle exocytosis from retinal rod bipolar cells** *Front Cell Neurosci* **17**:1306006 <https://doi.org/10.3389/fncel.2023.1306006> | Google Scholar
103. Wan Q-F, Heidelberger R (2011) **Synaptic release at mammalian bipolar cell terminals** *Vis Neurosci* **28**:109–119 <https://doi.org/10.1017/S0952523810000453> | Google Scholar
104. Warwick RA, Kaushansky N, Sarid N, Golan A, Rivlin-Etzion M (2018) **Inhomogeneous Encoding of the Visual Field in the Mouse Retina** *Curr Biol* **28**:655–665 <https://doi.org/10.1016/j.cub.2018.01.016> | Google Scholar
105. Wässle H, Puller C, Müller F, Haverkamp S (2009) **Cone contacts, mosaics, and territories of bipolar cells in the mouse retina** *J Neurosci* **29**:106–117 <https://doi.org/10.1523/JNEUROSCI.4442-08.2009> | Google Scholar

106. Werblin FS (1970) **Response of retinal cells to moving spots: intracellular recording in *Necturus maculosus*** *J Neurophysiol* **33**:342–350 <https://doi.org/10.1152/jn.1970.33.3.342> | [Google Scholar](#)
107. Werginz P, Raghuram V, Fried SI (2020) **Tailoring of the axon initial segment shapes the conversion of synaptic inputs into spiking output in OFF- α T retinal ganglion cells** *Sci Adv* **6**:eabb6642 <https://doi.org/10.1126/sciadv.abb6642> | [Google Scholar](#)
108. Wienbar S, Schwartz GW (2022) **Differences in spike generation instead of synaptic inputs determine the feature selectivity of two retinal cell types** *Neuron* **110**:2110–2123 <https://doi.org/10.1016/j.neuron.2022.04.012> | [Google Scholar](#)
109. Wu SM, Gao F, Maple BR (2000) **Functional architecture of synapses in the inner retina: segregation of visual signals by stratification of bipolar cell axon terminals** *J Neurosci* **20**:4462–4470 <https://doi.org/10.1523/JNEUROSCI.20-12-04462.2000> | [Google Scholar](#)
110. Yu W-Q, Swanstrom R, Sigulinsky CL, Ahlquist RM, Knecht S, Jones BW, Berson DM, Wong RO (2023) **Distinctive synaptic structural motifs link excitatory retinal interneurons to diverse postsynaptic partner types** *Cell Rep* **42**:112006 <https://doi.org/10.1016/j.celrep.2023.112006> | [Google Scholar](#)
111. Zhang Y, Kim I-J, Sanes JR, Meister M (2012) **The most numerous ganglion cell type of the mouse retina is a selective feature detector** *Proc Natl Acad Sci U S A* **109**:E2391–2398 <https://doi.org/10.1073/pnas.1211547109> | [Google Scholar](#)

Author information

Sidney P Kuo

Department of Neurobiology and Biophysics, University of Washington, Seattle, United States,
Department of Neuroscience, University of Minnesota, Minneapolis, United States

Wan-Qing Yu

Department of Neurobiology and Biophysics, University of Washington, Seattle, United States

Prerna Srivastava

Department of Biology, University of Victoria, Victoria, Canada

Haruhisa Okawa

Department of Neurobiology and Biophysics, University of Washington, Seattle, United States

Luca Della Santina

Department of Neurobiology and Biophysics, University of Washington, Seattle, United States

David M Berson

Department of Neuroscience, Brown University, Providence, United States

Gautam B Awatramani

Department of Biology, University of Victoria, Victoria, Canada
ORCID iD: [0000-0002-0610-5271](https://orcid.org/0000-0002-0610-5271)

Rachel OL Wong

Department of Neurobiology and Biophysics, University of Washington, Seattle, United States
ORCID iD: [0000-0001-5089-6296](https://orcid.org/0000-0001-5089-6296)

Fred Rieke

Department of Neurobiology and Biophysics, University of Washington, Seattle, United States
ORCID iD: [0000-0002-1052-2609](https://orcid.org/0000-0002-1052-2609)

For correspondence: rieke@uw.edu

Editors

Reviewing Editor

Henrique von Gersdorff

Oregon Health and Science University, Portland, United States of America

Senior Editor

Sacha Nelson

Brandeis University, Waltham, United States of America

Reviewer #1 (Public review):

Summary:

In the retina, parallel processing of cone photoreceptor output under bright light conditions dissects critical features of our visual environment, and fundamental to visual function. Cone photoreceptor signals are sampled by several types of bipolar cells and passed onto the ganglion cells. At the output of retinal processing, retinal ganglion cells send about 40 different codes of the visual scene to the brain for further processing. In this study, the authors focus on whether subtype-specific differences in the size of synaptic ribbon-associated vesicle pools of bipolar cells contribute to different retinal ganglion cell (RGC) responses.

Specifically, inputs to ON alpha RGCs producing transient versus sustained kinetics (ON-S vs. ON-T, respectively) are compared. The authors first demonstrate that ON-S vs. ON-T RGCs are readily identifiable in a whole mount preparation and respond differently to both static and to a spatially uniform, randomly fluctuating (Gaussian noise) light stimulus. Linear-nonlinear (LN) models were used to estimate the transformation between visual input and excitatory synaptic input for each RGCs; these models suggested the presence of transient versus sustained kinetics already in the excitatory inputs to ON-T and ON-S RGCs.

Indeed, the authors show that (glutamatergic) excitatory inputs to ON-S vs. ON-T RGCs are of distinct kinetics. The subtypes of bipolar cells providing input to ON-S are known (i.e., type 6 and 7), but the source of excitatory bipolar inputs to ON-T RGCs needed to be determined. In a tedious process, it is elegantly shown here that ON-T RGCs receive most of their excitatory inputs from type 5 and 6 bipolars. Interestingly, the temporal properties of light-evoked responses of type 5, 6 and 7 bipolars recorded from the somas were indistinguishable and rather sustained, suggesting that the origin of transient kinetics of excitatory inputs to ON-T RGCs suggested by the LN model might be found in the processing of visual signals at the bipolar cell axon terminal. Blocking GABA- or glycinergic inhibitory inputs did not alter the light-evoked excitatory input kinetics to ON-T and ON-S RGCs. Two-photon glutamate sensor imaging revealed significantly faster kinetics of light-evoked glutamate signals at ON-T versus ON-S RGCs, and that differences in glutamate release from presynaptic bipolar cells are retained without amacrine feedback to bipolar cells. Detailed EM analysis of bipolar cell

ribbon synapses onto ON-T and ON-S RGCs revealed fewer ribbon-associated vesicles at ON-T synapses, that is consistent with stronger paired-flash depression of light-evoked excitatory currents in ON-T RGCs versus ON-S RGCs. This study suggests that bipolar subtype-specific differences in the size of synaptic ribbon-associated vesicle pools contributes to transient versus sustained kinetics in RGCs.

Strengths:

The use of multiple, state-of-the-art tools and approaches to address the kinetics of bipolar to ganglion cell synapse in an identified circuit.

<https://doi.org/10.7554/eLife.98817.2.sa3>

Reviewer #2 (Public review):

Summary:

Goal of the study. The authors tried to pinpoint the origins of transient and sustained responses measured at retinal ganglion cells (rgcs), which is the output layer of the retina. Response characteristics of rgcs are used to group them into different types. The diversity of rgc types represents the ability of the retina to transform visual inputs into distinct output channels. They find that the physical dimensions of bipolar cell's synaptic ribbons (specialized release sites/active zones) vary across the different types of cone on-bpcs, in ways that they argue could facilitate transient or sustained release. This diversity of release output is what they argue underlies the differences in on-rgcs response characteristics, and ultimately represents a mechanism for creating parallel cone-driven channels.

Strengths:

The major strengths of the study are the anatomical approaches employed and the use of the "glutamate sniffer" to assay synaptic glutamate levels. The outline of the study is elegant and reflects the strengths of the authors.

Comments on revised version:

The authors have addressed my comments either through new experiments and/or with additional citations.

Explanation of the studies significance. I think the study provides a solid set of data, acquired through exceptional methodologies, and delivers a compelling hypothesis. This is an exceptionally talented group of systems level thinkers and experimentalists, who are now pointing to smaller scale biophysical principles of synaptic transmission.

<https://doi.org/10.7554/eLife.98817.2.sa2>

Reviewer #3 (Public review):

Summary:

Different types of retinal ganglion cell (RGC) have different temporal properties - most prominently a distinction between sustained vs. transient responses to contrast. This has been well established in multiple species, including mouse. In general, RGCs with dendrites that stratify close to the ganglion cell layer (GCL) are sustained; whereas those that stratify near the middle of the inner plexiform layer (IPL) are transient. This difference in RGC spiking responses aligns with similar differences in excitatory synaptic currents as well as with differences in glutamate release in the respective layers - shown previously and here,

with a glutamate sensor (iGluSnFR) expressed in the RGCs of interest. Differences in glutamate release were not explained by differences in the distinct presynaptic bipolar cells' voltage responses, which were quite similar to one another. Rather, the difference in transient vs. sustained responses seems to emerge at the bipolar cell axon terminals in the form of glutamate release. This difference in the temporal pattern of glutamate release was correlated with differences in the size of synaptic ribbons (larger in the bipolar cells with more sustained responses), which also correlated with a greater number of vesicles in the vicinity of the larger ribbons.

The main conclusion of the study relates to a correlation (because it is difficult to manipulate ribbon size or vesicle density experimentally): the bipolar cells with increased ribbon size/vesicle number would have a greater possibility of sustained release, which would be reflected in the postsynaptic RGC synaptic currents and RGC firing rates. This model proposes a mechanism for temporal channels that is independent of synaptic inhibition. Indeed, some experiments in the paper suggest that inhibition cannot explain the transient nature of glutamate release onto one of the RGC types. Still, it is surprising that such a diverse set of inhibitory interneurons in the retina would not play some role in diversifying the temporal properties of RGC responses.

Strengths:

- (1) The study uses a systematic approach to evaluating temporal properties of retinal ganglion cell (RGC) spiking outputs, excitatory synaptic inputs, presynaptic voltage responses, and presynaptic glutamate release. The combination of these experiments demonstrates an important step in the conversion from voltage to glutamate release in shaping response dynamics in RGCs.
- (2) The study uses a combination of electrophysiology, two-photon imaging and scanning block face EM to build a quantitative and coherent story about specific retinal circuits and their functional properties.

Weaknesses:

- (1) There were some interesting aspects of the study that were not completely resolved, and resolving some of these issues may go beyond the current study. For example, it was interesting that different extracellular media (Ames medium vs. ACSF) generated different degrees of transient vs. sustained responses in RGCs, but it was unclear how these media might have impacted ion channels at different levels of the circuit that could explain the effects on temporal tuning.
- (2) It was surprising that inhibition played such a small role in generating temporal tuning. The authors explored this further in the revision, which supported the original claim that inhibition plays a minor role in glutamate release dynamics from the bipolar cells under study.

<https://doi.org/10.7554/eLife.98817.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public Review):

Summary:

In the retina, parallel processing of cone photoreceptor output under bright light conditions dissects critical features of our visual environment and is fundamental to visual function. Cone photoreceptor signals are sampled by several types of bipolar cells and passed onto the ganglion cells. At the output of retinal processing, retinal ganglion cells send about 40 different codes of the visual scene to the brain for further processing. In this study, the authors focus on whether subtype-specific differences in the size of synaptic ribbon-associated vesicle pools of bipolar cells contribute to different retinal ganglion cell (RGC) responses. Specifically, inputs to ON alpha RGCs producing transient versus sustained kinetics (ON-S vs. ON-T, respectively) are compared. The authors first demonstrate that ON-S vs. ON-T RGCs are readily identifiable in a whole mount preparation and respond differently to both static and to a spatially uniform, randomly fluctuating (Gaussian noise) light stimulus. Linear-nonlinear (LN) models were used to estimate the transformation between visual input and excitatory synaptic input for each RGCs; these models suggested the presence of transient versus sustained kinetics already in the excitatory inputs to ON-T and ON-S RGCs. Indeed, the authors show that (glutamatergic) excitatory inputs to ON-S vs. ON-T RGCs are of distinct kinetics. The subtypes of bipolar cells providing input to ON-S are known (i.e., type 6 and 7), but the source of excitatory bipolar inputs to ON-T RGCs needed to be determined. In a tedious process, it is elegantly shown here that ON-T RGCs receive most of their excitatory inputs from type 5 and 6 bipolars. Interestingly, the temporal properties of light-evoked responses of type 5, 6, and 7 bipolars recorded from the somas were indistinguishable and rather sustained, suggesting that the origin of transient kinetics of excitatory inputs to ON-T RGCs suggested by the LN model might be found in the processing of visual signals at the bipolar cell axon terminal. Blocking GABA- or glycinergic inhibitory inputs did not alter the light-evoked excitatory input kinetics to ON-T and ON-S RGCs. Twophoton glutamate sensor imaging revealed significantly faster kinetics of light-evoked glutamate signals at ON-T versus ON-S RGCs. Detailed EM analysis of bipolar cell ribbon synapses onto ON-T and ON-S RGCs revealed fewer ribbon-associated vesicles at ON-T synapses, which is consistent with stronger paired-flash depression of lightevoked excitatory currents in ON-T RGCs versus ON-S RGCs. This study suggests that bipolar subtype-specific differences in the size of synaptic ribbon-associated vesicle pools contribute to transient versus sustained kinetics in RGCs.

Strengths:

The use of multiple, state-of-the-art tools and approaches to address the kinetics of bipolar to ganglion cell synapse in an identified circuit.

Weaknesses:

For the most part, the data in the paper support the conclusions, and the authors were careful to try to address questions in multiple ways. Two-photon glutamate sensor imaging experiment showing that blocking GABA- and glycinergic inhibition does not change the kinetics of light-evoked glutamate signals at ON-T RGCs would strengthen the conclusion that bipolar subtype-specific differences in the size of synaptic ribbon-associated vesicle pools contribute to transient versus sustained kinetics in RGCs.

Thank you for this suggestion. We have revised the text throughout to be careful not to imply that amacrine cells have no role in shaping EPSCs and spike output, but instead that the transience of the On-T responses persists without amacrine cells (see for example lines 91, 450-453, 514-518, 696-714). We have also added additional iGluSnFR experiments to the paper to further test this conclusion (new Figure 7). The new data shows that the transience of glutamate release from the On-T cells is retained when 1) spiking amacrine cell activity is suppressed by blocking voltage-gated Na^+ channels with TTX or 2) all amacrine cell activity is

suppressed by blocking AMPA receptors with NBQX. This does provide nice additional evidence that amacrine cells are not necessary for the sustained/transient distinction.

Reviewer #2 (Public Review):

Summary:

Goal of the study. The authors tried to pinpoint the origins of transient and sustained responses measured at retinal ganglion cells (rgcs), which is the output layer of the retina. Response characteristics of rgcs are used to group them into different types. The diversity of rgc types represents the ability of the retina to transform visual inputs into distinct output channels. They find that the physical dimensions of bipolar cell's synaptic ribbons (specialized release sites/active zones) vary across the different types of cone on-bpcs, in ways that they argue could facilitate transient or sustained release. This diversity of release output is what they argue underlies the differences in on-rgcs response characteristics, and ultimately represents a mechanism for creating parallel cone-driven channels.

Strengths:

The major strengths of the study are the anatomical approaches employed and the use of the "glutamate sniffer" to assay synaptic glutamate levels. The outline of the study is elegant and reflects the strengths of the authors.

Weaknesses:

The major weakness is that the ambitious outline is not matched with a complete set of results, and the set of physiological protocols is disjointed, not sufficient to bridge the systems-level question with the presynaptic release question.

Thank you for this comment as it provides an opportunity (here and in the paper) for us to clarify our main goal. We wanted to link the well-established distinction between transient and sustained retinal responses to anatomy. This required locating where this difference arises within the circuitry – which we show to be at least largely the bipolar output synapse – and then examining the structure of this synapse in detail. While we would certainly be interested in connecting our results to a biophysical description of the synapse, that was not the primary focus of our study and was not something we could add without substantial additional work.

Major comments on the results and suggestions.

The ribbon model of release has been explored for decades and needs to be further adapted to systems-level work. The study under consideration by Kuo et al. takes on this task. Unfortunately, the experimental design does not permit a level of control over presynaptic/bpc behavior that is comparable to earlier studies, nor do they manipulate release in ways that test the ribbon model (i.e., paired recordings or Ribeye-ko). Furthermore, the data needs additional evaluation, and the presentation and interpretations should draw on published biophysical and molecular studies.

As described above, our goal was to test several possible explanations for the difference between transient and sustained responses in OnT and OnS ganglion cells: (1) differences in the light responses of the bipolar cells that convey photoreceptor signals to the relevant ganglion cells; (2) shaping of bipolar transmitter release by presynaptic inhibition; (3) shaping of ganglion cell responses by postsynaptic inhibition or spike generation; (4) differences in feedforward bipolar synapses. We were surprised to find that the feedforward bipolar synapses play a central role in this difference, and your comment nicely prompts us

to relate this to the large literature on biophysical studies of release from ribbon synapses. We have made substantial revisions in the text to do this. This includes anticipating the importance of feedforward synaptic properties in the abstract and introduction (lines 36-37 and 61-64), pointers in the results (lines 539-548), and several new paragraphs in the discussion (starting on lines 751, 773 and 787). By showing that the transient/sustained differences originates largely at feedforward bipolar synapses, we set the stage for future work that shows how biophysical properties of the synapse shape physiological signals that traverse it.

To build a ribbon-centric context, consider recent literature that supports the assertion that ribbons play a role in forming AZ release sites and facilitating exocytosis. Reference Ribeye-ko studies. For example, ribbonless bpcas show an 80% reduction in release (Maxeiner et al EMBO J 2016), the ribbonless retina exhibits signaling deficits at the output layer (Okawa et al ...Rieke, ..Wong Nat Comm 2019), and ribbonless rods show an 80% reduction the readily releasable pool (RRP) of SVs (Grabner Moser, elife 2021). In addition, the authors could refer to whole-cell membrane capacitance studies on mammalian rods, cones, and bpcas, because the size of the RRP of SVs scales with the dimensions and numbers of ribbons (total ribbon footprint). For comparison, bipolars see the review by Wan and Heidelberger 2011. For a comparison of mammalian rods and cones, see, rods: Grabner and Moser (2021 eLife), Mueller.. Regus Leidig et al. (2019; J Neurosci) and cones Grabner ...DeVries (Nat Comm 2023). A comparison of cell types shows that the extent of release is (1) proportional to the total size of the ribbon footprint, and (2) less release is witnessed when ribbons are deleted (also see photo ablation studies by Snellman.... And Mehta..Zenisek, Nat Neurosci and Neuron).

Thank you for these pointers into the literature. We have included much of this work in the revised Discussion (see three paragraphs starting on line 751). The revised text focuses on the evidence that larger and more numerous ribbons lead to increased release. The direct evidence from previous work for this relationship supports our (indirect) conclusions in the current paper about the role of ribbon size and associated vesicle pools in transient vs sustained responses.

Ribbon morphology may change in an activity-dependent manner. The rod ribbon AZ has been reported to lengthen in the dark (Dembla et al 2020), and deletion of the ribbon shortens the length of the AZ (defined by Cav1,4 or RIM2); in addition, the Ribeye-ko AZs fail to change in size with light and dark conditioning. Furthermore, EM studies on rod and cone AZs in light and dark argue that the number of SVs at the base of the ribbon increases in the dark, when PRs are depolarized (see Figure 10, Babai et al 2016 JNeurosci). Lastly, using goldfish Mb1 on-bipolars, Hull et al (2006, J Neurophysiol) correlated an increase in release efficiency with an increase in ribbon numbers, which accompanied daylight. >> When release activity is high, ribbon AZ length increases (Dembla, rods), the number of docked SVs increases (Babai, rods cones), and the number of ribbons increases (Hull, diurnal Mb1s).

We have extensively revised the discussion section to include more discussion of ribbons, particularly emphasizing evidence supporting the general argument that larger ribbons support higher release rates. We focused on studies that provided direct links between release rates and ribbon size or number of ribbon-associated vesicles. This includes studies that pair electrophysiology and anatomy and those that measure the consequences of ablating ribbons,

The results under review, Kuo et al., were attained with SBF-SEM, which has the benefit of addressing large-volume questions as required here, yet it achieves lower spatial resolution than what is attained with TEM tomography and FIB-EM. Ideally, the EM

description would include SV size, and the density of ribbon-tethered SVs that are docked at the plasma membrane, because this is where the SVs fuse (additional non-ribbon release sites may also exist? Mehta ... Singer 2014 J Neurosci). Studies by Graydon et al 2011 and 2014 (both in J Neurosci), and Jean ... Moser et al 2018 (eLife) are good examples of quantitative estimates of SVs docking sites at ribbons. SBF-SEM does not allow for an assessment of SVs within 5 nm of the PM, but if the authors can identify the number of SVs that appear within the limit of resolution (10 to 15 nm) from the PM, then this data would be useful. Also, what dimension(s) of the large ribbons make them larger? Typically, ribbons are fixed in height (at least in the outer retina, 200 to 250 nm), but their length varies and the number ribbons per terminal varies. Is the larger ribbon size observed in type 6 bpcs do to longer ribbons, or taller ribbons? A longer ribbon likely has more docked SVs. An additional possibility is that more SVs are about the ribbon-PM footprint, either more densely packed and/or expanding laterally (see definitions in Jean....Moser, elife 2018).

We have included an additional analysis of ribbon surface area from our 3D SBFSEM reconstructions. As with the volume measurements included in the original submission, ribbon surface areas are distinct between type 5i and type 6 bipolar cells (Fig. S10A), ON-T RGCs on average receive input from ribbons with smaller surface area than ON-S RGCs (Fig. S10B), and ribbon surface area predicts the number of adjacent vesicles across bipolar cell types (Fig. S10C). We agree that a higher resolution view of presynaptic structures would be very helpful, but the resolution of our SBF-SEM data is limited (e.g. each pixel is 40 nm on a side). This resolution does not allow us to distinguish between vesicles at vs near the membrane.

In our observations, both length and height of the ribbons showed variability across individual bipolar cells. And ribbons in type 6 bipolar cells tended to be either longer and/or taller compared to those in type 5 cells. We agree that a longer ribbon may accommodate more docked SVs. A more definitive analysis would benefit from higher-resolution, isotropic 3D reconstructions of ribbons, which would allow more precise shape analysis and ,together with a detailed assessment of docked SVs at the ribbons.

The ribbon literature given above makes the argument that ribbons increase exocytotic output, and morphological studies suggest that release activity enhances 1) ribbon length (Dembla) and 2) the density of SVs near the PM (Babai). These findings could lead one to propose that type 6 bpcs (inputs to On-sustained) are more active than type 5i (feed into On-transient). Here Kuo et al. show that the bpcs have similar Vm (measured from the soma) in response to light stimulation. Does Vm predict release? Not entirely as the authors acknowledge, because: Cav channel properties, SV availability, and negative feedback are all downstream of bpc Vm. The only experiment performed to test downstream factors focused on negative feedback from amacrine. The data presented in Figures 5C-F led me to conclude the opposite of what the authors concluded. My impression is that the T-ON rgc exhibits strong disinhibition when GABA-blockers are applied (the initial phase is greatly increased in amplitude and broadened with the drug), which contrasts with the S-On rgc responses that show a change in the amplitude of the initial phase but not its width (taus would be nice). Here and in many places the authors refer to changes in release kinetics, without implementing a useful description of kinetics. For instance, take the cumulative current (charge) in Figure 5C and fit the control and drug traces to arrive at taus, and their respective amplitudes, and use these values to describe kinetic phases. One final point, the summary in Figure 5D has a p: 0.06, very close to the cutoff for significance, which begs for more than an n = 5. Given that previous studies have shown that bpc output is shaped by immediate msec GABA feedback, in ways that influence kinetic phases of release (..Mb1 bipolars, see Vigh et al 2005 Neuron), more attention to this matter is needed before the authors rule out

feedback inhibition in favor of ribbon size. If by chance, type 5i bpcs are under uniquely strong feedback inhibition, then ribbon size may result from less activity, not less output resulting from smaller ribbons.

The text surrounding Figure 5 led to some confusion, and we have revised that text and the figure for clarity. First, the data in that figure is entirely from On-T cells (the upper and lower panels show block of GABA and glycine receptors separately). Second, the observation that we make there is that block of inhibitory receptors increases the transience of the On-T excitatory input, rather than decreasing it as would be expected if the transience is created by presynaptic inhibition. We have added additional data and that increase in transience is now significant. Inhibitory block does substantially increase the amplitude of the postsynaptic response, and a likely origin of this change in response is inhibitory feedback to the bipolar synaptic terminal. We now indicate this in the text on page 13, lines 438-453.

The key result of this figure for our purposes here is that the transience of the excitatory input to the OffT cell remains with inhibitory input blocked. We have clarified throughout the text that our results indicate that inhibitory feedback is not necessary for the difference between transient release into On-T and sustained release onto On-S. This does not mean that inhibitory feedback does not shape the responses in other ways or contribute to the transient/sustained difference - just that for the specific stimuli we use that difference is retained without presynaptic inhibition. We have also added citations to past work showing that activity of amacrine cells can modulate bipolar transmitter release.

Whether strong feedback inhibition limits activity and therefore limits ribbon size in an activity-dependent way is an intriguing possibility. Indeed, addressing why ribbons are larger in type 6 bipolar cells vs. other bipolar types will be an interesting avenue of further study. However, it would be surprising if ribbon sizes changed during the acute pharmacological block conditions (~10-15 minutes) we employed in our study. Our point here is that there is an interesting correlation between presynaptic ribbon size and the kinetics of glutamate release. We do not think that the two possibilities stated in the last sentence (“...ribbon size may result from less activity, not less output resulting from smaller ribbons”) are mutually exclusive.

We have not further quantified the response kinetics in the experiments of Figure 5 as the large changes induced by the pharmacology (especially GABA receptor block) make it unclear how to interpret quantitative differences. In other places we have quantified kinetics through the STA or specified that our focus was more qualitative (i.e. transient vs sustained kinetics).

As mentioned above, the behavior of Cav channels is important here. This is difficult to address with voltage clamps from the soma, especially in the Vm range relevant to this study. Given that it has previously been modeled that the rod bpc to AII pathway adapts to prolonged depolarization of rbcs through downregulating Cav channel-mediated Ca²⁺ influx (GrimesRieke 2014 Neuron), it seems important for Kou et al to test if there is a difference in Cav regulation between type 6 and 5i bpcs. Ca²⁺ imaging with a GCaMP strategy (Baden....Lagnado Current Biology, 2011) or filling the presynapse with Ca dyes (see inner hair cells: Ozcete and Moser, EMBO J 2020) would allow for the correlation of [Ca]_{intra} with GluSnf signals (both local readouts).

This is a good suggestion but is outside the scope of our current paper. Our focus was on the circuit origin of the difference in response of the OnT and OnS responses rather than the specific biophysical mechanism. We are of course interested in the mechanism, but the additional experiments needed to pin that down would need to be a part of future experiments. The work here represents an important step in that direction as it greatly reduces the number of possible locations and mechanisms for the sustained/transient difference and hence serves to focus any future mechanistic investigations.

Stimulation protocol and presentation of Glutamate Sniffer data in Figure 6. In all of your figures where you state steady state as a % of pk amplitude, please indicate in the figure where you estimate steady state. Alternatively, if you take the cumulative dF/F signal, then you can fit the different kinetic phases. From the appearance of the data, the Sustained Glu signals look like square waves (Figure 6B ROI1-4), without a transient at onset, which is not predicted in your ribbon model that assumes different kinetic phases (1. depletion of docked SVs, and 2. refilling and repriming). The Transient responses (Figure 6B ROI5-8) are transient and more compatible with a depressing ribbon scheme. If you take the cumulative, for all of the On-S and compare it to all of the On-T responses, my guess is the cumulative dF/F will be 10 to 20 larger for the S-On. Would you conclude that bpc inputs to On-S (type 6) release 20fold more SVs per 4 seconds on a per ribbon basis, and does the surface area of the type 6 bpcs account for this difference? From Figures 8B and D, the volume of the ribbon is ~2 fold greater for type 6 vs 5i, but the Surface Area (both faces of ribbon) is more relevant to your model that claims ribbon size is the pivotal factor. If making cumulative traces, and comparisons on an absolute scale is unfounded, then we need to know how to compare different observations. The classic ribbon models always have a conversion factor such as the capacitance of an SV or q size that is used to derive SV numbers from total dCm or Qcontent. See Kimet al von Gersdorff, 2023, Cell Reports. Why not use the Gaussian noise stimulus in Fig 6 as in Figure 1 and 2?

For iGluSnFR recordings, steady-state responses were measured from the mean fluorescence over the last 1 sec of the light step (2 sec duration) response. We have included this information in the figure caption and in the Methods.

There is a good deal of variability in the iGluSnFR responses from one ROI to another, and the ROIs shown in the original submission had a less prominent transient component than many other ROIs. We have replaced this figure with another that is more representative of the average behavior across ROIs. The full range of behavior is captured in Figure 6C; it is clear across ROIs that glutamate release near ON-S dendrites shows both sustained and transient components. The new experiments in which we block amacrine cell activity also include a few more example ROIs from ON-S cells, and those also show both transient and sustained components.

Your suggestion to integrate the iGluSnFR signals to compare to our structural analysis of ribbons is interesting. However, we are hesitant to make a quantitative comparison between the two without further experiments to validate how the iGluSnFR signals we measure relate to release of single vesicles. For example, a quantitative measure of release based on the iGluSnFR experiments would require accounting for possible differences in the expression of the indicator - which could differ both in overall level and/or location relative to release sites.

This comment and one above highlight the importance of measures of ribbon surface area, which we now provide (Figure S10).

Figure 7. What is the recovery time for mammalian cones derived from ribbon-based models? There are estimates from membrane capacitance studies. Ground squirrel cones take 0.7 to 1 sec to recover the ultrafast, primed pool of SVs when probed with a paired-pulse protocol (Grabner ...DeVries 2016, Neuron). Their off-bpcs take anywhere from under 0.2 sec to a second to recover, which is a combination of many synaptic factors (Grabner ...DeVries Nat Comm 2023). Rod On bpcs take over a second (Singer Diamond 2006, reviewed Wan and Heidelberger 2011). In Figure 7B, the recovery time is ~150 ms for the responses measured at rgcs. This brief recovery time is incompatible with existing ribbon models of release. Whole-cell membrane capacitance measurements would be helpful here.

Thanks for drawing our attention to this issue. Indeed, we see a relatively rapid recovery in the paired-flash experiments. We now discuss this recovery time in the context of past measurements of recovery of responses in cones and bipolar cells (paragraph starting on line 773). There are many factors that could contribute to the relatively rapid recovery we observe - including synaptic factors such as those highlighted by Grabner et al., (2016) either at the cone-to-bipolar synapses or the bipolar-to-RGC synapses. We are certainly interested in a more detailed understanding of this issue, but the additional experiments are outside the scope of this paper.

Experimental Suggestion: Add GABA blockers and see if type 5i bpc responds with more release (GluSniff) and prolonged [Ca2+] intra (GCaMP). Compare this to type 6 bpc behavior with GABA/gly blockers. This will rule in or out whether feedback inhibition is involved.

Figure 7 in the revised manuscript includes two new experiments examining glutamate release (without the simultaneous measurement of bipolar cell intracellular calcium) while blocking (1) all/most amacrine cell-mediated inhibition via inclusion of NBQX in the bath solution, and (2) blocking spiking amacrine cells via inclusion of TTX in the bath solution. The transient vs sustained difference in light-evoked glutamate release around ON-T and ON-S RGC dendrites remained with amacrine activity suppressed. These new results are consistent with the anatomical and pharmacological data that were included in the initial submission of the manuscript (Fig. 5) that indicate presynaptic inhibition does not have a major role in shaping release kinetics at these synapses.

Reviewer #3 (Public Review):

Summary:

Different types of retinal ganglion cell (RGC) have different temporal properties - most prominently a distinction between sustained vs. transient responses to contrast. This has been well established in multiple species, including mice. In general, RGCs with dendrites that stratify close to the ganglion cell layer (GCL) are sustained; whereas those that stratify near the middle of the inner plexiform layer (IPL) are transient. This difference in RGC spiking responses aligns with similar differences in excitatory synaptic currents as well as with differences in glutamate release in the respective layers - shown previously and here, with a glutamate sensor (iGluSnFR) expressed in the RGCs of interest. Differences in glutamate release were not explained by differences in the distinct presynaptic bipolar cells' voltage responses, which were quite similar to one another. Rather, the difference in transient vs. sustained responses seems to emerge at the bipolar cell axon terminals in the form of glutamate release. This difference in the temporal pattern of glutamate release was correlated with differences in the size of synaptic ribbons (larger in the bipolar cells with more sustained responses), which also correlated with a greater number of vesicles in the vicinity of the larger ribbons.

The main conclusion of the study relates to a correlation (because it is difficult to manipulate ribbon size or vesicle density experimentally): the bipolar cells with increased ribbon size/vesicle number would have a greater possibility of sustained release, which would be reflected in the postsynaptic RGC synaptic currents and RGC firing rates. This model proposes a mechanism for temporal channels that is independent of synaptic inhibition. Indeed, some experiments in the paper suggest that inhibition cannot explain the transient nature of glutamate release onto one of the RGC types. Still, it is surprising that such a diverse set of inhibitory interneurons in the retina would not play some role in diversifying the temporal properties of RGC responses.

Strengths:

(1) The study uses a systematic approach to evaluating temporal properties of retinal ganglion cell (RGC) spiking outputs, excitatory synaptic inputs, presynaptic voltage responses, and presynaptic glutamate release. The combination of these experiments demonstrates an important step in the conversion from voltage to glutamate release in shaping response dynamics in RGCs.

(2) The study uses a combination of electrophysiology, two-photon imaging, and scanning block-face EM to build a quantitative and coherent story about specific retinal circuits and their functional properties.

Weaknesses:

(1) There were some interesting aspects of the study that were not completely resolved, and resolving some of these issues may go beyond the current study. For example, it was interesting that different extracellular media (Ames medium vs. ACSF) generated different degrees of transient vs. sustained responses in RGCs, but it was unclear how these media might have impacted ion channels at different levels of the circuit that could explain the effects on temporal tuning.

We do not have an explanation for the quantitative differences in response kinetics we observed in Ames' medium vs. ACSF. There are modest differences in calcium and magnesium concentration and a larger difference in potassium (2.5 mM in ACSF vs 3.6 mM in Ames). It would be interesting to test which of these (or other) differences accounts for the difference in response kinetics.

(2) It was surprising that inhibition played such a small role in generating temporal tuning. At the same time, there were some gaps in the investigation of inhibition (e.g., IPSCs were not measured in either of the RGC types; pharmacology was used to investigate responses only in the transient RGCs).

We were also surprised at this result. We have included additional data on inhibition in the revised manuscript. Figure S3 shows light-evoked IPSC data from both RGC types (Fig. S3) and Fig. 7 shows additional iGluSnFR measurements around both ON-T and ON-S RGC dendrites with inhibition blocked via bath application of NBQX (Fig. 7) and separately with inhibition from spiking amacrine cells blocked with TTX. These experiments provide additional evidence for the small role of inhibition. We attempted to measure the kinetics of excitatory input to ON-S cells with inhibition blocked, but we found that the excitatory input showed strong spontaneous oscillations under these conditions and the light responses were changed so drastically that we did not feel we could make a clear comparison with control conditions.

(3) There could be additional discussion and references to the literature describing several topics, including: temporal dynamics of glutamate release at different levels of the IPL; previous evidence that release sites from a single presynaptic neuron can differ in their temporal properties depending on the postsynaptic target; previous investigations of the role of inhibition in temporal tuning within retinal circuitry.

Thanks, we have included more discussion and references to the relevant literature as you have suggested in the recommendations to authors.

Reviewer #1 (Recommendations For The Authors):

The presented raw data of the pharmacological experiments show that SR95531 and TPMPA robustly increased both the amplitude and duration of the transient component of the light step-evoked excitatory currents, with slight, if any enhancement of the sustained component in ON-T RGCs Figure 5C. Statistical analysis of the population data

($n=5$) with Wilcoxon signed rank test yielded no significant difference ($\ln 363$). However, reanalyzing the data extracted from the graph (Figure 5D) revealed that the difference between the paired observations is normally distributed (Shapiro-Wilk normality test, $P=0.48$) allowing parametric statistics to be used, which provides higher statistical power. Accordingly, reanalyzing the presented data with paired Student's t -test data revealed significant differences ($P=0.01$) in the steady-state amplitude normalized to that of the peak, recorded in the presence of SR95531 and TPMPA. In other words, based on the (rough) analysis of the presented pharmacology data GABAergic feedback inhibition significantly contributes to shaping the transient portion of the light-evoked excitatory currents in ON-T RGCs, by making it more transient. I believe a similar analysis based on the actual data is necessary, and the results should be communicated either way. However, if warranted, two-photon glutamate sensor imaging experiments showing that blocking GABA- and glycinergic inhibition does not change the kinetics of light-evoked glutamate signals at ON-T RGCs should also be performed, as these would be critical in drawing a conclusion regarding the effect of feedback inhibition on glutamate release from bipolar cells.

Thanks for this feedback. We have added another cell to the data set in Fig. 5D. With this addition, SR95531/TPMPA application significantly increases the response transience of excitatory currents measured in ON-T RGCs compared to control. This enhanced transience in GABA_{A/C} receptor blockers is due to an increase in the amplitude of the initial peak component of the response (control peak amplitude: -833.7 ± 103.3 pA; SR95531+TPMPA peak amplitude: 2023 ± 372.7 pA; $p=0.03$, Wilcoxon signed rank test), with no change to the later sustained component (control plateau amplitude: -200.7 ± 14.71 pA; SR95531+TPMPA plateau amplitude: -290.9 ± 43.69 pA; $p=0.15$, Wilcoxon signed rank test).

We should clarify that this result indicates that GABAergic inhibition makes the excitatory inputs to ON-T RGCs less transient. Block of GABA receptors increased transience, thus intact GABAergic transmission appears to limit the initial peak of the response and therefore make excitatory currents more sustained. We unfortunately were not able to examine whether sustained excitatory currents in ON-S RGCs would become more transient using the same approach. In our hands, bath application of SR95531+TPMPA led to the generation of large-amplitude (>1 nA) oscillatory bursts of excitatory input that developed within 5 minutes and persisted for the duration of the incubation (up to ~ 30 min) in drugs. Further, presentation of light steps tended to induce variable amplitude responses, likely dependent on the presence of spontaneous bursts; when large amplitude responses were evoked, these typically oscillated for several seconds after the step.

To examine a potential role for presynaptic inhibition in transient vs. sustained bipolar cell output, we therefore chose to eliminate amacrine cell-mediated inhibition by bath application of the AMPA/kainate receptor antagonist NBQX in additional iGluSnFR measurements. This manipulation should leave ON bipolar cell responses intact while eliminating most amacrine cell-mediated responses (and OFF bipolar cell driven responses). In separate experiments, we also eliminated inhibition from spiking amacrine cells by bath application of TTX. As shown in new Fig. 7, sustained and transient responses persisted in distal versus proximal RGC dendrites, respectively. Compared to SR95531/TPMPA, bath application of NBQX was not associated with spontaneous bursts of glutamate release around ON-S dendrites. These results show that amacrine cell-mediated inhibition is not required for either sustained or transient glutamate release from bipolar cells that provide input to ON-S and ON-T RGCs.

Small points:

(1) The legend of Figure 1 (D) refers to shaded areas to show {plus minus} SEM, but no shade is visible (at least in my printout).

The SEM shading is there in Fig. 1D but is mostly obscured by the mean lines for the respective RGC types. We have added this to the figure caption.

(2) *I found the reported V_{rest} for the ON bipolar cells somewhat depolarized. Perhaps due to the uncompensated junction potentials?*

These measurements are indeed not corrected for the liquid junction potential (which is approximately -10.8 mV between K-gluconate internal and Ames' solution). We did not apply this correction since the appropriate value is not clear in perforated patch recordings as the intracellular chloride concentration is unknown (and can differ from that in the pipette solution). We have clarified this in the results text where we describe the V_{rest} values (lines 335-338).

(3) *It is Wilcoxon signed rank test, not Wilcoxon.*

Thanks for catching this. This has been corrected in the revised manuscript.

Reviewer #2 (Recommendations For The Authors):

Some amacrine express vesicular Glut-3 transporter and are reported to release glutamate (Marshak, Vis Neurosci 2016). Are Amacrine vGluT3 signals postsynaptic (within ~0.5 μ m) to cone bpc ribbons?

We did not characterize VgluT3-expressing amacrine cells in our SEM datasets. A recent study by Friedrichson et al. (Nat. Comm. 2024; PMID 38580652) using 3D SEM reconstructions found that Vglut3-amacrine are postsynaptic to both type 5i and type 6 bipolar cells, as well as other type 5/xbc bipolar cells (and receive >50% of their input from type 3a OFF bipolar cells).

How far apart are the postsynaptic targets from the ribbon release sites? The ribbons at type 5i bpc/On-T input appear separated from the dendrites of On-T rgcs (Figure 8C). At least further away than the type 6 bpc ribbons are from On-S rgc dendrites (Figure 8C). Distance may create a thresholding phenomenon, whereby only multivesicular bouts at the onset of depolarization are able to elevate synaptic Glu to levels needed to activate On-T GluRs. See Grabner et al Nat Comm 2023 for such scenarios in the outer retina.

This is an intriguing possibility, but we should point out that the presynaptic ribbons in Fig. 9C (former Fig. 8C) are similar distances (within the resolution of our reconstructions) from the ON-T and ON-S dendrites. We have increased the brightness of the dendrite segments for both RGC types in the resubmission figure; note that ON-T RGCs have spine-like protrusions that may not have been as apparent in the previously submitted version of our manuscript.

In Figures 1 and 2, Sustained responses look like the derivative of Transient responses, minus the negative going inflection. In addition, the sustained responses appear to have a lower threshold of activation than the transient On rgcs, because there are more bouts of action potentials (and membrane depol in V-clamp) with earlier onset in sustained than transients traces. It would be great if the GLuSniff data captured these differences. Take cumulative dF/F and see what the onset time is, or an initial tau if possible.

This is a good suggestion. However, we are reluctant to make detailed quantitative comparisons such as this without further validation of how the kinetics of the iGluSnFR signals relate to kinetics of glutamate release. A specific concern is that differences in the location and amount of iGluSnFR expression could impact any such comparisons.

A recent study by Kim et al von Gersdorff (Cell Reports, 2023) presents interesting phases of release in response to light flashes, measured from AIIIs, and complementary results from pairs of rbc-AIIs. The findings highlight the complexity of SV pools under well-controlled experiments. Could their results be explained as variations in rbc ribbon size through development, and possibly between rbc or within an rbc?

This certainly seems possible and would be consistent with the dependence of release on ribbon size that our results support. It would be interesting to see if there are clear anatomical correlates of that change in release properties.

Figure 5 is a pivotal point in the study, but my review has identified numerous weaknesses. The feedback inhibition onto bipolar cell terminals is likely to sculpt glutamate release, and the results do not convincingly rule out this possibility. The suggestions for improvements range from the data needing to be reanalyzed with regard to statistical tests, and/or adding a few more data points ($n = 5$) before concluding a p : 0.06 is insignificant.

We have added an additional recording to this data set. With $n = 6$ cells, there is now a statistically significant difference between ON-T RGC excitatory currents measured in control conditions versus during GABA_{A/C} receptor blockade. Please note that all the recordings shown in Figure 5C-F are from ON-T RGCs (the two panels show separately block of GABAergic and glycinergic receptors). We did not make it sufficiently clear that the original trend (now statistically significant) is opposite of that expected if presynaptic GABAergic inhibition contributes to response transience in ON-T RGCs. What we see is that excitatory synaptic inputs to ON-T RGCs become more transient (rather than more sustained) during GABA_{A/C} receptor blockade. We have revised the text in that section to make this point more clearly.

We have also included new data from iGluSnFR measurements showing that bath application of NBQX does not affect light step-evoked glutamate release kinetics at proximal (sustained) or distal (transient) RGC dendrites (control: steady-state amp. as % of peak amp. 13 ± 10 ; mean $\pm$ S.D.; $n = 189$ ROIs/4 FOVs for ON-T dendrites vs 40 ± 12 ; mean $\pm$ S.D.; $n = 287$ ROIs/8 FOVs for ON-S dendrites; NBQX: 6 ± 3 ; mean $\pm$ S.D.; $n = 112$ ROIs/1 FOV for ON-T dendrites vs 23 ± 9 ; mean $\pm$ S.D.; $n = 97$ ROIs/2 FOVs for ON-S dendrites; $*p < 0.001$). By blocking glutamate receptors on amacrine cells, NBQX (AMPA/KAR antagonist) eliminates all/most amacrine cell-mediated signaling in the retina and should therefore abolish presynaptic inhibitory input to bipolar cell terminals across the IPL. Taken together, our results indicate that presynaptic inhibition does not play a critical role in establishing transient versus sustained kinetics for the stimulus conditions we employed in our study.

There is a need to cite more recent literature on bipolar cell ribbons (e.g. see Wakeham et al., Front. Cell. Neurosci., 2023), in order to support experimental design and interpretation of the results. The authors should discuss their Ribeye-KO data from Okawa et al 2019 Nat Comm, Figure 7, in the context of their new iGluSnFR results.

Thank you for prompting us on this issue. We have expanded the discussion regarding ribbons and included more citations to the ribbon literature. That is largely in the three paragraphs starting on line 727.

One point deserves emphasis because it is central to the authors' ribbon model but not consistent with their data. The ribbon model as they put it, and as commonly stated, holds that a transient phase of release at the onset of depolarization indicates the depletion of the primed SVs, and the subsequent slower rate of release (steady state release in the authors' terms) reflects recruiting, priming, and release of new SVs. The

On-transient dendrite GluSnf responses agree with this multiphasic process, but the sustained responses show only an elevation in glutamate without a pronounced initial peak, creating a square-wave-shaped response (Figure 6B). This does not agree with the simple ribbon-based release model. I would expect the signals from the T- and S-on dendrites to have a comparable initial phase, while the sustained phase should be greater in amplitude for the S-on dendrites. More discussion may clarify possible mechanisms.

Thanks for pointing this out. The example iGluSnFR traces we originally included in the manuscript were not entirely representative in that they did not show much initial transient phase. Note there is a distribution of steady-state amplitudes for proximal dendrites in Fig. 6C; the examples are from ROIs from the upper end of the distribution. In the new Figure 7, we have included some additional examples that show both a clear transient and sustained component. The summary data in Figure 6C shows the distribution of sustained/transient ratios across ROIs.

Reviewer #3 (Recommendations For The Authors):

(1) It would be interesting to understand the differences in IPSCs in the two RGC types. Perhaps they are small in both types, which would explain their apparent lack of impact on temporal tuning. The authors may already have these data.

We did make measurements of noise-evoked IPSCs (as well as EPSCs) in a subset of ON-T and ON-S recordings. We have now included this data as Figure S3. There are slight differences in the kinetics of inhibition between RGC types (Fig. S3C) and there is a trend towards stronger inhibition (relative to excitation) in ON-T RGCs compared to ON-S RGCs (Fig. S3E), although there is not a statistically significant difference. In both cases excitatory synaptic currents are as large or larger than inhibitory currents, and this does not include the difference in driving force near spike threshold which will favor excitatory input by a factor of 2-3. Hence our data suggests that postsynaptic inhibition does not play a major role in generating the differential temporal spiking responses of ON-T and ON-S RGCs. However, additional experiments examining the relative contribution of excitation and inhibition to spiking output in these RGCs would be needed to reach a firm conclusion.

The pharmacological experiments in which we blocked inhibition (Fig. 5C-F, new Fig. 7) were designed to test the effect of presynaptic inhibition on bipolar cell output (voltage-clamp isolation of excitatory currents in Fig. 5; iGluSnFR measurements of glutamate release in Fig. 7). We do not mean to suggest that postsynaptic inhibition does not have any role in shaping the spiking behavior of these RGC types, but that transient vs. sustained kinetics are already present in the bipolar cell output and that presynaptic inhibition of bipolar cell terminals does not appear to account for this difference. We have revised the text throughout to be clearer on this point.

(2) It could be convincing to show transient/sustained differences between RGC types in dim light, where the response would depend on the rod bipolar/AII circuit. In this case, any difference in temporal properties would presumably be explained by differences that localize to the cone bipolar cell axon terminals. Indeed, is that the result in Figure 1B? This seems to be a dim stimulus presented on darkness, which may be driven through the rod bipolar pathway. The authors could then discuss the interpretation of this data in terms of the rod bipolar circuit.

Yes, Figure 1B is a dim light step ($\sim 30R^*/\text{rod/s}$) presented from darkness and the distinction between cells is clear down at still lower light levels that more effectively isolate signaling through the rod bipolar pathway. Thanks for making this point that observation of distinct temporal responses under scotopic conditions where signals suggests these differences must

arise at and/or downstream of cone bipolar cell output. We have included additional text (lines 361-365) in the results describing bipolar cell responses that raise this point.

(3) Glutamate release was already measured across the full IPL depth by Borghuis et al. (2013) and Franke et al. (2017). It would be appropriate to better motivate the current study based on these existing measurements.

We have clarified that these important studies provided important motivation for measuring excitatory synaptic input to ON-T vs. ON-S RGCs (lines 165-169).

(4) Line 212/213. It would be appropriate to add to the list of papers showing the different stratification of transient vs. sustained responses: Borghuis et al. (2013) and Beaudoin et al. (2019).

Thank you - these references have been added.

(5) Line 635-638. It would be useful to discuss papers by Pottackal et al. (2020, 2021), which suggested that a single presynaptic cell (starburst) can signal with different temporal properties depending on the postsynaptic target (other starburst vs. DSGCs). The mechanism was not completely resolved (i.e., it was not explained by differences in presynaptic Ca channels at the two synapse types), but it at least shows that neurotransmitter release can show different filtering depending on the postsynaptic target from the same presynaptic neuron. (This could also be at play for the type 6 bipolar cell inputs to ON-S vs. ON-T RGCs in the present study.)

We have added a reference to Pottackal et al 2021 in this section.

(6) Line 714. Should describe the procedure for embedding the tissue in agarose.

We have added more detail regarding agarose embedding for preparation of retinal slices in the methods.

(7) Line 775. Need a better description of the virus (not the construct), what serotype? Provide the Addgene number if available.

This has been added to the methods.

(8) Line 808. Was the SD for the gaussian really 50%? That would cut off a lot of the distribution, i.e., it would get clipped at 0.

Yes, the SD for Gaussian noise was 50%. This high contrast stimulus was used in part to achieve measurable signals from bipolar cells. You are correct that some of the distribution was clipped at 0 (it was also clipped at twice the mean to make sure that the distribution remained symmetrical). The clipping was accounted for during our LN analyses.

(9) The paper should discuss Swygart et al. (2024) results showing different spatial surround properties of neighboring synapses from a type 6 bipolar cell. Based on this result, it would seem very likely that amacrine cells could play a role in shaping the temporal processing of bipolar cell glutamate release as well. Indeed, spatial and temporal processing will not be completely independent in a typical experiment. For example, with the spot stimulus used in the present study, bipolar cells within the center versus the edge of the spot will have different balances of center/surround activation, which could potentially influence their temporal processing.

We have included discussion of results from Swygart et al 2024 in the section of the Discussion in which we point out differences in surround inhibition between ON-S and ON-T RGCs (lines 710-714). We agree that spatial and temporal processing are not completely independent. Our results with SR95531/TPMPA indicate ON-T RGCs receive stronger GABAergic surround inhibition than ON-S RGCs (Fig. S8). However, our results in Fig. 5C-D show GABAergic surround inhibition makes ON-T excitation more sustained rather than more transient. So even though bipolar cells presynaptic to ON-T RGCs receive stronger surround inhibition (Fig. S8), this inhibition does not establish the transient kinetics of glutamate release from these bipolar cells (in fact, it works to make release more sustained). Additional iGluSnFR experiments where we used NBQX to block all/most amacrine cell-mediated responses also suggest presynaptic inhibition does not have an important role in establishing differential glutamate release kinetics onto ON-S vs. ON-T RGC dendrites (Fig. 7).

(10) Cui et al. 2016 described ON-S Alpha as having a divisive suppression mechanism that explained the temporal properties of white-noise response better than a standard LN model. Do the authors think the divisive suppression reflects a property of the excitatory synapses independent of inhibition?

This is an interesting question, but one for which we don't have a good answer for now. As mentioned in some of the above responses and as we have tried to clarify in the manuscript, we do not mean to imply that there is no role for presynaptic inhibition in modulating bipolar cell output, including for the divisive suppression described by Cui et al. Rather, our point is that the distinction between transient and sustained excitatory input to ON-T and ON-S RGCs does not require presynaptic inhibition and is more likely an intrinsic property of the bipolar cell synapses.

(11) Do the authors mean to imply that the pool size at bipolar cell ribbon synapses could depend on the use of Ames vs. ACSF?

For now, we do not have a good answer as to why there are quantitative differences in response kinetics between Ames and ACSF. We have not done any experiments to investigate whether ribbon sizes or ribbon pools are different in the different solutions.

(12) More generally, different mean luminance levels could drive different levels of baseline glutamate release, which could alter the available pool of vesicles at bipolar cell ribbon synapses. Can we explain varying degrees of transient/sustained in the same cell at different levels of mean luminance based on this mechanism (e.g., Grimes et al., 2014)?

Yes, the emergence of a transient component of excitatory input to ON-S RGCs at ~ 100 R*/rod/s versus at scotopic levels (0.5 R*/rod/s) in Grimes et al. (2014) could be due to differences in the number of releasable vesicles (due to different type 6 bipolar cell axon terminal membrane potentials and hence differences in spontaneous release rates) at the different light levels.

We should note that although ON-T and ON-S RGCs exhibit some changes in transient/sustained kinetics across different light levels, the relative differences between these RGC types are preserved across light levels. We have included a statement about this in the text (lines 361-367).

(13) Figure 1. Have the authors considered performing the LN analysis of the firing responses, to compare the degree of rectification between the two RGC types?

This is a good suggestions. From an LN analysis of spiking responses, we do not observe a clear difference between the static nonlinearity component of the model for ON-T and ON-S

RGCs. Both RGC types are strongly rectified under our experimental conditions.

(14) Figure 5. Do the authors have the pharmacology data for the ON-S cells? There are examples of sustained EPSCs in amacrine cells that become more transient after blocking inhibition, which at least suggests that inhibition can play some role in the transient/sustained nature of glutamate release (Park et al., 2015, Figure 3). Perhaps ON-S cells likewise become more transient with inhibition blocked.

(The colored symbols in A were not visible in a printout. It would be useful to indicate the cell type (ON-T) in C and E).

As described above in the response to reviewer 1's recommendation for authors, we were not able to use SR95531/TPMPA for recordings from ON-S RGCs. Bath application of these drugs led to oscillatory bursts of excitatory input to ON-S RGCs. However, the lack of effect of bath-applied NBQX on the kinetics of glutamate release around either ON-T or ON-S RGC dendrites (new Fig. 7) suggests that presynaptic inhibition does not contribute to generating sustained excitation to ON-S RGCs (or transient excitation to ON-T RGCs).

We have corrected Fig. 5A to include the referenced colored symbols and have also edited Fig 5C and E to clarify that measurements in Fig. 5C-F are from ON-T RGCs.

(15) Figure 6 legend. Should be Kcng4-Cre, not KCNG-Cre. Also, it should make clear that this is cre-dependent expression of iGluSnFR. For C, were the statistics based on the number of FOVs?

Thanks for catching this, we have corrected Figure 6 legend. The methods section includes a description of how we achieved iGluSnFR expression on alpha RGC dendrites via a cre-dependent viral strategy in Kcng4-Cre mice. We have also clarified that the statistics are based on ROIs in Figure 6C.

(16) Figure 7, Flashes were apparently 400% contrast on a dim background. What was the background? Is there a rod component to the response in this case?

In Figure 7 (now Figure 8), the same background (~ 3300 R*/rod/s; 2000 P*/Scone/s) was used as in the Gaussian noise and step response experiments. At this light level, the response should be primarily be mediated by cones.

(17) Figure S1. The colors here differ from those in previous figures (Here, ON-T, magenta; ON-S, cyan). Is something mislabeled?

Thanks for catching this. We mistakenly swapped the labels in the legend for Fig. S1. The figure colors were correct, but we have corrected the legend in the revised manuscript.

(18) Figure S2. For the LN model for RGC synaptic currents, the ON-S are more rectified than some previous recordings (Cui et al., 2016). Is this perhaps explained by different light levels?

We aren't sure why ON-S excitatory currents are more strongly rectified in our recordings compared to Cui et al., 2016. Cui et al. used an ~ 20 -fold higher background light intensity ($\sim 40,000$ P*/cone/s vs. ~ 2000 P*/cone/s in our study), so different light levels may be a factor (although we should point out that rectification increases in these RGCs between scotopic to low photopic light levels (see Grimes et al., 2014 and Kuo et al., 2016)).

(19) The study is apparently comparing PV1 and PV2 described in Farrow et al. (2013; see Supplementary information for stratification analysis), which should be cited.

Thanks, we have corrected this oversight in the revised manuscript. We now cite Farrow et al and mention the connection to PV1 and PV2 in the first paragraph of Results (lines 104-108).

<https://doi.org/10.7554/eLife.98817.2.sa0>